Theory of Mathematics

Volume V: Commutative Algebra and Homological Methods

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Part I

Rings and Ideals

Chapter 1

Rings, Ideals and Quotients

Commutative algebra begins with rings, ideals, and quotients. Ideals are exactly the additive subgroups that can serve as kernels of ring homomorphisms, and quotient rings convert congruence conditions into algebraic objects.

Learning goals

After this chapter, the reader should be able to:

- verify that a subset is an ideal and compute ideals generated by explicit elements;
- construct quotient rings and identify their elements and operations concretely;
- compute kernels and images of ring homomorphisms and apply the first isomorphism theorem;
- translate ideal containment in a quotient through the correspondence theorem;
- compute sums and intersections of ideals in elementary rings;
- distinguish ring, subring, ideal, and quotient constructions by examples and counterexamples;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

1.1 Commutative rings

A commutative ring has associative addition and multiplication, a multiplicative identity, and commutative multiplication.

1.2 Ideals

An ideal is an additive subgroup stable under multiplication by arbitrary ring elements.

Example 1.1 (Quotient arithmetic)**.** In $R = \mathbb{Z}/12\mathbb{Z}$, the ideal (4) is $\{0, 4, 8\}$. The reduction map $R \to \mathbb{Z}/4\mathbb{Z}$ is surjective with kernel (4), so $R/(4) \cong \mathbb{Z}/4\mathbb{Z}$.

1.3 Generated ideals

The ideal generated by elements $a_1, \dots, a_n$ consists of finite combinations $r_1 a_1 + \cdots + r_n a_n$.

1.4 Sums and intersections

Sums give the smallest ideal containing two ideals, while intersections give their greatest common subideal.

1.5 Quotient rings

For an ideal $I \subset A$, the quotient A/I has cosets as elements and multiplication induced from A.

Example 1.2 (Polynomial quotient). In $k[x,y]/(x,y^2)$ every class has a unique representative $a + by$. The class of y is nonzero but satisfies $y^2 = 0$, so the quotient exhibits a nilpotent direction explicitly.

1.6 Ring homomorphisms

Kernels are ideals and images are subrings; quotienting by the kernel isolates the injective part of a map.

1.7 Correspondence of ideals

Ideals of A/I correspond to ideals of A containing I.

Example 1.3 (Evaluation kernel). Evaluation at 1 gives $\varphi : k[x] \to k$. Polynomial division shows $\ker \varphi = (x-1)$, and constants show surjectivity. Hence $k[x]/(x-1) \cong k$.

1.8 Examples

Polynomial rings, integer residue rings, coordinate-style quotients, and product rings provide the main laboratory.

1.9 Core structural results

Theorem 1.4 (First isomorphism theorem). *For a ring homomorphism $\varphi : A \to B$, there is a natural isomorphism $A/\ker \varphi \cong \operatorname{im} \varphi$.*

Proof. Send the class of a to $\varphi(a)$. Equality of images is exactly equality modulo the kernel, so the map is well-defined and injective; surjectivity onto the image is immediate. □

Theorem 1.5 (Correspondence theorem). *Ideals of A/I are in inclusion-preserving bijection with ideals $J \subset A$ satisfying $I \subseteq J$.*

Proof. Take inverse image under the quotient map in one direction and quotient J/I in the other. The two constructions are inverse by direct inspection. □

Theorem 1.6 (Third isomorphism theorem). *If $I \subseteq J \subseteq A$ are ideals, then $(A/I)/(J/I) \cong A/J$.*

Proof. Apply the first isomorphism theorem to the homomorphism $A/I \to A/J$ sending $a+I$ to $a+J$; its kernel is J/I. □

1.10 Worked examples

Example 1.7 (Commutative rings diagnostic)**.** Give a rigorous worked analysis of Commutative rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A commutative ring has associative addition and multiplication, a multiplicative identity, and commutative multiplication. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 1.8 (Ideals diagnostic)**.** Give a rigorous worked analysis of Ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An ideal is an additive subgroup stable under multiplication by arbitrary ring elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 1.9 (Generated ideals diagnostic)**.** Give a rigorous worked analysis of Generated ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The ideal generated by elements $a_1, \ldots, a_n$ consists of finite combinations $r_1a_1 + \cdots + r_na_n$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 1.10 (Sums and intersections diagnostic)**.** Give a rigorous worked analysis of Sums and intersections. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Sums give the smallest ideal containing two ideals, while intersections give their greatest common subideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

1.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 1.11 (Commutative rings diagnostic)**.** Give a rigorous worked analysis of Commutative rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A commutative ring has associative addition and multiplication, a multiplicative identity, and commutative multiplication. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 1.12 (Ideals diagnostic)**.** Give a rigorous worked analysis of Ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An ideal is an additive subgroup stable under multiplication by arbitrary ring elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 1.13 (Generated ideals diagnostic)**.** Give a rigorous worked analysis of Generated ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The ideal generated by elements $a_1, \ldots, a_n$ consists of finite combinations $r_1 a_1 + \cdots + r_n a_n$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 1.14 (Sums and intersections diagnostic)**.** Give a rigorous worked analysis of Sums and intersections. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Sums give the smallest ideal containing two ideals, while intersections give their greatest common subideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 1.15 (Quotient rings diagnostic)**.** Give a rigorous worked analysis of Quotient rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For an ideal $I \subset A$, the quotient A/I has cosets as elements and multiplication induced from A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 1.16 (Ring homomorphisms diagnostic)**.** Give a rigorous worked analysis of Ring homomorphisms. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Kernels are ideals and images are subrings; quotienting by the kernel isolates the injective part of a map. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 1.17 (Correspondence of ideals diagnostic)**.** Give a rigorous worked analysis of Correspondence of ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Ideals of A/I correspond to ideals of A containing I. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 1.18 (Examples diagnostic)**.** Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Polynomial rings, integer residue rings, coordinate-style quotients, and product rings provide the main laboratory. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 1.19 (First isomorphism theorem proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For a ring homomorphism $\varphi : A \to B$, there is a natural isomorphism $A/\ker\varphi \cong \operatorname{im}\varphi$.

Solution. Send the class of a to $\varphi(a)$. Equality of images is exactly equality modulo the kernel, so the map is well-defined and injective; surjectivity onto the image is immediate. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 1.20 (Correspondence theorem proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Ideals of A/I are in inclusion-preserving bijection with ideals $J \subset A$ satisfying $I \subseteq J$.

Solution. Take inverse image under the quotient map in one direction and quotient J/I in the other. The two constructions are inverse by direct inspection. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 1.21 (Third isomorphism theorem proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $I \subseteq J \subseteq A$ are ideals, then $(A/I)/(J/I) \cong A/J$.

Solution. Apply the first isomorphism theorem to the homomorphism $A/I \to A/J$ sending $a + I$ to $a + J$; its kernel is J/I. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 1.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Commutative algebra begins with rings, ideals, and quotients. Ideals are exactly the additive subgroups that can serve as kernels of ring homomorphisms, and quotient rings convert congruence conditions into algebraic objects. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

1.12 Exercises with complete solutions

Exercise 1.23. State and apply the central principle behind Commutative rings. Give one concrete algebraic consequence.

Hint. Check closure under subtraction and multiplication by arbitrary ring elements; both are needed for an ideal, not merely a subring. □

Solution. A commutative ring has associative addition and multiplication, a multiplicative identity, and commutative multiplication. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 1.24. State and apply the central principle behind Ideals. Give one concrete algebraic consequence.

Hint. To identify an ideal generated by elements, write a general finite ring-linear combination of those generators. □

Solution. An ideal is an additive subgroup stable under multiplication by arbitrary ring elements. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 1.25. State and apply the central principle behind Generated ideals. Give one concrete algebraic consequence.

Hint. For sums and intersections, prove both containments against the universal smallest/largest ideal descriptions. ☐

Solution. The ideal generated by elements $a_1, \ldots, a_n$ consists of finite combinations $r_1 a_1 + \cdots + r_n a_n$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 1.26. State and apply the central principle behind Sums and intersections. Give one concrete algebraic consequence.

Hint. In a quotient, two representatives are equal exactly when their difference lies in the ideal; use that before computing operations. ☐

Solution. Sums give the smallest ideal containing two ideals, while intersections give their greatest common subideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 1.27. State and apply the central principle behind Quotient rings. Give one concrete algebraic consequence.

Hint. Compute the kernel first when applying an isomorphism theorem; the quotient by that kernel is the canonical source of injectivity. ☐

Solution. For an ideal $I \subset A$, the quotient A/I has cosets as elements and multiplication induced from A. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 1.28. State and apply the central principle behind Ring homomorphisms. Give one concrete algebraic consequence.

Hint. For a ring map, distinguish image as a subring from kernel as an ideal before invoking quotient structure. ☐

Solution. Kernels are ideals and images are subrings; quotienting by the kernel isolates the injective part of a map. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 1.29. State and apply the central principle behind Correspondence of ideals. Give one concrete algebraic consequence.

Hint. Lift an ideal of A/I to its inverse image in A and check that it contains I; quotienting back should recover the original ideal. ☐

Solution. Ideals of A/I correspond to ideals of A containing I. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 1.30. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. Use polynomial, integer-residue, or product-ring examples to test whether an asserted ideal or quotient statement really holds. ☐

Solution. Polynomial rings, integer residue rings, coordinate-style quotients, and product rings provide the main laboratory. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

1.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 1.31 (Integer generated ideal)**.** Find $(18, 30)$ in $\mathbb{Z}$.

Hint. Use kernels, generated ideals, and quotient universal properties. ☐

Solution. $(18, 30) = (6)$. ☐

Exercise 1.32 (Residue quotient)**.** How many classes are in $\mathbb{Z}/(15)$?

Hint. Use kernels, generated ideals, and quotient universal properties. ☐

Solution. There are 15 classes. ☐

Exercise 1.33 (Polynomial reduction)**.** Reduce $x^4 + 2x + 3$ modulo (x^2).

Hint. Use kernels, generated ideals, and quotient universal properties. ☐

Solution. The class is $2x + 3$. ☐

Exercise 1.34 (Kernel)**.** Find the kernel of $\mathbb{Z} \to \mathbb{Z}/7\mathbb{Z}$.

Hint. Use kernels, generated ideals, and quotient universal properties. ☐

Solution. The kernel is $7\mathbb{Z}$. ☐

Exercise 1.35 (Generated ideal)**.** Describe $(x, 2)$ in $\mathbb{Z}[x]$.

Hint. Use kernels, generated ideals, and quotient universal properties. ☐

Solution. It consists of $xf(x) + 2g(x)$. ☐

Proofs

Exercise 1.36 (First isomorphism theorem proof)**.** Prove: For a ring homomorphism $\varphi : A \to B$, there is a natural isomorphism $A/\ker\varphi \cong \operatorname{im}\varphi$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Send the class of a to $\varphi(a)$. Equality of images is exactly equality modulo the kernel, so the map is well-defined and injective; surjectivity onto the image is immediate. ☐

Exercise 1.37 (Correspondence theorem proof)**.** Prove: Ideals of A/I are in inclusion-preserving bijection with ideals $J \subset A$ satisfying $I \subseteq J$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Take inverse image under the quotient map in one direction and quotient J/I in the other. The two constructions are inverse by direct inspection. ☐

Exercise 1.38 (Third isomorphism theorem proof)**.** Prove: If $I \subseteq J \subseteq A$ are ideals, then $(A/I)/(J/I) \cong A/J$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Apply the first isomorphism theorem to the homomorphism $A/I \to A/J$ sending $a + I$ to $a + J$; its kernel is J/I. ☐

Exercise 1.39 (Structural synthesis)**.** Prove a short corollary by combining First isomorphism theorem with Correspondence theorem. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes First isomorphism theorem and then applies Correspondence theorem; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 1.40 (Union test)**.** Test the claim: $2\mathbb{Z} \cup 3\mathbb{Z}$ is an ideal.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: 2 and 3 belong but 5 does not. ☐

Exercise 1.41 (Principal ideal test)**.** Test the claim: Every ideal of $k[x, y]$ is principal.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: (x, y) is not principal. ☐

Exercise 1.42 (Field quotient test)**.** Test the claim: $\mathbb{Z}/6\mathbb{Z}$ is a field.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: $2 \cdot 3 = 0$ modulo 6. ☐

Applications and investigations

Exercise 1.43 (Congruence model)**.** Explain quotient rings as arithmetic modulo n.

Hint. Translate the situation into kernels, generated ideals, and quotient universal properties. ☐

Solution. Equality of cosets is exactly congruence because differences lie in (n). ☐

Exercise 1.44 (Coordinate restriction)**.** Interpret $k[x, y]/(y)$.

Hint. Translate the situation into kernels, generated ideals, and quotient universal properties. ☐

Solution. It is the coordinate ring obtained by imposing the equation $y = 0$. ☐

Challenge problems

Exercise 1.45 (Local-global synthesis)**.** Connect the chapter ideas 'Commutative rings' and 'Examples' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. □

Solution. The argument begins with A commutative ring has associative addition and multiplication, a multiplicative identity, and commutative multiplication. It then uses the later viewpoint: Polynomial rings, integer residue rings, coordinate-style quotients, and product rings provide the main laboratory. The bridge is supplied by the structural theorems proved in the chapter. □

Exercise 1.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Chapter 2

Prime and Maximal Ideals

Prime and maximal ideals encode the irreducible quotient structures of a commutative ring. Prime ideals detect domains, maximal ideals detect fields, and maximal ideals are automatically prime.

Learning goals

After this chapter, the reader should be able to:

- test whether an ideal is prime or maximal using quotient-domain and quotient-field criteria;
- compute prime and maximal ideals in standard rings such as integers, polynomial rings, and products;
- prove that maximal ideals are prime and identify hypotheses needed for converses;
- analyze inverse images of prime ideals under ring homomorphisms;
- distinguish irreducible elements from prime ideals and prime elements in examples;
- construct chains of prime ideals that reveal basic dimension-theoretic behavior;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

2.1 Prime ideals

A proper ideal $\mathfrak{p}$ is prime when $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$.

2.2 Maximal ideals

A proper ideal $\mathfrak{m}$ is maximal when no proper ideal lies strictly between $\mathfrak{m}$ and the whole ring.

Example 2.1 (Prime but not maximal)**.** The zero ideal in $\mathbb{Z}$ is prime because $\mathbb{Z}$ is a domain, but it is not maximal since $(0) \subsetneq (2) \subsetneq \mathbb{Z}$.

2.3 Quotient criteria

Prime ideals correspond to domain quotients; maximal ideals correspond to field quotients.

2.4 Existence of maximal ideals

Every proper ideal lies in a maximal ideal, using Zorn's lemma.

2.5 Contraction

Inverse images of prime ideals under ring maps are prime when the inverse image remains proper.

Example 2.2 (Maximal evaluation ideal)**.** In $k[x, y]$, evaluation at (a, b) has kernel $(x-a, y-b)$ and image k. The quotient is a field, so the kernel is maximal and therefore prime.

2.6 Prime chains

Chains of prime ideals reflect increasing algebraic specialization and later feed into dimension theory.

2.7 Units and maximal ideals

An element is a unit iff it belongs to no maximal ideal.

Example 2.3 (Nonprime integer ideal)**.** The ideal (6) is not prime because $2 \cdot 3 \in (6)$ while neither factor belongs to (6). Equivalently, $\mathbb{Z}/6\mathbb{Z}$ is not a domain.

2.8 Examples

Prime and maximal ideals in $\mathbb{Z}$, polynomial rings, product rings, and localizations illustrate the definitions.

2.9 Core structural results

Theorem 2.4 (Prime quotient criterion)**.** *A proper ideal $\mathfrak{p}$ is prime iff $A/\mathfrak{p}$ is an integral domain.*

Proof. The zero-divisor condition in the quotient is exactly the statement that $ab \in \mathfrak{p}$ forces one factor into $\mathfrak{p}$. ☐

Theorem 2.5 (Maximal quotient criterion)**.** *A proper ideal $\mathfrak{m}$ is maximal iff $A/\mathfrak{m}$ is a field.*

Proof. Ideals in $A/\mathfrak{m}$ correspond to ideals of A containing $\mathfrak{m}$. A quotient is a field exactly when its only ideals are zero and the whole ring. ☐

Theorem 2.6 (Maximal implies prime)**.** *Every maximal ideal in a commutative ring with identity is prime.*

Proof. The quotient by a maximal ideal is a field, hence a domain; apply the prime quotient criterion. ☐

2.10 Worked examples

Example 2.7 (Prime ideals diagnostic)**.** Give a rigorous worked analysis of Prime ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A proper ideal $\mathfrak{p}$ is prime when $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 2.8 (Maximal ideals diagnostic)**.** Give a rigorous worked analysis of Maximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A proper ideal $\mathfrak{m}$ is maximal when no proper ideal lies strictly between $\mathfrak{m}$ and the whole ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 2.9 (Quotient criteria diagnostic)**.** Give a rigorous worked analysis of Quotient criteria. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Prime ideals correspond to domain quotients; maximal ideals correspond to field quotients. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 2.10 (Existence of maximal ideals diagnostic)**.** Give a rigorous worked analysis of Existence of maximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every proper ideal lies in a maximal ideal, using Zorn's lemma. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

2.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 2.11 (Prime ideals diagnostic)**.** Give a rigorous worked analysis of Prime ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A proper ideal $\mathfrak{p}$ is prime when $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 2.12 (Maximal ideals diagnostic)**.** Give a rigorous worked analysis of Maximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A proper ideal $\mathfrak{m}$ is maximal when no proper ideal lies strictly between $\mathfrak{m}$ and the whole ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 2.13 (Quotient criteria diagnostic)**.** Give a rigorous worked analysis of Quotient criteria. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Prime ideals correspond to domain quotients; maximal ideals correspond to field quotients. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 2.14 (Existence of maximal ideals diagnostic)**.** Give a rigorous worked analysis of Existence of maximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every proper ideal lies in a maximal ideal, using Zorn's lemma. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 2.15 (Contraction diagnostic)**.** Give a rigorous worked analysis of Contraction. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Inverse images of prime ideals under ring maps are prime when the inverse image remains proper. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 2.16 (Prime chains diagnostic)**.** Give a rigorous worked analysis of Prime chains. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Chains of prime ideals reflect increasing algebraic specialization and later feed into dimension theory. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 2.17 (Units and maximal ideals diagnostic)**.** Give a rigorous worked analysis of Units and maximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element is a unit iff it belongs to no maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 2.18 (Examples diagnostic)**.** Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Prime and maximal ideals in $\mathbb{Z}$, polynomial rings, product rings, and localizations illustrate the definitions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 2.19 (Prime quotient criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A proper ideal $\mathfrak{p}$ is prime iff $A/\mathfrak{p}$ is an integral domain.

Solution. The zero-divisor condition in the quotient is exactly the statement that $ab \in \mathfrak{p}$ forces one factor into $\mathfrak{p}$. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 2.20 (Maximal quotient criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A proper ideal $\mathfrak{m}$ is maximal iff $A/\mathfrak{m}$ is a field.

Solution. Ideals in $A/\mathfrak{m}$ correspond to ideals of A containing $\mathfrak{m}$. A quotient is a field exactly when its only ideals are zero and the whole ring. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 2.21 (Maximal implies prime proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Every maximal ideal in a commutative ring with identity is prime.

Solution. The quotient by a maximal ideal is a field, hence a domain; apply the prime quotient criterion. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 2.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Prime and maximal ideals encode the irreducible quotient structures of a commutative ring. Prime ideals detect domains, maximal ideals detect fields, and maximal ideals are automatically prime. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

2.12 Exercises with complete solutions

Exercise 2.23. State and apply the central principle behind Prime ideals. Give one concrete algebraic consequence.

Hint. Use A/p being a domain as the fastest prime-ideal test when the quotient is easy to understand. □

Solution. A proper ideal $\mathfrak{p}$ is prime when $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 2.24. State and apply the central principle behind Maximal ideals. Give one concrete algebraic consequence.

Hint. Use A/m being a field as the fastest maximal-ideal test; then compare with the prime criterion. □

Solution. A proper ideal $\mathfrak{m}$ is maximal when no proper ideal lies strictly between $\mathfrak{m}$ and the whole ring. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 2.25. State and apply the central principle behind Quotient criteria. Give one concrete algebraic consequence.

Hint. In Z, reduce prime and maximal ideal questions to the generator n and factorization of n. □

Solution. Prime ideals correspond to domain quotients; maximal ideals correspond to field quotients. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 2.26. State and apply the central principle behind Existence of maximal ideals. Give one concrete algebraic consequence.

Hint. In a polynomial ring over a field, identify when a principal quotient is a domain or field from irreducibility of the polynomial. □

Solution. Every proper ideal lies in a maximal ideal, using Zorn's lemma. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 2.27. State and apply the central principle behind Contraction. Give one concrete algebraic consequence.

Hint. For inverse images, compose the quotient map with the homomorphism and use that preimages of prime ideals are prime when the target quotient is a domain. ☐

Solution. Inverse images of prime ideals under ring maps are prime when the inverse image remains proper. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 2.28. State and apply the central principle behind Prime chains. Give one concrete algebraic consequence.

Hint. To show maximal implies prime, use the field structure of the quotient rather than a direct ideal argument. ☐

Solution. Chains of prime ideals reflect increasing algebraic specialization and later feed into dimension theory. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 2.29. State and apply the central principle behind Units and maximal ideals. Give one concrete algebraic consequence.

Hint. For a product ring, inspect coordinate idempotents; prime ideals must kill exactly one factor in the basic finite-product case. ☐

Solution. An element is a unit iff it belongs to no maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 2.30. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. To build a prime chain, choose nested ideals whose quotient domains are transparent rather than guessing from element factorization alone. ☐

Solution. Prime and maximal ideals in $\mathbb{Z}$, polynomial rings, product rings, and localizations illustrate the definitions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

2.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 2.31 (Prime integer ideal)**.** Decide whether (7) is prime in $\mathbb{Z}$.

Hint. Use prime and maximal quotient criteria. ☐

Solution. Yes, because $\mathbb{Z}/7\mathbb{Z}$ is a domain. ☐

Exercise 2.32 (Maximal integer ideal)**.** Decide whether (7) is maximal in $\mathbb{Z}$.

Hint. Use prime and maximal quotient criteria. ☐

Solution. Yes, because the quotient is a field. ☐

Exercise 2.33 (Polynomial maximal ideal)**.** Decide whether $(x-2)$ is maximal in $\mathbb{Q}[x]$.

Hint. Use prime and maximal quotient criteria. ☐

Solution. Yes, the quotient is $\mathbb{Q}$. ☐

Exercise 2.34 (Prime failure)**.** Decide whether (xy) is prime in $k[x,y]$.

Hint. Use prime and maximal quotient criteria. ☐

Solution. No: the classes of x and y are nonzero zero divisors. ☐

Exercise 2.35 (Prime chain)**.** Give a prime chain in $k[x,y]$.

Hint. Use prime and maximal quotient criteria. ☐

Solution. $(0) \subset (x) \subset (x,y)$. ☐

Proofs

Exercise 2.36 (Prime quotient criterion proof)**.** Prove: A proper ideal $\mathfrak{p}$ is prime iff $A/\mathfrak{p}$ is an integral domain.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. The zero-divisor condition in the quotient is exactly the statement that $ab \in \mathfrak{p}$ forces one factor into $\mathfrak{p}$. ☐

Exercise 2.37 (Maximal quotient criterion proof)**.** Prove: A proper ideal $\mathfrak{m}$ is maximal iff $A/\mathfrak{m}$ is a field.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Ideals in $A/\mathfrak{m}$ correspond to ideals of A containing $\mathfrak{m}$. A quotient is a field exactly when its only ideals are zero and the whole ring. ☐

Exercise 2.38 (Maximal implies prime proof)**.** Prove: Every maximal ideal in a commutative ring with identity is prime.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. The quotient by a maximal ideal is a field, hence a domain; apply the prime quotient criterion. ☐

Exercise 2.39 (Structural synthesis)**.** Prove a short corollary by combining Prime quotient criterion with Maximal quotient criterion. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes Prime quotient criterion and then applies Maximal quotient criterion; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 2.40 (Prime maximality)**.** Test the claim: Every prime ideal is maximal.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: (x) is prime but not maximal in $k[x, y]$. ☐

Exercise 2.41 (Zero prime)**.** Test the claim: The zero ideal is prime in every ring.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: it is prime exactly in domains. ☐

Exercise 2.42 (Preimage maximality)**.** Test the claim: The inverse image of a maximal ideal is always maximal.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False without surjectivity; use $\mathbb{Z} \hookrightarrow \mathbb{Q}$. ☐

Applications and investigations

Exercise 2.43 (Affine point)**.** Interpret $(x - a, y - b)$ geometrically.

Hint. Translate the situation into prime and maximal quotient criteria. ☐

Solution. It records evaluation at the affine point (a, b). ☐

Exercise 2.44 (Irreducible component)**.** Explain why prime ideals model irreducible algebraic pieces.

Hint. Translate the situation into prime and maximal quotient criteria. ☐

Solution. The quotient domain has no nontrivial product equal to zero. ☐

Challenge problems

Exercise 2.45 (Local-global synthesis)**.** Connect the chapter ideas 'Prime ideals' and 'Examples' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. ☐

Solution. The argument begins with A proper ideal $\mathfrak{p}$ is prime when $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. It then uses the later viewpoint: Prime and maximal ideals in $\mathbb{Z}$, polynomial rings, product rings, and localizations illustrate the definitions. The bridge is supplied by the structural theorems proved in the chapter. ☐

Exercise 2.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. ☐

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. ☐

Chapter 3

Radicals and Nilpotents

Radicals and nilpotents measure how equations can vanish after taking powers. The radical of an ideal forgets multiplicity-like information and retains the underlying prime behavior.

Learning goals

After this chapter, the reader should be able to:

- compute radicals of explicit ideals and the nilradical of standard quotient rings;
- prove that the nilradical is the intersection of all prime ideals;
- test whether an ideal is radical by analyzing powers or quotient nilpotents;
- identify nilpotent elements in quotient rings from defining relations;
- relate radical containment to vanishing of powers of ideals or elements;
- construct examples separating prime, primary-looking, radical, and nonradical ideals;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

3.1 Nilpotent elements

An element a is nilpotent if $a^n = 0$ for some positive integer n.

3.2 Nilradical

The nilradical $\sqrt{(0)}$ is the ideal of all nilpotent elements.

Example 3.1 (Radical of an integer ideal)**.** For $I = (12) \subset \mathbb{Z}$, an integer has a power in I exactly when it is divisible by both 2 and 3. Thus $\sqrt{(12)} = (6)$.

3.3 Radical of an ideal

$\sqrt{I} = \{a : a^n \in I \text{ for some } n \geq 1\}$.

3.4 Radical ideals

An ideal is radical when $a^n \in I$ implies $a \in I$.

3.5 Quotient interpretation

$\sqrt{I}/I$ is the nilradical of A/I.

Example 3.2 (Nilpotent quotient)**.** In $k[x]/(x^3)$ the class of x is nonzero and nilpotent. The nilradical is (x) and the reduced quotient is k.

3.6 Prime containment

The radical of an ideal is the intersection of all prime ideals containing it.

3.7 Reduced rings

A ring is reduced when its nilradical is zero.

Example 3.3 (Monomial radical)**.** For $I = (x^2, y^3) \subset k[x, y]$, both x and y lie in $\sqrt{I}$, and a polynomial with nonzero constant term cannot have a power in I. Hence $\sqrt{I} = (x, y)$.

3.8 Examples

Powers of principal ideals, square-zero extensions, and polynomial quotients show how radicals discard nilpotent thickening.

3.9 Core structural results

Theorem 3.4 (Radical is an ideal)**.** *For every ideal I, the set $\sqrt{I}$ is an ideal containing I.*

Proof. Closure under multiplication is immediate. If $a^m, b^n \in I$, then every term in $(a+b)^{m+n}$ contains either at least m copies of a or at least n copies of b, hence lies in I. □

Theorem 3.5 (Quotient nilradical identity)**.** *The nilradical of A/I is $\sqrt{I}/I$.*

Proof. The class of a is nilpotent modulo I exactly when some power a^n lies in I. □

Theorem 3.6 (Radical as intersection of primes)**.** *$\sqrt{I}$ equals the intersection of all prime ideals containing I.*

Proof. One inclusion follows because primes are radical. For the reverse, if $a \notin \sqrt{I}$, localize the quotient A/I at powers of a and choose a maximal ideal there; its contraction is a prime containing I but avoiding a. □

3.10 Worked examples

Example 3.7 (Nilpotent elements diagnostic)**.** Give a rigorous worked analysis of Nilpotent elements. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An element a is nilpotent if $a^n = 0$ for some positive integer n. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 3.8 (Nilradical diagnostic)**.** Give a rigorous worked analysis of Nilradical. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The nilradical $\sqrt{(0)}$ is the ideal of all nilpotent elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 3.9 (Radical of an ideal diagnostic)**.** Give a rigorous worked analysis of Radical of an ideal. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\sqrt{I} = \{a : a^n \in I \text{ for some } n \geq 1\}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 3.10 (Radical ideals diagnostic)**.** Give a rigorous worked analysis of Radical ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An ideal is radical when $a^n \in I$ implies $a \in I$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

3.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 3.11 (Nilpotent elements diagnostic)**.** Give a rigorous worked analysis of Nilpotent elements. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element a is nilpotent if $a^n = 0$ for some positive integer n. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 3.12 (Nilradical diagnostic)**.** Give a rigorous worked analysis of Nilradical. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The nilradical $\sqrt{(0)}$ is the ideal of all nilpotent elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 3.13 (Radical of an ideal diagnostic)**.** Give a rigorous worked analysis of Radical of an ideal. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\sqrt{I} = \{a : a^n \in I \text{ for some } n \geq 1\}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 3.14 (Radical ideals diagnostic)**.** Give a rigorous worked analysis of Radical ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An ideal is radical when $a^n \in I$ implies $a \in I$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 3.15 (Quotient interpretation diagnostic)**.** Give a rigorous worked analysis of Quotient interpretation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\sqrt{I}/I$ is the nilradical of A/I. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 3.16 (Prime containment diagnostic)**.** Give a rigorous worked analysis of Prime containment. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The radical of an ideal is the intersection of all prime ideals containing it. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 3.17 (Reduced rings diagnostic)**.** Give a rigorous worked analysis of Reduced rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A ring is reduced when its nilradical is zero. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 3.18 (Examples diagnostic)**.** Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Powers of principal ideals, square-zero extensions, and polynomial quotients show how radicals discard nilpotent thickening. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 3.19 (Radical is an ideal proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For every ideal I, the set $\sqrt{I}$ is an ideal containing I.

Solution. Closure under multiplication is immediate. If $a^m, b^n \in I$, then every term in $(a+b)^{m+n}$ contains either at least m copies of a or at least n copies of b, hence lies in I. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 3.20 (Quotient nilradical identity proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: The nilradical of A/I is $\sqrt{I}/I$.

Solution. The class of a is nilpotent modulo I exactly when some power a^n lies in I. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 3.21 (Radical as intersection of primes proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: $\sqrt{I}$ equals the intersection of all prime ideals containing I.

Solution. One inclusion follows because primes are radical. For the reverse, if $a \notin \sqrt{I}$, localize the quotient A/I at powers of a and choose a maximal ideal there; its contraction is a prime containing I but avoiding a. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 3.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Radicals and nilpotents measure how equations can vanish after taking powers. The radical of an ideal forgets multiplicity-like information and retains the underlying prime behavior. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

3.12 Exercises with complete solutions

Exercise 3.23. State and apply the central principle behind Nilpotent elements. Give one concrete algebraic consequence.

Hint. An element lies in sqrt(I) exactly when some positive power lies in I; test powers before searching for generators of the radical. □

Solution. An element a is nilpotent if $a^n = 0$ for some positive integer n. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 3.24. State and apply the central principle behind Nilradical. Give one concrete algebraic consequence.

Hint. Compute the nilradical as sqrt((0)); in a quotient, nilpotents come from elements whose powers land in the defining ideal. □

Solution. The nilradical $\sqrt{(0)}$ is the ideal of all nilpotent elements. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 3.25. State and apply the central principle behind Radical of an ideal. Give one concrete algebraic consequence.

Hint. To prove the nilradical is contained in every prime, place x^n in the prime and use primality repeatedly to force x into it. □

Solution. $\sqrt{I} = \{a : a^n \in I \text{ for some } n \geq 1\}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 3.26. State and apply the central principle behind Radical ideals. Give one concrete algebraic consequence.

Hint. For the reverse containment, use a prime ideal avoiding a nonnilpotent element; this is the nontrivial direction of the intersection theorem. □

Solution. An ideal is radical when $a^n \in I$ implies $a \in I$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 3.27. State and apply the central principle behind Quotient interpretation. Give one concrete algebraic consequence.

Hint. An ideal I is radical exactly when A/I has no nonzero nilpotents; pass to the quotient when possible. □

Solution. $\sqrt{I}/I$ is the nilradical of A/I. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 3.28. State and apply the central principle behind Prime containment. Give one concrete algebraic consequence.

Hint. For principal monomial ideals, compare exponents: the radical keeps the underlying prime factors and forgets multiplicities. ☐

Solution. The radical of an ideal is the intersection of all prime ideals containing it. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 3.29. State and apply the central principle behind Reduced rings. Give one concrete algebraic consequence.

Hint. Check whether cancellation or quotient relations create nilpotents before classifying a ring as reduced. ☐

Solution. A ring is reduced when its nilradical is zero. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 3.30. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. Separate 'radical ideal' from 'prime ideal': every prime is radical, but intersections of distinct primes supply standard nonprime radical examples. ☐

Solution. Powers of principal ideals, square-zero extensions, and polynomial quotients show how radicals discard nilpotent thickening. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

3.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 3.31 (Integer radical)**.** Compute $\sqrt{(72)}$ in $\mathbb{Z}$.

Hint. Use radicals, nilpotents, and reduced quotients. ☐

Solution. The radical is (6). ☐

Exercise 3.32 (Nilpotency index)**.** Find the nilpotency index of x in $k[x]/(x^4)$.

Hint. Use radicals, nilpotents, and reduced quotients. ☐

Solution. It is 4. ☐

Exercise 3.33 (Reduced quotient)**.** Reduce $k[x]/(x^5)$.

Hint. Use radicals, nilpotents, and reduced quotients. □

Solution. The reduced quotient is k. □

Exercise 3.34 (Radical membership)**.** Is $xy \in \sqrt{(x^2y^2)}$?

Hint. Use radicals, nilpotents, and reduced quotients. □

Solution. Yes, because $(xy)^2 = x^2y^2$. □

Exercise 3.35 (Product nilradical)**.** Find the nilradical of $k \times k[t]/(t^2)$.

Hint. Use radicals, nilpotents, and reduced quotients. □

Solution. It is $0 \times (t)$. □

Proofs

Exercise 3.36 (Radical is an ideal proof)**.** Prove: For every ideal I, the set $\sqrt{I}$ is an ideal containing I.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. □

Solution. Closure under multiplication is immediate. If $a^m, b^n \in I$, then every term in $(a+b)^{m+n}$ contains either at least m copies of a or at least n copies of b, hence lies in I. □

Exercise 3.37 (Quotient nilradical identity proof)**.** Prove: The nilradical of A/I is $\sqrt{I}/I$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. □

Solution. The class of a is nilpotent modulo I exactly when some power a^n lies in I. □

Exercise 3.38 (Radical as intersection of primes proof)**.** Prove: $\sqrt{I}$ equals the intersection of all prime ideals containing I.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. □

Solution. One inclusion follows because primes are radical. For the reverse, if $a \notin \sqrt{I}$, localize the quotient A/I at powers of a and choose a maximal ideal there; its contraction is a prime containing I but avoiding a. □

Exercise 3.39 (Structural synthesis)**.** Prove a short corollary by combining Radical is an ideal with Quotient nilradical identity. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. □

Solution. A valid synthesis first invokes Radical is an ideal and then applies Quotient nilradical identity; the conclusion follows after checking the shared hypotheses. □

Counterexamples and hypothesis tests

Exercise 3.40 (Radical ideal)**.** Test the claim: (x^2) is radical in $k[x]$.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False because x^2 lies in the ideal but x does not. ☐

Exercise 3.41 (Reduced domain)**.** Test the claim: Every reduced ring is a domain.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: $k \times k$ is reduced with zero divisors. ☐

Exercise 3.42 (Zero divisor nilpotent)**.** Test the claim: Every zero divisor is nilpotent.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: $(1, 0)$ in $k \times k$ is idempotent. ☐

Applications and investigations

Exercise 3.43 (Reduced structure)**.** Explain $R/\sqrt{(0)}$.

Hint. Translate the situation into radicals, nilpotents, and reduced quotients. ☐

Solution. It removes exactly nilpotent elements. ☐

Exercise 3.44 (Zero sets)**.** Why does replacing I by $\sqrt{I}$ preserve common zeros?

Hint. Translate the situation into radicals, nilpotents, and reduced quotients. ☐

Solution. An element and every positive power have the same zero set. ☐

Challenge problems

Exercise 3.45 (Local-global synthesis)**.** Connect the chapter ideas 'Nilpotent elements' and 'Examples' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. ☐

Solution. The argument begins with An element a is nilpotent if $a^n = 0$ for some positive integer n. It then uses the later viewpoint: Powers of principal ideals, square-zero extensions, and polynomial quotients show how radicals discard nilpotent thickening. The bridge is supplied by the structural theorems proved in the chapter. ☐

Exercise 3.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. ☐

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. ☐

Chapter 4

Chinese Remainder Theory

Chinese remainder theory decomposes rings modulo comaximal ideals into products. It is both a structural theorem and an explicit congruence-solving method.

Learning goals

After this chapter, the reader should be able to:

- verify comaximality of ideals and apply the Chinese remainder theorem;
- construct explicit isomorphisms between quotient rings and products of quotients;
- solve simultaneous congruences by lifting idempotents or Bezout relations;
- extend the two-ideal theorem to finite pairwise-comaximal families;
- compute kernels of diagonal quotient maps and identify intersections of ideals;
- distinguish direct-product decompositions that arise from comaximal ideals from unrelated product presentations;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

4.1 Comaximal ideals

Ideals I, J are comaximal when $I + J = A$.

4.2 Two-ideal CRT

For comaximal I, J, the quotient by $I \cap J$ maps isomorphically to $A/I \times A/J$.

Example 4.1 (Integer CRT)**.** The congruences $x \equiv 2 \pmod 3$ and $x \equiv 4 \pmod 5$ have the unique solution $x \equiv 14 \pmod{15}$. This is the concrete form of $\mathbb{Z}/15 \cong \mathbb{Z}/3 \times \mathbb{Z}/5$.

4.3 Product equals intersection

For comaximal ideals, $I \cap J = IJ$.

4.4 Finite families

Pairwise comaximal finite families give product decompositions modulo the intersection or product.

4.5 Explicit idempotents

CRT decompositions produce orthogonal idempotents selecting the different factors.

Example 4.2 (Orthogonal idempotents)**.** In $\mathbb{Z}/12 \cong \mathbb{Z}/3\times\mathbb{Z}/4$, the classes 4 and 9 correspond to $(1, 0)$ and $(0, 1)$. They are orthogonal idempotents and sum to 1.

4.6 Integer congruences

The classical integer CRT is the special case of principal ideals in $\mathbb{Z}$.

4.7 Polynomial interpolation form

Comaximal polynomial ideals encode simultaneous residue conditions.

Example 4.3 (Polynomial CRT)**.** Since $(x) + (x-1) = k[x]$, one has $k[x]/(x(x-1)) \cong k \times k$. A polynomial class is determined by its values at 0 and 1.

4.8 Decomposition philosophy

Product rings are algebraic manifestations of independent components.

4.9 Core structural results

Theorem 4.4 (Two-ideal Chinese remainder theorem)**.** *If $I + J = A$, then $A/(I \cap J) \cong A/I \times A/J$.*

Proof. The natural map has kernel $I \cap J$. For surjectivity choose $u \in I, v \in J$ with $u + v = 1$; the element $bv + au$ has prescribed residues a modulo I and b modulo J. □

Theorem 4.5 (Intersection-product equality)**.** *If $I + J = A$, then $I \cap J = IJ$.*

Proof. The inclusion $IJ \subseteq I \cap J$ is automatic. If $x \in I \cap J$ and $1 = u + v$ with $u \in I, v \in J$, then $x = xu + xv$ lies in IJ. □

Theorem 4.6 (Finite CRT)**.** *If $I_1, \dots, I_n$ are pairwise comaximal, then $A/(I_1 \cdots I_n) \cong \prod_i A/I_i$.*

Proof. Induct on n, using that the product of the first $n-1$ ideals is comaximal with I_n, then apply the two-ideal theorem and the product-intersection identity. □

4.10 Worked examples

Example 4.7 (Comaximal ideals diagnostic)**.** Give a rigorous worked analysis of Comaximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Ideals I, J are comaximal when $I + J = A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 4.8 (Two-ideal CRT diagnostic)**.** Give a rigorous worked analysis of Two-ideal CRT. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For comaximal I, J, the quotient by $I \cap J$ maps isomorphically to $A/I \times A/J$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 4.9 (Product equals intersection diagnostic)**.** Give a rigorous worked analysis of Product equals intersection. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For comaximal ideals, $I \cap J = IJ$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 4.10 (Finite families diagnostic)**.** Give a rigorous worked analysis of Finite families. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Pairwise comaximal finite families give product decompositions modulo the intersection or product. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

4.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 4.11 (Comaximal ideals diagnostic)**.** Give a rigorous worked analysis of Comaximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Ideals I, J are comaximal when $I + J = A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 4.12 (Two-ideal CRT diagnostic)**.** Give a rigorous worked analysis of Two-ideal CRT. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For comaximal I, J, the quotient by $I \cap J$ maps isomorphically to $A/I \times A/J$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 4.13 (Product equals intersection diagnostic)**.** Give a rigorous worked analysis of Product equals intersection. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For comaximal ideals, $I \cap J = IJ$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 4.14 (Finite families diagnostic)**.** Give a rigorous worked analysis of Finite families. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Pairwise comaximal finite families give product decompositions modulo the intersection or product. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 4.15 (Explicit idempotents diagnostic)**.** Give a rigorous worked analysis of Explicit idempotents. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. CRT decompositions produce orthogonal idempotents selecting the different factors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 4.16 (Integer congruences diagnostic)**.** Give a rigorous worked analysis of Integer congruences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The classical integer CRT is the special case of principal ideals in $\mathbb{Z}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 4.17 (Polynomial interpolation form diagnostic)**.** Give a rigorous worked analysis of Polynomial interpolation form. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Comaximal polynomial ideals encode simultaneous residue conditions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 4.18 (Decomposition philosophy diagnostic)**.** Give a rigorous worked analysis of Decomposition philosophy. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Product rings are algebraic manifestations of independent components. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 4.19 (Two-ideal Chinese remainder theorem proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $I + J = A$, then $A/(I \cap J) \cong A/I \times A/J$.

Solution. The natural map has kernel $I \cap J$. For surjectivity choose $u \in I, v \in J$ with $u + v = 1$; the element $bv + au$ has prescribed residues a modulo I and b modulo J. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 4.20 (Intersection-product equality proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $I + J = A$, then $I \cap J = IJ$.

Solution. The inclusion $IJ \subseteq I \cap J$ is automatic. If $x \in I \cap J$ and $1 = u + v$ with $u \in I, v \in J$, then $x = xu + xv$ lies in IJ. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 4.21 (Finite CRT proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $I_1, \dots, I_n$ are pairwise comaximal, then $A/(I_1 \cdots I_n) \cong \prod_i A/I_i$.

Solution. Induct on n, using that the product of the first $n-1$ ideals is comaximal with I_n, then apply the two-ideal theorem and the product-intersection identity. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 4.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Chinese remainder theory decomposes rings modulo comaximal ideals into products. It is both a structural theorem and an explicit congruence-solving method. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. ☐

4.12 Exercises with complete solutions

Exercise 4.23. State and apply the central principle behind Comaximal ideals. Give one concrete algebraic consequence.

Hint. Prove comaximality by finding a+b=1 with a in I and b in J; this identity drives the explicit CRT decomposition. ☐

Solution. Ideals I, J are comaximal when $I + J = A$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 4.24. State and apply the central principle behind Two-ideal CRT. Give one concrete algebraic consequence.

Hint. The kernel of A -> A/I x A/J is I intersect J; compute it before invoking the first isomorphism theorem. ☐

Solution. For comaximal I, J, the quotient by $I \cap J$ maps isomorphically to $A/I \times A/J$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 4.25. State and apply the central principle behind Product equals intersection. Give one concrete algebraic consequence.

Hint. When I+J=A, prove I intersect J=IJ by inserting 1=a+b into an arbitrary element of the intersection. ☐

Solution. For comaximal ideals, $I \cap J = IJ$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 4.26. State and apply the central principle behind Finite families. Give one concrete algebraic consequence.

Hint. To construct a simultaneous solution, build idempotent-like coefficients from a Bezout relation and combine the prescribed residues. ☐

Solution. Pairwise comaximal finite families give product decompositions modulo the intersection or product. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 4.27. State and apply the central principle behind Explicit idempotents. Give one concrete algebraic consequence.

Hint. For finitely many ideals, verify pairwise comaximality and use induction rather than assuming the two-ideal proof automatically scales. ☐

Solution. CRT decompositions produce orthogonal idempotents selecting the different factors. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 4.28. State and apply the central principle behind Integer congruences. Give one concrete algebraic consequence.

Hint. In Z, translate comaximality to gcd(m,n)=1 before solving congruences. ☐

Solution. The classical integer CRT is the special case of principal ideals in $\mathbb{Z}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 4.29. State and apply the central principle behind Polynomial interpolation form. Give one concrete algebraic consequence.

Hint. Check surjectivity of the product quotient map by lifting one coordinate at a time with the CRT coefficients. ☐

Solution. Comaximal polynomial ideals encode simultaneous residue conditions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 4.30. State and apply the central principle behind Decomposition philosophy. Give one concrete algebraic consequence.

Hint. Do not replace intersections by products without comaximality; test a simple counterexample such as powers of one prime ideal. ☐

Solution. Product rings are algebraic manifestations of independent components. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

4.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 4.31 (Two congruences)**.** Solve $x \equiv 1 \pmod 4$ and $x \equiv 2 \pmod 3$.

Hint. Use comaximal ideals and Chinese remainder decompositions. ☐

Solution. $x \equiv 5 \pmod{12}$. ☐

Exercise 4.32 (CRT decomposition)**.** Decompose $\mathbb{Z}/35\mathbb{Z}$.

Hint. Use comaximal ideals and Chinese remainder decompositions. ☐

Solution. $\mathbb{Z}/35 \cong \mathbb{Z}/5 \times \mathbb{Z}/7$. ☐

Exercise 4.33 (Comaximality)**.** Are (x) and $(x-2)$ comaximal when 2 is invertible?

Hint. Use comaximal ideals and Chinese remainder decompositions. ☐

Solution. Yes. ☐

Exercise 4.34 (Intersection)**.** Compute $(3) \cap (5)$ in $\mathbb{Z}$.

Hint. Use comaximal ideals and Chinese remainder decompositions. ☐

Solution. It is (15). ☐

Exercise 4.35 (Idempotent)**.** Find the class mod 15 corresponding to $(1,0)$ mod $(3,5)$.

Hint. Use comaximal ideals and Chinese remainder decompositions. ☐

Solution. The class 10. ☐

Proofs

Exercise 4.36 (Two-ideal Chinese remainder theorem proof)**.** Prove: If $I + J = A$, then $A/(I \cap J) \cong A/I \times A/J$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. The natural map has kernel $I \cap J$. For surjectivity choose $u \in I, v \in J$ with $u + v = 1$; the element $bv + au$ has prescribed residues a modulo I and b modulo J. ☐

Exercise 4.37 (Intersection-product equality proof)**.** Prove: If $I + J = A$, then $I \cap J = IJ$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. The inclusion $IJ \subseteq I \cap J$ is automatic. If $x \in I \cap J$ and $1 = u + v$ with $u \in I, v \in J$, then $x = xu + xv$ lies in IJ. ☐

Exercise 4.38 (Finite CRT proof)**.** Prove: If $I_1, \ldots, I_n$ are pairwise comaximal, then $A/(I_1 \cdots I_n) \cong \prod_i A/I_i$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Induct on n, using that the product of the first $n-1$ ideals is comaximal with I_n, then apply the two-ideal theorem and the product-intersection identity. ☐

Exercise 4.39 (Structural synthesis)**.** Prove a short corollary by combining Two-ideal Chinese remainder theorem with Intersection-product equality. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes Two-ideal Chinese remainder theorem and then applies Intersection-product equality; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 4.40 (Noncomaximal CRT)**.** Test the claim: $\mathbb{Z}/8 \cong \mathbb{Z}/4 \times \mathbb{Z}/2$.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: the ideals are not comaximal. ☐

Exercise 4.41 (Intersection product)**.** Test the claim: $I \cap J = IJ$ for all ideals.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: take $I = J = (2)$ in $\mathbb{Z}$. ☐

Exercise 4.42 (Repeated factors)**.** Test the claim: CRT works unchanged with repeated equal proper ideals.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: the diagonal map is not surjective. ☐

Applications and investigations

Exercise 4.43 (Parallel arithmetic)**.** Why can CRT split modular arithmetic into components?

Hint. Translate the situation into comaximal ideals and Chinese remainder decompositions. ☐

Solution. The product-ring isomorphism makes operations componentwise. ☐

Exercise 4.44 (Interpolation)**.** Interpret CRT for distinct linear polynomial factors.

Hint. Translate the situation into comaximal ideals and Chinese remainder decompositions. ☐

Solution. It prescribes independent values at distinct points. ☐

Challenge problems

Exercise 4.45 (Local-global synthesis)**.** Connect the chapter ideas 'Comaximal ideals' and 'Decomposition philosophy' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. ☐

Solution. The argument begins with Ideals I, J are comaximal when $I + J = A$. It then uses the later viewpoint: Product rings are algebraic manifestations of independent components. The bridge is supplied by the structural theorems proved in the chapter. ☐

Exercise 4.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Part II

Localization

Chapter 5

Multiplicative Systems

Multiplicative systems specify which elements are to become invertible. They are the input data for localization and control which ideals, primes, and module elements survive.

Learning goals

After this chapter, the reader should be able to:

- verify that a subset is multiplicatively closed and determine when it contains zero;
- construct saturation sets and identify elements annihilated after localization;
- compare localization at powers of one element with localization at a general multiplicative system;
- identify universal properties satisfied by maps inverting a prescribed multiplicative set;
- compute equality conditions between fractions using the localization equivalence relation;
- choose multiplicative systems adapted to primes, elements, or geometric open sets;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

5.1 Multiplicative subsets

A multiplicative set S contains 1 and is closed under products; usually $0 \notin S$ for nontrivial localization.

5.2 Saturation

The saturation of S consists of elements that become invertible whenever all of S does.

Example 5.1 (Powers of two)**.** The set $S = \{1, 2, 4, 8, \ldots\}$ is multiplicatively closed. Inverting it produces $\mathbb{Z}[1/2]$, where 2 is a unit but 3 need not be.

5.3 Examples

Powers of one element, complements of prime ideals, and complements of unions of primes are standard multiplicative sets.

5.4 Disjointness from ideals

An ideal disjoint from S can survive as a proper ideal after localization.

5.5 Prime avoidance viewpoint

Maximal ideals among ideals disjoint from S are prime.

Example 5.2 (Prime complement)**.** If P is prime, then $R \setminus P$ is multiplicatively closed. Otherwise two elements outside P could multiply into P, contradicting primality.

5.6 Denominator calculus

Fractions identify numerators after multiplying by suitable elements of S.

5.7 Universal goal

Localization is characterized by making S invertible with no extra relations beyond those forced.

Example 5.3 (Denominator saturation)**.** For $S = \{1, x, x^2, \dots\} \subset k[x, y]$, every ideal meeting S becomes the unit ideal after localization, while ideals disjoint from S can survive.

5.8 Geometric preview

Choosing $S = \{1, f, f^2, \dots\}$ corresponds to restricting attention to where f is invertible.

5.9 Core structural results

Theorem 5.4 (Maximal disjoint ideal is prime)**.** *If I is maximal among ideals disjoint from a multiplicative set S, then I is prime.*

Proof. If $ab \in I$ with $a, b \notin I$, then both $I + (a)$ and $I + (b)$ meet S. Multiplying corresponding elements of S yields an element of $I + (ab) \subseteq I$, contradicting disjointness. $\square$

Theorem 5.5 (Complement of prime is multiplicative)**.** *If $\mathfrak{p}$ is prime, then $A \setminus \mathfrak{p}$ is multiplicatively closed.*

Proof. If $a, b \notin \mathfrak{p}$ but $ab \in \mathfrak{p}$, primality would force one factor into $\mathfrak{p}$. $\square$

Theorem 5.6 (Powers form a multiplicative set)**.** *For any $f \in A$, $S = \{1, f, f^2, \dots\}$ is multiplicative.*

Proof. The product $f^m f^n = f^{m+n}$ remains in the set and $1 = f^0$ is included. $\square$

5.10 Worked examples

Example 5.7 (Multiplicative subsets diagnostic)**.** Give a rigorous worked analysis of Multiplicative subsets. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A multiplicative set S contains 1 and is closed under products; usually $0 \notin S$ for nontrivial localization. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 5.8 (Saturation diagnostic)**.** Give a rigorous worked analysis of Saturation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The saturation of S consists of elements that become invertible whenever all of S does. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 5.9 (Examples diagnostic)**.** Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Powers of one element, complements of prime ideals, and complements of unions of primes are standard multiplicative sets. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 5.10 (Disjointness from ideals diagnostic)**.** Give a rigorous worked analysis of Disjointness from ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An ideal disjoint from S can survive as a proper ideal after localization. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

5.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 5.11 (Multiplicative subsets diagnostic)**.** Give a rigorous worked analysis of Multiplicative subsets. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A multiplicative set S contains 1 and is closed under products; usually $0 \notin S$ for nontrivial localization. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 5.12 (Saturation diagnostic)**.** Give a rigorous worked analysis of Saturation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The saturation of S consists of elements that become invertible whenever all of S does. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 5.13 (Examples diagnostic)**.** Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Powers of one element, complements of prime ideals, and complements of unions of primes are standard multiplicative sets. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 5.14 (Disjointness from ideals diagnostic)**.** Give a rigorous worked analysis of Disjointness from ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An ideal disjoint from S can survive as a proper ideal after localization. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 5.15 (Prime avoidance viewpoint diagnostic)**.** Give a rigorous worked analysis of Prime avoidance viewpoint. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Maximal ideals among ideals disjoint from S are prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 5.16 (Denominator calculus diagnostic)**.** Give a rigorous worked analysis of Denominator calculus. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Fractions identify numerators after multiplying by suitable elements of S. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 5.17 (Universal goal diagnostic)**.** Give a rigorous worked analysis of Universal goal. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization is characterized by making S invertible with no extra relations beyond those forced. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 5.18 (Geometric preview diagnostic)**.** Give a rigorous worked analysis of Geometric preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Choosing $S = \{1, f, f^2, \dots\}$ corresponds to restricting attention to where f is invertible. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 5.19 (Maximal disjoint ideal is prime proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If I is maximal among ideals disjoint from a multiplicative set S, then I is prime.

Solution. If $ab \in I$ with $a, b \notin I$, then both $I + (a)$ and $I + (b)$ meet S. Multiplying corresponding elements of S yields an element of $I + (ab) \subseteq I$, contradicting disjointness. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 5.20 (Complement of prime is multiplicative proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $\mathfrak{p}$ is prime, then $A \setminus \mathfrak{p}$ is multiplicatively closed.

Solution. If $a, b \notin \mathfrak{p}$ but $ab \in \mathfrak{p}$, primality would force one factor into $\mathfrak{p}$. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 5.21 (Powers form a multiplicative set proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For any $f \in A$, $S = \{1, f, f^2, \dots\}$ is multiplicative.

Solution. The product $f^m f^n = f^{m+n}$ remains in the set and $1 = f^0$ is included. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 5.22 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Multiplicative systems specify which elements are to become invertible. They are the input data for localization and control which ideals, primes, and module elements survive. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

5.12 Exercises with complete solutions

Exercise 5.23. State and apply the central principle behind Multiplicative subsets. Give one concrete algebraic consequence.

Hint. Verify 1 lies in S and that products stay in S; if 0 lies in S, the localization collapses to the zero ring. □

Solution. A multiplicative set S contains 1 and is closed under products; usually $0 \notin S$ for nontrivial localization. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 5.24. State and apply the central principle behind Saturation. Give one concrete algebraic consequence.

Hint. To compare fractions, use the actual equivalence relation: some s in S must annihilate the cross-multiplication difference. □

Solution. The saturation of S consists of elements that become invertible whenever all of S does. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 5.25. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. An element a/1 becomes zero exactly when some s in S annihilates a; look for that witness. □

Solution. Powers of one element, complements of prime ideals, and complements of unions of primes are standard multiplicative sets. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 5.26. State and apply the central principle behind Disjointness from ideals. Give one concrete algebraic consequence.

Hint. For localization at powers of f, rewrite every denominator as f^n and keep track of which powers are allowed. □

Solution. An ideal disjoint from S can survive as a proper ideal after localization. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 5.27. State and apply the central principle behind Prime avoidance viewpoint. Give one concrete algebraic consequence.

Hint. To use the universal property, first check the target map sends every element of S to a unit. □

Solution. Maximal ideals among ideals disjoint from S are prime. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 5.28. State and apply the central principle behind Denominator calculus. Give one concrete algebraic consequence.

Hint. Saturation records elements that become invertible after localization; test by multiplying into S. □

Solution. Fractions identify numerators after multiplying by suitable elements of S. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 5.29. State and apply the central principle behind Universal goal. Give one concrete algebraic consequence.

Hint. For a prime p, use $S = A \setminus p$ so precisely the elements outside p become units. □

Solution. Localization is characterized by making S invertible with no extra relations beyond those forced. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 5.30. State and apply the central principle behind Geometric preview. Give one concrete algebraic consequence.

Hint. Choose S from the algebraic goal: powers of one element for D(f), complement of a prime for local behavior, or a larger system for simultaneous inversion. □

Solution. Choosing $S = \{1, f, f^2, \dots\}$ corresponds to restricting attention to where f is invertible. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

5.13 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 5.31 (Closure)**.** Is the set of nonzero integers multiplicatively closed?

Hint. Use multiplicative systems and denominator control. ☐

Solution. Yes. ☐

Exercise 5.32 (Generated system)**.** List $1, 6, 6^2, 6^3$.

Hint. Use multiplicative systems and denominator control. ☐

Solution. $1, 6, 36, 216$. ☐

Exercise 5.33 (Prime complement)**.** Is $\mathbb{Z} \setminus (5)$ multiplicatively closed?

Hint. Use multiplicative systems and denominator control. ☐

Solution. Yes. ☐

Exercise 5.34 (Zero denominator)**.** What if $0 \in S$?

Hint. Use multiplicative systems and denominator control. ☐

Solution. The localization is the zero ring. ☐

Exercise 5.35 (Meeting ideal)**.** If $I \cap S \neq \varnothing$, compute $S^{-1}I$.

Hint. Use multiplicative systems and denominator control. ☐

Solution. It is $S^{-1}R$. ☐

Proofs

Exercise 5.36 (Maximal disjoint ideal is prime proof)**.** Prove: If I is maximal among ideals disjoint from a multiplicative set S, then I is prime.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. If $ab \in I$ with $a, b \notin I$, then both $I + (a)$ and $I + (b)$ meet S. Multiplying corresponding elements of S yields an element of $I + (ab) \subseteq I$, contradicting disjointness. ☐

Exercise 5.37 (Complement of prime is multiplicative proof)**.** Prove: If $\mathfrak{p}$ is prime, then $A \setminus \mathfrak{p}$ is multiplicatively closed.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. If $a, b \notin \mathfrak{p}$ but $ab \in \mathfrak{p}$, primality would force one factor into $\mathfrak{p}$. ☐

Exercise 5.38 (Powers form a multiplicative set proof)**.** Prove: For any $f \in A$, $S = \{1, f, f^2, \dots\}$ is multiplicative.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. The product $f^m f^n = f^{m+n}$ remains in the set and $1 = f^0$ is included. ☐

Exercise 5.39 (Structural synthesis)**.** Prove a short corollary by combining Maximal disjoint ideal is prime with Complement of prime is multiplicative. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes Maximal disjoint ideal is prime and then applies Complement of prime is multiplicative; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 5.40 (Ideal complement)**.** Test the claim: The complement of every proper ideal is multiplicatively closed.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False; the ideal must be prime. ☐

Exercise 5.41 (Nonunits)**.** Test the claim: The set of nonunits is always a multiplicative system.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. Not as a general structural replacement for prime complements; local-ring behavior is special. ☐

Exercise 5.42 (Prime visibility)**.** Test the claim: A prime meeting S survives as a proper prime after localization.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: it extends to the whole localized ring. ☐

Applications and investigations

Exercise 5.43 (Denominator design)**.** What does choosing S control?

Hint. Translate the situation into multiplicative systems and denominator control. ☐

Solution. It specifies exactly which elements are forced to become units. ☐

Exercise 5.44 (Local study)**.** Why use $S = R \setminus P$?

Hint. Translate the situation into multiplicative systems and denominator control. ☐

Solution. It focuses algebra at P by inverting everything outside it. ☐

Challenge problems

Exercise 5.45 (Local-global synthesis)**.** Connect the chapter ideas 'Multiplicative subsets' and 'Geometric preview' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. ☐

Solution. The argument begins with A multiplicative set S contains 1 and is closed under products; usually $0 \notin S$ for nontrivial localization. It then uses the later viewpoint: Choosing $S = \{1, f, f^2, \dots\}$ corresponds to restricting attention to where f is invertible. The bridge is supplied by the structural theorems proved in the chapter. □

Exercise 5.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Chapter 6

Localization of Rings

Localization of rings adjoins inverses to a chosen multiplicative system. It is governed by a universal property and induces a precise correspondence between primes of the localization and primes of the original ring disjoint from the denominators.

Learning goals

After this chapter, the reader should be able to:

- construct localized rings and simplify explicit fractions;
- apply the universal property of localization to extend ring homomorphisms uniquely;
- compute prime ideals of a localization from primes disjoint from the multiplicative set;
- determine when an element becomes zero or a unit after localization;
- compare localization at a prime, maximal ideal, and single element;
- prove that localization commutes with quotients in the standard compatible form;

Conceptual roadmap

$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$

6.1 Fraction construction

Elements of $S^{-1}A$ are fractions a/s modulo the relation generated by cross-multiplication after multiplying by a further denominator.

6.2 Ring operations

Fractions add and multiply by the usual formulas.

Example 6.1 (Integer localization)**.** Localizing $\mathbb{Z}$ at powers of 6 gives $\mathbb{Z}[1/6]$. Fractions with denominators dividing powers of 6 occur, while $1/5$ does not.

6.3 Canonical map

The map $A \to S^{-1}A$ sends a to $a/1$ and makes every element of S invertible.

6.4 Universal property

Any ring map from A sending S to units factors uniquely through $S^{-1}A$.

6.5 Zero criterion

A fraction a/s is zero iff some $t \in S$ annihilates a.

Example 6.2 (Principal localization)**.** In $k[x,y]_x$, x is a unit, so (x,y) becomes the whole ring. The ideal (y) remains proper and records the part visible where x is invertible.

6.6 Localized ideals

Extension sends I to $S^{-1}I$, while contraction returns ideals saturated with respect to S.

6.7 Prime correspondence

Prime ideals of $S^{-1}A$ correspond to prime ideals of A disjoint from S.

Example 6.3 (Prime localization)**.** The ring $\mathbb{Z}_{(5)}$ consists of a/b with $5 \nmid b$. Its unique maximal ideal consists of fractions whose numerator is divisible by 5.

6.8 Localization at an element

A_f is obtained by inverting powers of one element f.

6.9 Localization at a prime

$A_{\mathfrak{p}}$ inverts everything outside $\mathfrak{p}$ and is a local ring.

6.10 Core structural results

Theorem 6.4 (Universal property of localization)**.** *If $\varphi : A \to B$ sends every $s \in S$ to a unit, there is a unique map $\widetilde{\varphi} : S^{-1}A \to B$ with $\widetilde{\varphi}(a/s) = \varphi(a)\varphi(s)^{-1}$.*

Proof. The formula respects the localization equivalence relation and is forced by the requirement that denominators become inverses, giving existence and uniqueness. ☐

Theorem 6.5 (Prime correspondence)**.** *Prime ideals of $S^{-1}A$ are in bijection with prime ideals $\mathfrak{p} \subset A$ satisfying $\mathfrak{p} \cap S = \varnothing$.*

Proof. Contraction of a prime avoids units, hence avoids S. For a prime disjoint from S, its extension consists of fractions with numerator in the prime and is prime; extension and contraction are inverse. ☐

Theorem 6.6 (Localization commutes with quotient)**.** *For an ideal I, $S^{-1}(A/I) \cong S^{-1}A/S^{-1}I$ after replacing S by its image in A/I.*

Proof. Both rings satisfy the same universal property: maps out correspond to maps from A killing I and inverting S. ☐

6.11 Worked examples

Example 6.7 (Fraction construction diagnostic)**.** Give a rigorous worked analysis of Fraction construction. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Elements of $S^{-1}A$ are fractions a/s modulo the relation generated by cross-multiplication after multiplying by a further denominator. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 6.8 (Ring operations diagnostic)**.** Give a rigorous worked analysis of Ring operations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Fractions add and multiply by the usual formulas. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 6.9 (Canonical map diagnostic)**.** Give a rigorous worked analysis of Canonical map. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The map $A \to S^{-1}A$ sends a to $a/1$ and makes every element of S invertible. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 6.10 (Universal property diagnostic)**.** Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Any ring map from A sending S to units factors uniquely through $S^{-1}A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

6.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 6.11 (Fraction construction diagnostic)**.** Give a rigorous worked analysis of Fraction construction. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Elements of $S^{-1}A$ are fractions a/s modulo the relation generated by cross-multiplication after multiplying by a further denominator. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 6.12 (Ring operations diagnostic)**.** Give a rigorous worked analysis of Ring operations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Fractions add and multiply by the usual formulas. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 6.13 (Canonical map diagnostic)**.** Give a rigorous worked analysis of Canonical map. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The map $A \to S^{-1}A$ sends a to $a/1$ and makes every element of S invertible. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 6.14 (Universal property diagnostic)**.** Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Any ring map from A sending S to units factors uniquely through $S^{-1}A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 6.15 (Zero criterion diagnostic)**.** Give a rigorous worked analysis of Zero criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A fraction a/s is zero iff some $t \in S$ annihilates a. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 6.16 (Localized ideals diagnostic)**.** Give a rigorous worked analysis of Localized ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Extension sends I to $S^{-1}I$, while contraction returns ideals saturated with respect to S. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 6.17 (Prime correspondence diagnostic)**.** Give a rigorous worked analysis of Prime correspondence. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Prime ideals of $S^{-1}A$ correspond to prime ideals of A disjoint from S. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 6.18 (Localization at an element diagnostic)**.** Give a rigorous worked analysis of Localization at an element. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A_f is obtained by inverting powers of one element f. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 6.19 (Universal property of localization proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $\varphi : A \to B$ sends every $s \in S$ to a unit, there is a unique map $\widetilde{\varphi} : S^{-1}A \to B$ with $\widetilde{\varphi}(a/s) = \varphi(a)\varphi(s)^{-1}$.

Solution. The formula respects the localization equivalence relation and is forced by the requirement that denominators become inverses, giving existence and uniqueness. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 6.20 (Prime correspondence proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Prime ideals of $S^{-1}A$ are in bijection with prime ideals $\mathfrak{p} \subset A$ satisfying $\mathfrak{p} \cap S = \varnothing$.

Solution. Contraction of a prime avoids units, hence avoids S. For a prime disjoint from S, its extension consists of fractions with numerator in the prime and is prime; extension and contraction are inverse. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 6.21 (Localization commutes with quotient proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For an ideal I, $S^{-1}(A/I) \cong S^{-1}A/S^{-1}I$ after replacing S by its image in A/I.

Solution. Both rings satisfy the same universal property: maps out correspond to maps from A killing I and inverting S. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 6.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Localization of rings adjoins inverses to a chosen multiplicative system. It is governed by a universal property and induces a precise correspondence between primes of the localization and primes of the original ring disjoint from the denominators. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. ☐

6.13 Exercises with complete solutions

Exercise 6.23. State and apply the central principle behind Fraction construction. Give one concrete algebraic consequence.

Hint. Simplify a/s by multiplying numerator and denominator only by elements allowed by the localization equivalence. ☐

Solution. Elements of $S^{-1}A$ are fractions a/s modulo the relation generated by cross-multiplication after multiplying by a further denominator. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 6.24. State and apply the central principle behind Ring operations. Give one concrete algebraic consequence.

Hint. To prove a map from $S^{-1}A$ exists, define $a/s \mapsto \varphi(a)\varphi(s)^{-1}$ and check it is independent of representatives. ☐

Solution. Fractions add and multiply by the usual formulas. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 6.25. State and apply the central principle behind Canonical map. Give one concrete algebraic consequence.

Hint. A prime of $S^{-1}A$ contracts to a prime of A disjoint from S; extend a disjoint prime and check the two operations are inverse. ☐

Solution. The map $A \to S^{-1}A$ sends a to $a/1$ and makes every element of S invertible. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 6.26. State and apply the central principle behind Universal property. Give one concrete algebraic consequence.

Hint. To see whether a/1 is a unit, ask whether some multiple of a lies in S. ☐

Solution. Any ring map from A sending S to units factors uniquely through $S^{-1}A$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 6.27. State and apply the central principle behind Zero criterion. Give one concrete algebraic consequence.

Hint. For localization at p, every element outside p is a unit and pA_p is the unique maximal ideal. ☐

Solution. A fraction a/s is zero iff some $t \in S$ annihilates a. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 6.28. State and apply the central principle behind Localized ideals. Give one concrete algebraic consequence.

Hint. Use $S^{-1}(A/I) \cong S^{-1}A/S^{-1}I$ only after checking the image of S does not force the quotient to collapse unexpectedly. ☐

Solution. Extension sends I to $S^{-1}I$, while contraction returns ideals saturated with respect to S. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 6.29. State and apply the central principle behind Prime correspondence. Give one concrete algebraic consequence.

Hint. Localization at f can be computed as adjoining an inverse with relation ft=1; use this presentation when explicit algebra helps. ☐

Solution. Prime ideals of $S^{-1}A$ correspond to prime ideals of A disjoint from S. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 6.30. State and apply the central principle behind Localization at an element. Give one concrete algebraic consequence.

Hint. Check equality locally by cross multiplication with an S-witness rather than cancelling denominators as if the ring were a domain. ☐

Solution. A_f is obtained by inverting powers of one element f. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

6.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 6.31 (Unit)**.** Is 2 a unit in $\mathbb{Z}[1/2]$?

Hint. Use fraction localization and its universal property. □

Solution. Yes, inverse $1/2$. □

Exercise 6.32 (Localized ideal)**.** Compute $(2)\mathbb{Z}[1/2]$.

Hint. Use fraction localization and its universal property. □

Solution. It is the whole ring. □

Exercise 6.33 (Prime denominators)**.** Describe denominators in $\mathbb{Z}_{(7)}$.

Hint. Use fraction localization and its universal property. □

Solution. They are integers not divisible by 7. □

Exercise 6.34 (Polynomial inverse)**.** Find the inverse of x^3 in $k[x,y]_x$.

Hint. Use fraction localization and its universal property. □

Solution. It is x^{-3}. □

Exercise 6.35 (Zero fraction)**.** When is $a/s = 0$?

Hint. Use fraction localization and its universal property. □

Solution. When some denominator $u \in S$ satisfies $ua = 0$. □

Proofs

Exercise 6.36 (Universal property of localization proof)**.** Prove: If $\varphi : A \to B$ sends every $s \in S$ to a unit, there is a unique map $\widetilde{\varphi} : S^{-1}A \to B$ with $\widetilde{\varphi}(a/s) = \varphi(a)\varphi(s)^{-1}$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. □

Solution. The formula respects the localization equivalence relation and is forced by the requirement that denominators become inverses, giving existence and uniqueness. □

Exercise 6.37 (Prime correspondence proof)**.** Prove: Prime ideals of $S^{-1}A$ are in bijection with prime ideals $\mathfrak{p} \subset A$ satisfying $\mathfrak{p} \cap S = \varnothing$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. □

Solution. Contraction of a prime avoids units, hence avoids S. For a prime disjoint from S, its extension consists of fractions with numerator in the prime and is prime; extension and contraction are inverse. □

Exercise 6.38 (Localization commutes with quotient proof)**.** Prove: For an ideal I, $S^{-1}(A/I) \cong S^{-1}A/S^{-1}I$ after replacing S by its image in A/I.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. □

Solution. Both rings satisfy the same universal property: maps out correspond to maps from A killing I and inverting S. ☐

Exercise 6.39 (Structural synthesis)**.** Prove a short corollary by combining Universal property of localization with Prime correspondence. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes Universal property of localization and then applies Prime correspondence; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 6.40 (Injectivity)**.** Test the claim: $R \to S^{-1}R$ is always injective.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: denominator torsion can map to zero. ☐

Exercise 6.41 (Prime extension)**.** Test the claim: Every prime extends to a proper prime.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False if the prime meets S. ☐

Exercise 6.42 (Collapse)**.** Test the claim: A nonzero ring can never localize to zero.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False if 0 is inverted. ☐

Applications and investigations

Exercise 6.43 (Principal open)**.** Interpret R_f geometrically.

Hint. Translate the situation into fraction localization and its universal property. ☐

Solution. It describes the region where f is nonzero. ☐

Exercise 6.44 (Arithmetic near p)**.** Why use $\mathbb{Z}_{(p)}$?

Hint. Translate the situation into fraction localization and its universal property. ☐

Solution. All integers prime to p become units. ☐

Challenge problems

Exercise 6.45 (Local-global synthesis)**.** Connect the chapter ideas 'Fraction construction' and 'Localization at a prime' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. ☐

Solution. The argument begins with Elements of $S^{-1}A$ are fractions a/s modulo the relation generated by cross-multiplication after multiplying by a further denominator. It then uses the later viewpoint: $A_{\mathfrak{p}}$ inverts everything outside $\mathfrak{p}$ and is a local ring. The bridge is supplied by the structural theorems proved in the chapter. ☐

Exercise 6.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Chapter 7

Localization of Modules

Localization of modules extends fraction methods from rings to linear algebra over rings. It is exact, commutes with finite algebraic constructions, and detects local behavior of modules.

Learning goals

After this chapter, the reader should be able to:

- construct localized modules and simplify fractions of module elements;
- determine when a localized module element vanishes;
- prove that localization is an exact functor on modules;
- compute localized kernels, images, quotients, and finite presentations;
- test module identities locally at prime ideals and recover global consequences where valid;
- distinguish module localization from tensoring with an unrelated algebra by using the canonical isomorphism with localized rings;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

7.1 Module fractions

$S^{-1}M$ consists of symbols m/s modulo the expected denominator equivalence.

7.2 Scalar structure

$S^{-1}M$ is naturally a module over $S^{-1}A$.

Example 7.1 (Localized torsion)**.** Localizing $\mathbb{Z}/12$ at powers of 2 kills the 4-primary part and leaves the 3-primary information. This is a concrete instance of denominators annihilating torsion.

7.3 Universal property

Maps from M into modules where S acts invertibly factor uniquely through $S^{-1}M$.

7.4 Tensor description

There is a natural isomorphism $S^{-1}M \cong S^{-1}A \otimes_A M$.

7.5 Exactness

Localization preserves exact sequences.

Example 7.2 (Module vanishing)**.** For $M = R/(f)$, localization at powers of f gives $M_f = 0$: the relation $f \cdot 1 = 0$ and invertibility of f force $1 = 0$.

7.6 Submodules and quotients

Localization commutes with quotients and sends submodules to localized submodules.

7.7 Vanishing criterion

An element $m/1$ vanishes locally iff some denominator in S annihilates m.

Example 7.3 (Exact sequence localization)**.** Localizing $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$ at powers of 2 turns the first map into an isomorphism and the localized cokernel into zero.

7.8 Finite generation

Localization preserves finite generation and finite presentation.

7.9 Local detection

Many module properties can be tested after localizing at all prime or maximal ideals.

7.10 Core structural results

Theorem 7.4 (Tensor description of localization)**.** *There is a natural isomorphism $S^{-1}A \otimes_A M \cong S^{-1}M$ sending $(a/s) \otimes m$ to am/s.*

Proof. The formula is balanced and induces a homomorphism. Its inverse sends m/s to $(1/s) \otimes m$; the two composites are identities. □

Theorem 7.5 (Exactness of localization)**.** *If $0 \to M' \to M \to M'' \to 0$ is exact, then $0 \to S^{-1}M' \to S^{-1}M \to S^{-1}M'' \to 0$ is exact.*

Proof. Surjectivity is immediate on fractions. For injectivity or kernel equality, if a localized element maps to zero, some denominator annihilates the relevant numerator, allowing the numerator to be moved into the preceding module before localization. □

Theorem 7.6 (Localization commutes with quotient)**.** *For $N \subset M$, $S^{-1}(M/N) \cong S^{-1}M/S^{-1}N$.*

Proof. Apply exactness to $0 \to N \to M \to M/N \to 0$. □

7.11 Worked examples

Example 7.7 (Module fractions diagnostic)**.** Give a rigorous worked analysis of Module fractions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $S^{-1}M$ consists of symbols m/s modulo the expected denominator equivalence. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 7.8 (Scalar structure diagnostic)**.** Give a rigorous worked analysis of Scalar structure. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $S^{-1}M$ is naturally a module over $S^{-1}A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 7.9 (Universal property diagnostic)**.** Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Maps from M into modules where S acts invertibly factor uniquely through $S^{-1}M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 7.10 (Tensor description diagnostic)**.** Give a rigorous worked analysis of Tensor description. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. There is a natural isomorphism $S^{-1}M \cong S^{-1}A \otimes_A M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

7.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 7.11 (Module fractions diagnostic)**.** Give a rigorous worked analysis of Module fractions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $S^{-1}M$ consists of symbols m/s modulo the expected denominator equivalence. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 7.12 (Scalar structure diagnostic)**.** Give a rigorous worked analysis of Scalar structure. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $S^{-1}M$ is naturally a module over $S^{-1}A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 7.13 (Universal property diagnostic)**.** Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Maps from M into modules where S acts invertibly factor uniquely through $S^{-1}M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 7.14 (Tensor description diagnostic)**.** Give a rigorous worked analysis of Tensor description. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. There is a natural isomorphism $S^{-1}M \cong S^{-1}A \otimes_A M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 7.15 (Exactness diagnostic)**.** Give a rigorous worked analysis of Exactness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization preserves exact sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 7.16 (Submodules and quotients diagnostic)**.** Give a rigorous worked analysis of Submodules and quotients. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization commutes with quotients and sends submodules to localized submodules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 7.17 (Vanishing criterion diagnostic)**.** Give a rigorous worked analysis of Vanishing criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element $m/1$ vanishes locally iff some denominator in S annihilates m. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 7.18 (Finite generation diagnostic)**.** Give a rigorous worked analysis of Finite generation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization preserves finite generation and finite presentation. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 7.19 (Tensor description of localization proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: There is a natural isomorphism $S^{-1}A \otimes_A M \cong S^{-1}M$ sending $(a/s) \otimes m$ to am/s.

Solution. The formula is balanced and induces a homomorphism. Its inverse sends m/s to $(1/s) \otimes m$; the two composites are identities. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 7.20 (Exactness of localization proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $0 \to M' \to M \to M'' \to 0$ is exact, then $0 \to S^{-1}M' \to S^{-1}M \to S^{-1}M'' \to 0$ is exact.

Solution. Surjectivity is immediate on fractions. For injectivity or kernel equality, if a localized element maps to zero, some denominator annihilates the relevant numerator, allowing the numerator to be moved into the preceding module before localization. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 7.21 (Localization commutes with quotient proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $N \subset M$, $S^{-1}(M/N) \cong S^{-1}M/S^{-1}N$.

Solution. Apply exactness to $0 \to N \to M \to M/N \to 0$. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 7.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Localization of modules extends fraction methods from rings to linear algebra over rings. It is exact, commutes with finite algebraic constructions, and detects local behavior of modules. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. ☐

7.13 Exercises with complete solutions

Exercise 7.23. State and apply the central principle behind Module fractions. Give one concrete algebraic consequence.

Hint. Represent module fractions as m/s and use the same denominator-equivalence idea as for rings, with annihilation by an element of S. ☐

Solution. $S^{-1}M$ consists of symbols m/s modulo the expected denominator equivalence. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 7.24. State and apply the central principle behind Scalar structure. Give one concrete algebraic consequence.

Hint. m/1 is zero in $S^{-1}M$ exactly when some s in S satisfies sm=0; this is the basic local-vanishing test. ☐

Solution. $S^{-1}M$ is naturally a module over $S^{-1}A$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 7.25. State and apply the central principle behind Universal property. Give one concrete algebraic consequence.

Hint. To prove exactness after localization, clear denominators and use exactness before localization to find the needed preimage. ☐

Solution. Maps from M into modules where S acts invertibly factor uniquely through $S^{-1}M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 7.26. State and apply the central principle behind Tensor description. Give one concrete algebraic consequence.

Hint. Use $S^{-1}M \cong S^{-1}A \otimes_A M$ when tensor-product tools simplify the argument. ☐

Solution. There is a natural isomorphism $S^{-1}M \cong S^{-1}A \otimes_A M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 7.27. State and apply the central principle behind Exactness. Give one concrete algebraic consequence.

Hint. For a quotient, localize the short exact sequence $0 \to N \to M \to M/N \to 0$ and read off $S^{-1}(M/N)$. ☐

Solution. Localization preserves exact sequences. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 7.28. State and apply the central principle behind Submodules and quotients. Give one concrete algebraic consequence.

Hint. To test an equality of submodules locally, compare their localizations at every prime and then invoke the local-global criterion only when its hypotheses apply. ☐

Solution. Localization commutes with quotients and sends submodules to localized submodules. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 7.29. State and apply the central principle behind Vanishing criterion. Give one concrete algebraic consequence.

Hint. Finite presentations localize by localizing the presentation matrix entrywise. ☐

Solution. An element $m/1$ vanishes locally iff some denominator in S annihilates m. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 7.30. State and apply the central principle behind Finite generation. Give one concrete algebraic consequence.

Hint. Do not infer that local vanishing at one prime gives global vanishing; determine which primes must be checked. ☐

Solution. Localization preserves finite generation and finite presentation. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

7.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 7.31 (Zero module)**.** Compute $S^{-1}0$.

Hint. Use localization of modules and exactness. ☐

Solution. It is zero. ☐

Exercise 7.32 (Killed quotient)**.** Compute $(R/(x))_x$.

Hint. Use localization of modules and exactness. ☐

Solution. It is zero. ☐

Exercise 7.33 (Free module)**.** Compute $S^{-1}(R^3)$.

Hint. Use localization of modules and exactness. ☐

Solution. It is $(S^{-1}R)^3$. ☐

Exercise 7.34 (Element vanishing)**.** When does $m/1$ vanish?

Hint. Use localization of modules and exactness. ☐

Solution. When some $s \in S$ annihilates m. ☐

Exercise 7.35 (Quotient localization)**.** Identify $S^{-1}(M/N)$.

Hint. Use localization of modules and exactness. ☐

Solution. It is $S^{-1}M/S^{-1}N$. ☐

Proofs

Exercise 7.36 (Tensor description of localization proof)**.** Prove: There is a natural isomorphism $S^{-1}A \otimes_A M \cong S^{-1}M$ sending $(a/s) \otimes m$ to am/s.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. The formula is balanced and induces a homomorphism. Its inverse sends m/s to $(1/s) \otimes m$; the two composites are identities. ☐

Exercise 7.37 (Exactness of localization proof)**.** Prove: If $0 \to M' \to M \to M'' \to 0$ is exact, then $0 \to S^{-1}M' \to S^{-1}M \to S^{-1}M'' \to 0$ is exact.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Surjectivity is immediate on fractions. For injectivity or kernel equality, if a localized element maps to zero, some denominator annihilates the relevant numerator, allowing the numerator to be moved into the preceding module before localization. ☐

Exercise 7.38 (Localization commutes with quotient proof)**.** Prove: For $N \subset M$, $S^{-1}(M/N) \cong S^{-1}M/S^{-1}N$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Apply exactness to $0 \to N \to M \to M/N \to 0$. ☐

Exercise 7.39 (Structural synthesis)**.** Prove a short corollary by combining Tensor description of localization with Exactness of localization. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes Tensor description of localization and then applies Exactness of localization; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 7.40 (Nonzero preservation)**.** Test the claim: Localization preserves every nonzero module.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: $(R/(f))_f = 0$. ☐

Exercise 7.41 (Module injectivity)**.** Test the claim: $M \to S^{-1}M$ is always injective.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False when M has S-torsion. ☐

Exercise 7.42 (New support)**.** Test the claim: Localization can create support at primes meeting S.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False; those primes disappear. ☐

Applications and investigations

Exercise 7.43 (Local checking)**.** Why check modules at all prime localizations?

Hint. Translate the situation into localization of modules and exactness. ☐

Solution. A module is zero exactly when all prime localizations vanish. ☐

Exercise 7.44 (Restriction)**.** Interpret M_f.

Hint. Translate the situation into localization of modules and exactness. ☐

Solution. It is module data restricted to the principal open where f is invertible. ☐

Challenge problems

Exercise 7.45 (Local-global synthesis)**.** Connect the chapter ideas 'Module fractions' and 'Local detection' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. ☐

Solution. The argument begins with $S^{-1}M$ consists of symbols m/s modulo the expected denominator equivalence. It then uses the later viewpoint: Many module properties can be tested after localizing at all prime or maximal ideals. The bridge is supplied by the structural theorems proved in the chapter. ☐

Exercise 7.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Chapter 8

Local Rings and Localization at Primes

Local rings concentrate algebra around one maximal ideal. Localization at a prime produces the canonical local ring attached to that prime, and unit tests become especially simple.

Learning goals

After this chapter, the reader should be able to:

- verify that a ring is local and identify its unique maximal ideal;
- compute localizations at prime ideals and describe their units and maximal ideals;
- prove the unit criterion that elements outside the maximal ideal are invertible in a local ring;
- use Nakayama-type reasoning in elementary finitely generated module situations where available;
- translate global ideal questions into local questions at primes;
- construct examples of local rings arising from polynomial, quotient, and localization constructions;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

8.1 Local rings

A local ring has exactly one maximal ideal.

8.2 Nonunits

In a local ring, the nonunits form the unique maximal ideal.

Example 8.1 (Integers at a prime)**.** In $\mathbb{Z}_{(5)}$, a/b is a unit exactly when $5 \nmid a$. The fractions with numerator divisible by 5 form the unique maximal ideal.

8.3 Unit criterion

An element is a unit precisely when it lies outside the maximal ideal.

8.4 Localization at primes

For a prime $\mathfrak{p}$, $A_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.

8.5 Residue field

The field $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is the fraction field of $A/\mathfrak{p}$ with respect to nonzero classes.

Example 8.2 (Polynomial local ring)**.** The ring $k[x]_{(x)}$ consists of f/g with $g(0) \neq 0$. Its maximal ideal is generated by x, and a fraction is a unit exactly when its residue at $x = 0$ is nonzero.

8.6 Nakayama preview

Local rings make finite-generation arguments rigid because the maximal ideal controls reduction modulo the residue field.

8.7 Local homomorphisms

A homomorphism of local rings is local when the maximal ideal pulls back to the maximal ideal.

Example 8.3 (Two-variable local ring)**.** In $k[x, y]_{(x,y)}$, every polynomial with nonzero constant term becomes a unit. The maximal ideal is generated by the images of x and y.

8.8 Examples

Discrete valuation rings, localized polynomial rings, and power-series rings provide standard local examples.

8.9 Local-global philosophy

Properties of modules and ideals are often checked prime by prime using these local rings.

8.10 Core structural results

Theorem 8.4 (Nonunits characterize local rings)**.** *A commutative ring is local iff its nonunits form an ideal; in that case this ideal is the unique maximal ideal.*

Proof. Every proper ideal consists of nonunits. If nonunits form an ideal, it is proper and contains every proper ideal, hence is uniquely maximal. Conversely, in a local ring every element outside the maximal ideal generates the unit ideal and is therefore a unit. □

Theorem 8.5 (Prime localization is local)**.** *For a prime $\mathfrak{p}$, the ring $A_{\mathfrak{p}}$ is local with unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.*

Proof. Prime ideals of the localization correspond to primes of A contained in $\mathfrak{p}$. The extension of $\mathfrak{p}$ contains every such prime and is therefore the unique maximal ideal. □

Theorem 8.6 (Residue field identification). *The residue field of $A_{\mathfrak{p}}$ is canonically isomorphic to the fraction field of $A/\mathfrak{p}$.*

Proof. Modulo $\mathfrak{p}A_{\mathfrak{p}}$, numerators and denominators descend to $A/\mathfrak{p}$, and every denominator outside $\mathfrak{p}$ becomes a nonzero class, hence invertible in the fraction field. □

8.11 Worked examples

Example 8.7 (Local rings diagnostic). Give a rigorous worked analysis of Local rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A local ring has exactly one maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 8.8 (Nonunits diagnostic). Give a rigorous worked analysis of Nonunits. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. In a local ring, the nonunits form the unique maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 8.9 (Unit criterion diagnostic). Give a rigorous worked analysis of Unit criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An element is a unit precisely when it lies outside the maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 8.10 (Localization at primes diagnostic). Give a rigorous worked analysis of Localization at primes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For a prime $\mathfrak{p}$, $A_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

8.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 8.11 (Local rings diagnostic). Give a rigorous worked analysis of Local rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A local ring has exactly one maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 8.12 (Nonunits diagnostic). Give a rigorous worked analysis of Nonunits. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. In a local ring, the nonunits form the unique maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 8.13 (Unit criterion diagnostic)**.** Give a rigorous worked analysis of Unit criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element is a unit precisely when it lies outside the maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 8.14 (Localization at primes diagnostic)**.** Give a rigorous worked analysis of Localization at primes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For a prime $\mathfrak{p}$, $A_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 8.15 (Residue field diagnostic)**.** Give a rigorous worked analysis of Residue field. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The field $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is the fraction field of $A/\mathfrak{p}$ with respect to nonzero classes. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 8.16 (Nakayama preview diagnostic)**.** Give a rigorous worked analysis of Nakayama preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Local rings make finite-generation arguments rigid because the maximal ideal controls reduction modulo the residue field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 8.17 (Local homomorphisms diagnostic)**.** Give a rigorous worked analysis of Local homomorphisms. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A homomorphism of local rings is local when the maximal ideal pulls back to the maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 8.18 (Examples diagnostic)**.** Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Discrete valuation rings, localized polynomial rings, and power-series rings provide standard local examples. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 8.19 (Nonunits characterize local rings proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A commutative ring is local iff its nonunits form an ideal; in that case this ideal is the unique maximal ideal.

Solution. Every proper ideal consists of nonunits. If nonunits form an ideal, it is proper and contains every proper ideal, hence is uniquely maximal. Conversely, in a local ring every element outside the maximal ideal generates the unit ideal and is therefore a unit. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 8.20 (Prime localization is local proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For a prime $\mathfrak{p}$, the ring $A_{\mathfrak{p}}$ is local with unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.

Solution. Prime ideals of the localization correspond to primes of A contained in $\mathfrak{p}$. The extension of $\mathfrak{p}$ contains every such prime and is therefore the unique maximal ideal. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 8.21 (Residue field identification proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: The residue field of $A_{\mathfrak{p}}$ is canonically isomorphic to the fraction field of $A/\mathfrak{p}$.

Solution. Modulo $\mathfrak{p}A_{\mathfrak{p}}$, numerators and denominators descend to $A/\mathfrak{p}$, and every denominator outside $\mathfrak{p}$ becomes a nonzero class, hence invertible in the fraction field. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 8.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Local rings concentrate algebra around one maximal ideal. Localization at a prime produces the canonical local ring attached to that prime, and unit tests become especially simple. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

8.13 Exercises with complete solutions

Exercise 8.23. State and apply the central principle behind Local rings. Give one concrete algebraic consequence.

Hint. In a local ring, show every element outside the maximal ideal is a unit by using maximality of the principal ideal it generates together with the maximal ideal. □

Solution. A local ring has exactly one maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 8.24. State and apply the central principle behind Nonunits. Give one concrete algebraic consequence.

Hint. To identify the maximal ideal of A_p, localize p itself and show every fraction outside pA_p is invertible. □

Solution. In a local ring, the nonunits form the unique maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 8.25. State and apply the central principle behind Unit criterion. Give one concrete algebraic consequence.

Hint. A ring is local iff the nonunits form an ideal; test this criterion when the maximal ideal is not obvious. ☐

Solution. An element is a unit precisely when it lies outside the maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 8.26. State and apply the central principle behind Localization at primes. Give one concrete algebraic consequence.

Hint. For a quotient of a local ring, track the image of the maximal ideal and check when it remains the unique maximal ideal. ☐

Solution. For a prime $\mathfrak{p}$, $A_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 8.27. State and apply the central principle behind Residue field. Give one concrete algebraic consequence.

Hint. Localizing at a maximal ideal produces a local ring; describe its residue field as A_m/mA_m. ☐

Solution. The field $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is the fraction field of $A/\mathfrak{p}$ with respect to nonzero classes. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 8.28. State and apply the central principle behind Nakayama preview. Give one concrete algebraic consequence.

Hint. Use Nakayama only for finitely generated modules and a Jacobson-radical/maximal-ideal hypothesis; identify that hypothesis explicitly. ☐

Solution. Local rings make finite-generation arguments rigid because the maximal ideal controls reduction modulo the residue field. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 8.29. State and apply the central principle behind Local homomorphisms. Give one concrete algebraic consequence.

Hint. To prove a finitely generated module is zero from M=mM, reduce generators modulo m and apply Nakayama. ☐

Solution. A homomorphism of local rings is local when the maximal ideal pulls back to the maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 8.30. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. Translate a global statement to A_p or M_p, prove it at each relevant prime, then use a local-global criterion rather than assuming local truth glues automatically. ☐

Solution. Discrete valuation rings, localized polynomial rings, and power-series rings provide standard local examples. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

8.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 8.31 (Unit test)**.** Is $3/7$ a unit in $\mathbb{Z}_{(5)}$?

Hint. Use local rings, maximal ideals, and unit criteria. ☐

Solution. Yes. ☐

Exercise 8.32 (Nonunit test)**.** Is $10/3$ a unit in $\mathbb{Z}_{(5)}$?

Hint. Use local rings, maximal ideals, and unit criteria. ☐

Solution. No. ☐

Exercise 8.33 (Maximal ideal)**.** Find the maximal ideal of $k[x]_{(x)}$.

Hint. Use local rings, maximal ideals, and unit criteria. ☐

Solution. It is $(x)k[x]_{(x)}$. ☐

Exercise 8.34 (Residue field)**.** Find the residue field of $\mathbb{Z}_{(p)}$.

Hint. Use local rings, maximal ideals, and unit criteria. ☐

Solution. It is $\mathbb{F}_p$. ☐

Exercise 8.35 (Polynomial residue)**.** Find $k[x,y]_{(x,y)}/(x,y)$.

Hint. Use local rings, maximal ideals, and unit criteria. ☐

Solution. It is k. ☐

Proofs

Exercise 8.36 (Nonunits characterize local rings proof)**.** Prove: A commutative ring is local iff its nonunits form an ideal; in that case this ideal is the unique maximal ideal.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Every proper ideal consists of nonunits. If nonunits form an ideal, it is proper and contains every proper ideal, hence is uniquely maximal. Conversely, in a local ring every element outside the maximal ideal generates the unit ideal and is therefore a unit. ☐

Exercise 8.37 (Prime localization is local proof)**.** Prove: For a prime $\mathfrak{p}$, the ring $A_{\mathfrak{p}}$ is local with unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Prime ideals of the localization correspond to primes of A contained in $\mathfrak{p}$. The extension of $\mathfrak{p}$ contains every such prime and is therefore the unique maximal ideal. ☐

Exercise 8.38 (Residue field identification proof)**.** Prove: The residue field of $A_{\mathfrak{p}}$ is canonically isomorphic to the fraction field of $A/\mathfrak{p}$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Modulo $\mathfrak{p}A_{\mathfrak{p}}$, numerators and denominators descend to $A/\mathfrak{p}$, and every denominator outside $\mathfrak{p}$ becomes a nonzero class, hence invertible in the fraction field. ☐

Exercise 8.39 (Structural synthesis)**.** Prove a short corollary by combining Nonunits characterize local rings with Prime localization is local. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes Nonunits characterize local rings and then applies Prime localization is local; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 8.40 (Two maximals)**.** Test the claim: A ring with two distinct maximal ideals is local.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False. ☐

Exercise 8.41 (Prime spectrum)**.** Test the claim: A local ring has no nonmaximal prime ideals.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False. ☐

Exercise 8.42 (Unit residue)**.** Test the claim: An element in the maximal ideal can be a unit.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False in a local ring. ☐

Applications and investigations

Exercise 8.43 (Near one prime)**.** Why are local rings adapted to local algebra?

Hint. Translate the situation into local rings, maximal ideals, and unit criteria. ☐

Solution. They invert everything away from one chosen prime. ☐

Exercise 8.44 (Residue measurement)**.** What does the residue field measure?

Hint. Translate the situation into local rings, maximal ideals, and unit criteria. ☐

Solution. It records scalars at the chosen local point. ☐

Challenge problems

Exercise 8.45 (Local-global synthesis)**.** Connect the chapter ideas 'Local rings' and 'Local-global philosophy' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. □

Solution. The argument begins with A local ring has exactly one maximal ideal. It then uses the later viewpoint: Properties of modules and ideals are often checked prime by prime using these local rings. The bridge is supplied by the structural theorems proved in the chapter. □

Exercise 8.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Part III

Modules and Tensor Products

Chapter 9

Modules and Exact Sequences

Modules generalize vector spaces by allowing scalars from an arbitrary ring. Exact sequences record how modules are assembled from submodules and quotients and become the basic grammar of homological algebra.

Learning goals

After this chapter, the reader should be able to:

- verify exactness of sequences by computing kernels and images at each term;
- construct short exact sequences from submodules and quotients;
- apply the snake-lemma-style kernel/cokernel viewpoint in elementary diagrams where developed;
- identify splitting maps and determine when a short exact sequence splits;
- compute kernels, cokernels, images, and coimages of module homomorphisms;
- distinguish left exact, right exact, and exact behavior of basic module functors;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

9.1 Modules

An A-module is an abelian group with compatible scalar multiplication by A.

9.2 Submodules and quotients

Submodules are additive subgroups stable under scalars; quotient modules encode congruence modulo a submodule.

Example 9.1 (Kernel and cokernel)**.** Multiplication by 6 gives a short exact sequence $0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/6 \to 0$. The kernel is zero and the cokernel is the residue module.

9.3 Module maps

Kernels, images, and cokernels are modules and satisfy the familiar first isomorphism theorem.

9.4 Exact sequences

A sequence is exact when the image at each term equals the next kernel.

9.5 Short exact sequences

$0 \to M' \to M \to M'' \to 0$ expresses M' as a submodule and M'' as the corresponding quotient.

Example 9.2 (Matrix exactness)**.** The inclusion $k^2 \to k^3$ into the first two coordinates has cokernel k. Projection onto the third coordinate completes a short exact sequence.

9.6 Splittings

A short exact sequence splits when the middle module is isomorphic to a direct sum of its ends.

9.7 Direct sums

Direct sums collect finitely supported tuples and implement coproducts in the module category.

Example 9.3 (Split sequence)**.** The sequence $0 \to A \to A \oplus B \to B \to 0$ splits via $b \mapsto (0, b)$, exhibiting the middle module as a direct sum.

9.8 Diagram arguments

Exact sequences organize kernel-image calculations and support diagram lemmas later in homological algebra.

9.9 Examples

Ideals, quotient modules, free modules, torsion abelian groups, and vector spaces all fit the same language.

9.10 Core structural results

Theorem 9.4 (First isomorphism theorem for modules)**.** *For an A-linear map $f : M \to N$, $M/\ker f \cong \operatorname{im} f$.*

Proof. Send $m + \ker f$ to $f(m)$. The same kernel argument as for groups and rings gives a well-defined bijective linear map onto the image. □

Theorem 9.5 (Splitting criterion)**.** *A short exact sequence $0 \to M' \xrightarrow{i} M \xrightarrow{p} M'' \to 0$ splits iff p has a section, equivalently iff i has a retraction.*

Proof. A section s gives $M = i(M') \oplus s(M'')$. Conversely a direct-sum decomposition supplies both projection and inclusion maps realizing the section and retraction. □

Theorem 9.6 (Exactness of direct sums)**.** *Taking arbitrary direct sums of short exact sequences of modules preserves exactness.*

Proof. All maps act componentwise. An element of a direct sum has finite support, so kernel and image membership can be checked component by component. □

9.11 Worked examples

Example 9.7 (Modules diagnostic)**.** Give a rigorous worked analysis of Modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An A-module is an abelian group with compatible scalar multiplication by A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 9.8 (Submodules and quotients diagnostic)**.** Give a rigorous worked analysis of Submodules and quotients. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Submodules are additive subgroups stable under scalars; quotient modules encode congruence modulo a submodule. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 9.9 (Module maps diagnostic)**.** Give a rigorous worked analysis of Module maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Kernels, images, and cokernels are modules and satisfy the familiar first isomorphism theorem. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 9.10 (Exact sequences diagnostic)**.** Give a rigorous worked analysis of Exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A sequence is exact when the image at each term equals the next kernel. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

9.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 9.11 (Modules diagnostic)**.** Give a rigorous worked analysis of Modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An A-module is an abelian group with compatible scalar multiplication by A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 9.12 (Submodules and quotients diagnostic)**.** Give a rigorous worked analysis of Submodules and quotients. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Submodules are additive subgroups stable under scalars; quotient modules encode congruence modulo a submodule. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 9.13 (Module maps diagnostic)**.** Give a rigorous worked analysis of Module maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Kernels, images, and cokernels are modules and satisfy the familiar first isomorphism theorem. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 9.14 (Exact sequences diagnostic)**.** Give a rigorous worked analysis of Exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A sequence is exact when the image at each term equals the next kernel. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 9.15 (Short exact sequences diagnostic)**.** Give a rigorous worked analysis of Short exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $0 \to M' \to M \to M'' \to 0$ expresses M' as a submodule and M'' as the corresponding quotient. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 9.16 (Splittings diagnostic)**.** Give a rigorous worked analysis of Splittings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A short exact sequence splits when the middle module is isomorphic to a direct sum of its ends. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 9.17 (Direct sums diagnostic)**.** Give a rigorous worked analysis of Direct sums. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Direct sums collect finitely supported tuples and implement coproducts in the module category. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 9.18 (Diagram arguments diagnostic)**.** Give a rigorous worked analysis of Diagram arguments. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Exact sequences organize kernel-image calculations and support diagram lemmas later in homological algebra. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 9.19 (First isomorphism theorem for modules proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For an A-linear map $f : M \to N$, $M/\ker f \cong \operatorname{im} f$.

Solution. Send $m + \ker f$ to $f(m)$. The same kernel argument as for groups and rings gives a well-defined bijective linear map onto the image. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 9.20 (Splitting criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A short exact sequence $0 \to M' \xrightarrow{i} M \xrightarrow{p} M'' \to 0$ splits iff p has a section, equivalently iff i has a retraction.

Solution. A section s gives $M = i(M') \oplus s(M'')$. Conversely a direct-sum decomposition supplies both projection and inclusion maps realizing the section and retraction. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 9.21 (Exactness of direct sums proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Taking arbitrary direct sums of short exact sequences of modules preserves exactness.

Solution. All maps act componentwise. An element of a direct sum has finite support, so kernel and image membership can be checked component by component. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 9.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Modules generalize vector spaces by allowing scalars from an arbitrary ring. Exact sequences record how modules are assembled from submodules and quotients and become the basic grammar of homological algebra. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. ☐

9.13 Exercises with complete solutions

Exercise 9.23. State and apply the central principle behind Modules. Give one concrete algebraic consequence.

Hint. At each term of a sequence, compute image of the incoming map and kernel of the outgoing map; exactness means equality of those submodules. ☐

Solution. An A-module is an abelian group with compatible scalar multiplication by A. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 9.24. State and apply the central principle behind Submodules and quotients. Give one concrete algebraic consequence.

Hint. For 0 -> N -> M -> Q -> 0, verify injectivity, surjectivity, and that Q is the quotient M/N up to the induced isomorphism. ☐

Solution. Submodules are additive subgroups stable under scalars; quotient modules encode congruence modulo a submodule. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 9.25. State and apply the central principle behind Module maps. Give one concrete algebraic consequence.

Hint. A splitting of the surjection gives $M \cong N \oplus Q$; construct the isomorphism explicitly from inclusion and section. ☐

Solution. Kernels, images, and cokernels are modules and satisfy the familiar first isomorphism theorem. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 9.26. State and apply the central principle behind Exact sequences. Give one concrete algebraic consequence.

Hint. A splitting of the inclusion gives a retraction; use its kernel as the complementary summand. ☐

Solution. A sequence is exact when the image at each term equals the next kernel. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 9.27. State and apply the central principle behind Short exact sequences. Give one concrete algebraic consequence.

Hint. Compute cokernel as target modulo image, not merely as a set-theoretic complement. ☐

Solution. $0 \to M' \to M \to M'' \to 0$ expresses M' as a submodule and M'' as the corresponding quotient. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 9.28. State and apply the central principle behind Splittings. Give one concrete algebraic consequence.

Hint. To test functor exactness, apply the functor to a short exact sequence and inspect where injectivity or surjectivity is lost. ☐

Solution. A short exact sequence splits when the middle module is isomorphic to a direct sum of its ends. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 9.29. State and apply the central principle behind Direct sums. Give one concrete algebraic consequence.

Hint. For diagram arguments, chase one element at a time through commuting squares and write down where exactness is used. ☐

Solution. Direct sums collect finitely supported tuples and implement coproducts in the module category. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 9.30. State and apply the central principle behind Diagram arguments. Give one concrete algebraic consequence.

Hint. Do not confuse exactness of a complex with splitting; a short exact sequence can be nonsplit. ☐

Solution. Exact sequences organize kernel-image calculations and support diagram lemmas later in homological algebra. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

9.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 9.31 (Kernel)**.** Find the kernel of multiplication by 4 on $\mathbb{Z}$.

Hint. Use kernels, cokernels, and exact sequences. ☐

Solution. It is zero. ☐

Exercise 9.32 (Cokernel)**.** Find its cokernel.

Hint. Use kernels, cokernels, and exact sequences. ☐

Solution. It is $\mathbb{Z}/4$. ☐

Exercise 9.33 (Injection)**.** Is multiplication by 2 on $\mathbb{Z}$ injective?

Hint. Use kernels, cokernels, and exact sequences. ☐

Solution. Yes. ☐

Exercise 9.34 (Matrix rank)**.** Find the rank of the projection $k^3 \to k^2$.

Hint. Use kernels, cokernels, and exact sequences. ☐

Solution. The rank is 2. ☐

Exercise 9.35 (Section)**.** Give a section of $A \oplus B \to B$.

Hint. Use kernels, cokernels, and exact sequences. ☐

Solution. $b \mapsto (0, b)$. ☐

Proofs

Exercise 9.36 (First isomorphism theorem for modules proof)**.** Prove: For an A-linear map $f : M \to N$, $M/\ker f \cong \operatorname{im} f$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Send $m + \ker f$ to $f(m)$. The same kernel argument as for groups and rings gives a well-defined bijective linear map onto the image. ☐

Exercise 9.37 (Splitting criterion proof)**.** Prove: A short exact sequence $0 \to M' \xrightarrow{i} M \xrightarrow{p} M'' \to 0$ splits iff p has a section, equivalently iff i has a retraction.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. A section s gives $M = i(M') \oplus s(M'')$. Conversely a direct-sum decomposition supplies both projection and inclusion maps realizing the section and retraction. ☐

Exercise 9.38 (Exactness of direct sums proof)**.** Prove: Taking arbitrary direct sums of short exact sequences of modules preserves exactness.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. All maps act componentwise. An element of a direct sum has finite support, so kernel and image membership can be checked component by component. ☐

Exercise 9.39 (Structural synthesis)**.** Prove a short corollary by combining First isomorphism theorem for modules with Splitting criterion. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes First isomorphism theorem for modules and then applies Splitting criterion; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 9.40 (Injective surjective)**.** Test the claim: Every injective module map is surjective.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: multiplication by 2 on $\mathbb{Z}$. ☐

Exercise 9.41 (Surjective injective)**.** Test the claim: Every surjection is injective.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: $\mathbb{Z} \to \mathbb{Z}/2$. ☐

Exercise 9.42 (Zero composite)**.** Test the claim: $g \circ f = 0$ implies exactness at the middle term.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: it gives only image contained in kernel. ☐

Applications and investigations

Exercise 9.43 (Presentations)**.** How do exact sequences encode generators and relations?

Hint. Translate the situation into kernels, cokernels, and exact sequences. ☐

Solution. A free surjection supplies generators and its kernel records relations. ☐

Exercise 9.44 (Failure measures)**.** What do kernel and cokernel measure?

Hint. Translate the situation into kernels, cokernels, and exact sequences. ☐

Solution. They measure failure of injectivity and surjectivity. ☐

Challenge problems

Exercise 9.45 (Local-global synthesis)**.** Connect the chapter ideas 'Modules' and 'Examples' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. □

Solution. The argument begins with An A-module is an abelian group with compatible scalar multiplication by A. It then uses the later viewpoint: Ideals, quotient modules, free modules, torsion abelian groups, and vector spaces all fit the same language. The bridge is supplied by the structural theorems proved in the chapter. □

Exercise 9.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Chapter 10

Tensor Products

The tensor product converts bilinear maps into linear maps. It controls base change, localization, flatness, and many multilinear constructions throughout algebra and geometry.

Learning goals

After this chapter, the reader should be able to:

- construct tensor products from bilinear maps using their universal property;
- compute elementary tensor products of cyclic modules and quotients;
- use tensor relations to simplify pure tensors and detect zero tensors;
- prove right exactness of tensor product and identify where injectivity may fail;
- apply base-change identities involving tensor products of algebras and modules;
- distinguish tensor product from direct product, direct sum, and ordinary multiplication;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

10.1 Balanced bilinear maps

An A-balanced map satisfies $b(am, n) = b(m, an)$ in addition to additivity in each variable.

10.2 Tensor product

$M \otimes_A N$ is generated by symbols $m \otimes n$ modulo additivity and balancing relations.

Example 10.1 (Cyclic tensor product)**.** For positive integers m, n, $\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n)$. Thus $\mathbb{Z}/6 \otimes \mathbb{Z}/15 \cong \mathbb{Z}/3$.

10.3 Universal property

Balanced bilinear maps out of $M \times N$ correspond uniquely to linear maps out of $M \otimes_A N$.

10.4 Basic identities

$A \otimes_A M \cong M$ and direct sums commute with tensor products.

10.5 Quotient tensors

$(A/I) \otimes_A M \cong M/IM$.

Example 10.2 (Tensor with a quotient)**.** For an ideal I and module M, $(R/I)\otimes_R M \cong M/IM$. Tensor relations force elements of I to act as zero.

10.6 Associativity and symmetry

Canonical isomorphisms identify iterated tensor products and swap factors over a commutative ring.

10.7 Right exactness

Tensoring preserves cokernels and surjections, but not necessarily injections.

Example 10.3 (Extension of scalars)**.** For a field extension K/k, $K \otimes_k k^n \cong K^n$. A k-basis becomes a K-basis after scalar extension.

10.8 Localization

$S^{-1}A \otimes_A M \cong S^{-1}M$.

10.9 Examples

Cyclic groups, quotient modules, field extensions, and polynomial base change exhibit torsion and dimension phenomena.

10.10 Multilinear extension

Exterior and symmetric powers are built by further quotienting tensor powers.

10.11 Core structural results

Theorem 10.4 (Universal property)**.** *For every balanced bilinear map $b : M \times N \to P$, there is a unique linear map $\widetilde{b} : M \otimes_A N \to P$ with $\widetilde{b}(m \otimes n) = b(m, n)$.*

Proof. Construct the tensor product as the free module on pairs modulo the subgroup generated by additivity and balancing relations. The relations are exactly those killed by every balanced bilinear map. □

Theorem 10.5 (Quotient tensor identity)**.** *For an ideal $I \subset A$, $(A/I) \otimes_A M \cong M/IM$.*

Proof. Send $(a + I) \otimes m$ to $am + IM$. The map is balanced and surjective; the universal properties on both sides show that its kernel is precisely the relations generated by IM. □

Theorem 10.6 (Right exactness of tensor)**.** *If $M' \to M \to M'' \to 0$ is exact, then $M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$ is exact.*

Proof. The tensor product is a left adjoint in either variable, so it preserves cokernels; directly, generators in the kernel of the final map differ by tensors coming from the preceding module. □

10.12 Worked examples

Example 10.7 (Balanced bilinear maps diagnostic)**.** Give a rigorous worked analysis of Balanced bilinear maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An A-balanced map satisfies $b(am, n) = b(m, an)$ in addition to additivity in each variable. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 10.8 (Tensor product diagnostic)**.** Give a rigorous worked analysis of Tensor product. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $M \otimes_A N$ is generated by symbols $m \otimes n$ modulo additivity and balancing relations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 10.9 (Universal property diagnostic)**.** Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Balanced bilinear maps out of $M \times N$ correspond uniquely to linear maps out of $M \otimes_A N$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 10.10 (Basic identities diagnostic)**.** Give a rigorous worked analysis of Basic identities. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $A \otimes_A M \cong M$ and direct sums commute with tensor products. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

10.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 10.11 (Balanced bilinear maps diagnostic)**.** Give a rigorous worked analysis of Balanced bilinear maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An A-balanced map satisfies $b(am, n) = b(m, an)$ in addition to additivity in each variable. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 10.12 (Tensor product diagnostic)**.** Give a rigorous worked analysis of Tensor product. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $M \otimes_A N$ is generated by symbols $m \otimes n$ modulo additivity and balancing relations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 10.13 (Universal property diagnostic)**.** Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Balanced bilinear maps out of $M \times N$ correspond uniquely to linear maps out of $M \otimes_A N$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 10.14 (Basic identities diagnostic)**.** Give a rigorous worked analysis of Basic identities. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $A \otimes_A M \cong M$ and direct sums commute with tensor products. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 10.15 (Quotient tensors diagnostic)**.** Give a rigorous worked analysis of Quotient tensors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $(A/I) \otimes_A M \cong M/IM$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 10.16 (Associativity and symmetry diagnostic)**.** Give a rigorous worked analysis of Associativity and symmetry. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Canonical isomorphisms identify iterated tensor products and swap factors over a commutative ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 10.17 (Right exactness diagnostic)**.** Give a rigorous worked analysis of Right exactness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Tensoring preserves cokernels and surjections, but not necessarily injections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 10.18 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $S^{-1}A \otimes_A M \cong S^{-1}M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 10.19 (Universal property proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For every balanced bilinear map $b : M \times N \to P$, there is a unique linear map $\widetilde{b} : M \otimes_A N \to P$ with $\widetilde{b}(m \otimes n) = b(m, n)$.

Solution. Construct the tensor product as the free module on pairs modulo the subgroup generated by additivity and balancing relations. The relations are exactly those killed by every balanced bilinear map. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 10.20 (Quotient tensor identity proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For an ideal $I \subset A$, $(A/I) \otimes_A M \cong M/IM$.

Solution. Send $(a + I) \otimes m$ to $am + IM$. The map is balanced and surjective; the universal properties on both sides show that its kernel is precisely the relations generated by IM. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 10.21 (Right exactness of tensor proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $M' \to M \to M'' \to 0$ is exact, then $M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$ is exact.

Solution. The tensor product is a left adjoint in either variable, so it preserves cokernels; directly, generators in the kernel of the final map differ by tensors coming from the preceding module. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 10.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. The tensor product converts bilinear maps into linear maps. It controls base change, localization, flatness, and many multilinear constructions throughout algebra and geometry. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

10.14 Exercises with complete solutions

Exercise 10.23. State and apply the central principle behind Balanced bilinear maps. Give one concrete algebraic consequence.

Hint. Use the universal property: a bilinear map M x N -> P factors uniquely through M tensor N. □

Solution. An A-balanced map satisfies $b(am, n) = b(m, an)$ in addition to additivity in each variable. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 10.24. State and apply the central principle behind Tensor product. Give one concrete algebraic consequence.

Hint. Apply relations (m+m') tensor n, m tensor (n+n'), and (am) tensor n = m tensor (an) to simplify tensors. □

Solution. $M \otimes_A N$ is generated by symbols $m \otimes n$ modulo additivity and balancing relations. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 10.25. State and apply the central principle behind Universal property. Give one concrete algebraic consequence.

Hint. For cyclic modules, start from $A/I \otimes_A N \cong N/IN$ and reduce the computation to a quotient. □

Solution. Balanced bilinear maps out of $M \times N$ correspond uniquely to linear maps out of $M \otimes_A N$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 10.26. State and apply the central principle behind Basic identities. Give one concrete algebraic consequence.

Hint. To prove right exactness, tensor a presentation and identify the resulting cokernel. ☐

Solution. $A \otimes_A M \cong M$ and direct sums commute with tensor products. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 10.27. State and apply the central principle behind Quotient tensors. Give one concrete algebraic consequence.

Hint. Injectivity can fail after tensoring; look for torsion killed by multiplication before and after tensoring. ☐

Solution. $(A/I) \otimes_A M \cong M/IM$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 10.28. State and apply the central principle behind Associativity and symmetry. Give one concrete algebraic consequence.

Hint. Use pure tensors only as generators; not every tensor is itself pure. ☐

Solution. Canonical isomorphisms identify iterated tensor products and swap factors over a commutative ring. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 10.29. State and apply the central principle behind Right exactness. Give one concrete algebraic consequence.

Hint. For base change, reassociate tensors carefully and keep track of which ring acts at each stage. ☐

Solution. Tensoring preserves cokernels and surjections, but not necessarily injections. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 10.30. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. Distinguish direct sum from tensor product by their universal properties: one is additive coproduct, the other linearizes bilinear maps. ☐

Solution. $S^{-1}A \otimes_A M \cong S^{-1}M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

10.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 10.31 (Unit tensor)**.** Compute $R \otimes_R M$.

Hint. Use tensor products and their universal property. ☐

Solution. It is M. ☐

Exercise 10.32 (Zero tensor)**.** Compute $M \otimes_R 0$.

Hint. Use tensor products and their universal property. ☐

Solution. It is zero. ☐

Exercise 10.33 (Cyclic tensor)**.** Compute $\mathbb{Z}/4 \otimes \mathbb{Z}/6$.

Hint. Use tensor products and their universal property. ☐

Solution. It is $\mathbb{Z}/2$. ☐

Exercise 10.34 (Coprime tensor)**.** Compute $\mathbb{Z}/4 \otimes \mathbb{Z}/9$.

Hint. Use tensor products and their universal property. ☐

Solution. It is zero. ☐

Exercise 10.35 (Quotient tensor)**.** Compute $(R/I) \otimes_R (R/J)$.

Hint. Use tensor products and their universal property. ☐

Solution. It is $R/(I + J)$. ☐

Proofs

Exercise 10.36 (Universal property proof)**.** Prove: For every balanced bilinear map $b : M \times N \to P$, there is a unique linear map $\widetilde{b} : M \otimes_A N \to P$ with $\widetilde{b}(m \otimes n) = b(m, n)$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Construct the tensor product as the free module on pairs modulo the subgroup generated by additivity and balancing relations. The relations are exactly those killed by every balanced bilinear map. ☐

Exercise 10.37 (Quotient tensor identity proof)**.** Prove: For an ideal $I \subset A$, $(A/I) \otimes_A M \cong M/IM$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Send $(a + I) \otimes m$ to $am + IM$. The map is balanced and surjective; the universal properties on both sides show that its kernel is precisely the relations generated by IM. ☐

Exercise 10.38 (Right exactness of tensor proof)**.** Prove: If $M' \to M \to M'' \to 0$ is exact, then $M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$ is exact.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. The tensor product is a left adjoint in either variable, so it preserves cokernels; directly, generators in the kernel of the final map differ by tensors coming from the preceding module. ☐

Exercise 10.39 (Structural synthesis)**.** Prove a short corollary by combining Universal property with Quotient tensor identity. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes Universal property and then applies Quotient tensor identity; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 10.40 (Left exactness)**.** Test the claim: Tensoring always preserves injections.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: tensor multiplication by 2 with $\mathbb{Z}/2$. ☐

Exercise 10.41 (Pure tensor)**.** Test the claim: A pure tensor is zero only if one factor is zero.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: use $\mathbb{Z}/2 \otimes \mathbb{Z}/3 = 0$. ☐

Exercise 10.42 (Dimension rule)**.** Test the claim: Dimension multiplication holds for arbitrary modules.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False; it is a vector-space formula under suitable finiteness. ☐

Applications and investigations

Exercise 10.43 (Scalar extension)**.** Why is $B \otimes_A M$ extension of scalars?

Hint. Translate the situation into tensor products and their universal property. ☐

Solution. It is universal for reinterpreting the A-action through B. ☐

Exercise 10.44 (Affine products)**.** Why do tensor products of algebras model product coordinates?

Hint. Translate the situation into tensor products and their universal property. ☐

Solution. They combine independent coordinate functions over the base field. ☐

Challenge problems

Exercise 10.45 (Local-global synthesis)**.** Connect the chapter ideas 'Balanced bilinear maps' and 'Multilinear extension' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. □

Solution. The argument begins with An A-balanced map satisfies $b(am, n) = b(m, an)$ in addition to additivity in each variable. It then uses the later viewpoint: Exterior and symmetric powers are built by further quotienting tensor powers. The bridge is supplied by the structural theorems proved in the chapter. □

Exercise 10.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Chapter 11

Quotients and Base Change

Quotients and base change describe how algebraic relations behave after changing the scalar ring. Tensor products are the mechanism that transports modules, presentations, and exact sequences to the new base.

Learning goals

After this chapter, the reader should be able to:

- compute tensor products after quotienting rings or modules;
- derive standard base-change isomorphisms from universal properties;
- compare extension and restriction of scalars along a ring homomorphism;
- track ideals and modules through quotient and base-change constructions;
- identify when tensoring preserves or destroys injectivity in quotient sequences;
- use localization as a special case of base change to simplify calculations;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

11.1 Scalar extension

For $A \to B$, the base-changed module is $B \otimes_A M$.

11.2 Restriction of scalars

Every B-module is an A-module by composing scalar actions with $A \to B$.

Example 11.1 (Base change of a quotient)**.** For $A \to B$ and $I \subset A$, one has $B \otimes_A A/I \cong B/IB$. Equations defining I are transported to the new coefficient ring.

11.3 Quotient base change

$B \otimes_A (A/I)$ is naturally B/IB.

11.4 Submodule caution

Base change need not preserve injections unless the new scalar module is flat.

11.5 Presentations

Base changing a presentation replaces its matrices by the same matrices with entries mapped into B.

Example 11.2 (Field extension splitting)**.** Base changing $\mathbb{R}[x]/(x^2+1)$ to $\mathbb{C}$ gives $\mathbb{C}[x]/((x-i)(x+i))$. A polynomial irreducible over the original field can split after base change.

11.6 Localization as base change

Localization is scalar extension along $A \to S^{-1}A$.

11.7 Residue-field fibers

$M \otimes_A \kappa(\mathfrak{p})$ measures the fiber of a module at a prime.

Example 11.3 (Flat base change)**.** For $0 \to N \to M \to M/N \to 0$, tensoring with a flat A-algebra B preserves the injection and gives a short exact base-changed sequence.

11.8 Iterated base change

For $A \to B \to C$, there is a canonical isomorphism $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$.

11.9 Compatibility with quotients

Cokernels commute with base change because tensor is right exact.

11.10 Core structural results

Theorem 11.4 (Quotient under base change)**.** *For $A \to B$ and ideal $I \subset A$, $B \otimes_A A/I \cong B/IB$.*

Proof. Apply the quotient tensor identity to the A-module B; the submodule IB is generated by images of elements of I. □

Theorem 11.5 (Transitivity of base change)**.** *For $A \to B \to C$ and an A-module M, $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$.*

Proof. Send $c \otimes (b \otimes m)$ to $cb \otimes m$. Balancedness gives a well-defined map, and the inverse sends $c \otimes m$ to $c \otimes (1 \otimes m)$. □

Theorem 11.6 (Base change of a cokernel)**.** *If $M' \to M \to Q \to 0$ is exact, then $B \otimes_A M' \to B \otimes_A M \to B \otimes_A Q \to 0$ is exact.*

Proof. This is the right exactness of tensor product with the A-module B. □

11.11 Worked examples

Example 11.7 (Scalar extension diagnostic)**.** Give a rigorous worked analysis of Scalar extension. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For $A \to B$, the base-changed module is $B \otimes_A M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 11.8 (Restriction of scalars diagnostic)**.** Give a rigorous worked analysis of Restriction of scalars. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every B-module is an A-module by composing scalar actions with $A \to B$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 11.9 (Quotient base change diagnostic)**.** Give a rigorous worked analysis of Quotient base change. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $B \otimes_A (A/I)$ is naturally B/IB. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 11.10 (Submodule caution diagnostic)**.** Give a rigorous worked analysis of Submodule caution. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Base change need not preserve injections unless the new scalar module is flat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

11.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 11.11 (Scalar extension diagnostic)**.** Give a rigorous worked analysis of Scalar extension. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For $A \to B$, the base-changed module is $B \otimes_A M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 11.12 (Restriction of scalars diagnostic)**.** Give a rigorous worked analysis of Restriction of scalars. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every B-module is an A-module by composing scalar actions with $A \to B$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 11.13 (Quotient base change diagnostic)**.** Give a rigorous worked analysis of Quotient base change. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $B \otimes_A (A/I)$ is naturally B/IB. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 11.14 (Submodule caution diagnostic)**.** Give a rigorous worked analysis of Submodule caution. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Base change need not preserve injections unless the new scalar module is flat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 11.15 (Presentations diagnostic)**.** Give a rigorous worked analysis of Presentations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Base changing a presentation replaces its matrices by the same matrices with entries mapped into B. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 11.16 (Localization as base change diagnostic)**.** Give a rigorous worked analysis of Localization as base change. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization is scalar extension along $A \to S^{-1}A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 11.17 (Residue-field fibers diagnostic)**.** Give a rigorous worked analysis of Residue-field fibers. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $M \otimes_A \kappa(\mathfrak{p})$ measures the fiber of a module at a prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 11.18 (Iterated base change diagnostic)**.** Give a rigorous worked analysis of Iterated base change. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For $A \to B \to C$, there is a canonical isomorphism $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 11.19 (Quotient under base change proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $A \to B$ and ideal $I \subset A$, $B \otimes_A A/I \cong B/IB$.

Solution. Apply the quotient tensor identity to the A-module B; the submodule IB is generated by images of elements of I. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 11.20 (Transitivity of base change proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $A \to B \to C$ and an A-module M, $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$.

Solution. Send $c \otimes (b \otimes m)$ to $cb \otimes m$. Balancedness gives a well-defined map, and the inverse sends $c \otimes m$ to $c \otimes (1 \otimes m)$. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 11.21 (Base change of a cokernel proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $M' \to M \to Q \to 0$ is exact, then $B \otimes_A M' \to B \otimes_A M \to B \otimes_A Q \to 0$ is exact.

Solution. This is the right exactness of tensor product with the A-module B. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 11.22 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Quotients and base change describe how algebraic relations behave after changing the scalar ring. Tensor products are the mechanism that transports modules, presentations, and exact sequences to the new base. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. ☐

11.13 Exercises with complete solutions

Exercise 11.23. State and apply the central principle behind Scalar extension. Give one concrete algebraic consequence.

Hint. Use $(A/I) \otimes_A M \cong M/IM$ as the basic quotient/base-change identity. ☐

Solution. For $A \to B$, the base-changed module is $B \otimes_A M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 11.24. State and apply the central principle behind Restriction of scalars. Give one concrete algebraic consequence.

Hint. For ring maps $A \to B \to C$, reassociate $C \otimes_B (B \otimes_A M)$ to $C \otimes_A M$. ☐

Solution. Every B-module is an A-module by composing scalar actions with $A \to B$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 11.25. State and apply the central principle behind Quotient base change. Give one concrete algebraic consequence.

Hint. Extension of scalars sends M to $B \otimes_A M$; restriction of scalars forgets the B-action but keeps the underlying abelian group/module. ☐

Solution. $B \otimes_A (A/I)$ is naturally B/IB. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 11.26. State and apply the central principle behind Submodule caution. Give one concrete algebraic consequence.

Hint. When tensoring a quotient sequence, expect right exactness and check separately whether the left injection survives. ☐

Solution. Base change need not preserve injections unless the new scalar module is flat. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 11.27. State and apply the central principle behind Presentations. Give one concrete algebraic consequence.

Hint. Local base change is just $B = S^{-1}A$; translate general tensor identities to localization to test them. ☐

Solution. Base changing a presentation replaces its matrices by the same matrices with entries mapped into B. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 11.28. State and apply the central principle behind Localization as base change. Give one concrete algebraic consequence.

Hint. Track the image of an ideal under extension carefully: IB is generated by images of elements of I, not an arbitrary set-theoretic image. ☐

Solution. Localization is scalar extension along $A \to S^{-1}A$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 11.29. State and apply the central principle behind Residue-field fibers. Give one concrete algebraic consequence.

Hint. For quotient then base change, compare $B \otimes_A A/I$ with B/IB. ☐

Solution. $M \otimes_A \kappa(\mathfrak{p})$ measures the fiber of a module at a prime. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 11.30. State and apply the central principle behind Iterated base change. Give one concrete algebraic consequence.

Hint. Use universal properties to construct the natural map first, then prove it is an isomorphism rather than guessing the formula. ☐

Solution. For $A \to B \to C$, there is a canonical isomorphism $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

11.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 11.31 (Quotient base change)**.** Compute $B \otimes_A A/I$.

Hint. Use quotients and base change. ☐

Solution. It is B/IB. ☐

Exercise 11.32 (Polynomial extension)**.** Compute $K \otimes_k k[x]$.

Hint. Use quotients and base change. ☐

Solution. It is $K[x]$. □

Exercise 11.33 (Free module)**.** Compute $B \otimes_A A^n$.

Hint. Use quotients and base change. □

Solution. It is B^n. □

Exercise 11.34 (Nested quotient)**.** Compute $(A/I)/(J/I)$.

Hint. Use quotients and base change. □

Solution. It is A/J. □

Exercise 11.35 (Module quotient)**.** Compute $(A/I) \otimes_A M$.

Hint. Use quotients and base change. □

Solution. It is M/IM. □

Proofs

Exercise 11.36 (Quotient under base change proof)**.** Prove: For $A \to B$ and ideal $I \subset A$, $B \otimes_A A/I \cong B/IB$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. □

Solution. Apply the quotient tensor identity to the A-module B; the submodule IB is generated by images of elements of I. □

Exercise 11.37 (Transitivity of base change proof)**.** Prove: For $A \to B \to C$ and an A-module M, $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. □

Solution. Send $c \otimes (b \otimes m)$ to $cb \otimes m$. Balancedness gives a well-defined map, and the inverse sends $c \otimes m$ to $c \otimes (1 \otimes m)$. □

Exercise 11.38 (Base change of a cokernel proof)**.** Prove: If $M' \to M \to Q \to 0$ is exact, then $B \otimes_A M' \to B \otimes_A M \to B \otimes_A Q \to 0$ is exact.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. □

Solution. This is the right exactness of tensor product with the A-module B. □

Exercise 11.39 (Structural synthesis)**.** Prove a short corollary by combining Quotient under base change with Transitivity of base change. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. □

Solution. A valid synthesis first invokes Quotient under base change and then applies Transitivity of base change; the conclusion follows after checking the shared hypotheses. □

Counterexamples and hypothesis tests

Exercise 11.40 (Injectivity)**.** Test the claim: Every base change preserves injections.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False without flatness. ☐

Exercise 11.41 (Irreducibility)**.** Test the claim: Irreducible polynomials remain irreducible after every field extension.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: $x^2 + 1$ from $\mathbb{R}$ to $\mathbb{C}$. ☐

Exercise 11.42 (Extension contraction)**.** Test the claim: Ideal extension and contraction are always inverse.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False for arbitrary ring maps. ☐

Applications and investigations

Exercise 11.43 (Solving equations)**.** Why extend the coefficient field?

Hint. Translate the situation into quotients and base change. ☐

Solution. It can reveal roots and factorizations while retaining canonical base-change data. ☐

Exercise 11.44 (Fiber module)**.** Interpret $M \otimes_A \kappa(P)$.

Hint. Translate the situation into quotients and base change. ☐

Solution. It is the fiber of M at the prime P. ☐

Challenge problems

Exercise 11.45 (Local-global synthesis)**.** Connect the chapter ideas 'Scalar extension' and 'Compatibility with quotients' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. ☐

Solution. The argument begins with For $A \to B$, the base-changed module is $B \otimes_A M$. It then uses the later viewpoint: Cokernels commute with base change because tensor is right exact. The bridge is supplied by the structural theorems proved in the chapter. ☐

Exercise 11.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. ☐

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. ☐

Chapter 12

Hom and Finitely Presented Modules

Hom and finite presentation control maps out of modules and the stability of algebraic data under limits and base change. Finitely presented modules are those described by finitely many generators and finitely many relations.

Learning goals

After this chapter, the reader should be able to:

- compute Hom modules for finitely generated free and cyclic modules;
- distinguish finitely generated from finitely presented modules by explicit presentations;
- use finite presentation to justify commuting Hom with filtered colimits or localization in the chapter's setting;
- construct presentation matrices for modules and interpret generators and relations;
- apply the Hom universal property and naturality in concrete calculations;
- identify which finiteness hypothesis is needed in a proposed Hom/base-change statement;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

12.1 Hom modules

For A-modules M, N, $\mathrm{Hom}_A(M, N)$ is an A-module under pointwise operations.

12.2 Contravariance and covariance

Hom is contravariant in its first argument and covariant in its second.

Example 12.1 (Hom from a cyclic module)**.** A map $\mathbb{Z}/n \to M$ is determined by the image of 1, which must be killed by n. Hence $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n, M) \cong M[n]$.

12.3 Left exactness

$\mathrm{Hom}_A(M, -)$ preserves kernels but not generally surjections.

12.4 Finite generation

A module is finitely generated if a finite free module surjects onto it.

12.5 Finite presentation

A module is finitely presented if it is the cokernel of a map between finite free modules.

Example 12.2 (Presentation matrix)**.** The cokernel of $\mathrm{diag}(x, y) : R^2 \to R^2$ is $R/(x) \oplus R/(y)$. The diagonal matrix records two independent relations.

12.6 Generators and relations

A finite presentation packages finitely many generators together with finitely many module relations.

12.7 Hom and filtered colimits

Finite presentation is characterized by commuting of Hom out of the module with filtered colimits.

Example 12.3 (Hom and localization)**.** For finitely presented M, localization commutes with Hom in the first variable: the finite presentation reduces the claim to kernels between finite powers.

12.8 Base change of presentations

Finite presentations remain finite after scalar extension.

12.9 Localization of Hom

For finitely presented M, localization commutes with $\mathrm{Hom}_A(M, N)$.

12.10 Matrix viewpoint

Maps between finite free modules are matrices, so finite presentations translate module questions into matrix algebra.

12.11 Core structural results

Theorem 12.4 (Finite presentation criterion)**.** *M is finitely presented iff there exists an exact sequence $A^m \to A^n \to M \to 0$ with finite m, n.*

Proof. This is the definition translated into generators and relations: the second map chooses finitely many generators and the image of the first supplies finitely many generators for all relations. □

Theorem 12.5 (Hom is left exact)**.** *For* $0 \to N' \to N \to N''$*, the sequence* $0 \to \operatorname{Hom}(M, N') \to \operatorname{Hom}(M, N) \to \operatorname{Hom}(M, N'')$ *is exact.*

Proof. A map into N lands in N' exactly when its composite with $N \to N''$ is zero. Injectivity follows because $N' \to N$ is injective. □

Theorem 12.6 (Localization of Hom for finitely presented modules)**.** *If M is finitely presented, then* $S^{-1} \operatorname{Hom}_A(M, N) \cong \operatorname{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N)$*.*

Proof. Choose a finite presentation of M, apply Hom to it, localize the resulting kernel description, and use exactness of localization together with the evident formula for finite free modules. □

12.12 Worked examples

Example 12.7 (Hom modules diagnostic)**.** Give a rigorous worked analysis of Hom modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For A-modules M, N, $\operatorname{Hom}_A(M, N)$ is an A-module under pointwise operations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 12.8 (Contravariance and covariance diagnostic)**.** Give a rigorous worked analysis of Contravariance and covariance. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Hom is contravariant in its first argument and covariant in its second. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 12.9 (Left exactness diagnostic)**.** Give a rigorous worked analysis of Left exactness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\operatorname{Hom}_A(M, -)$ preserves kernels but not generally surjections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 12.10 (Finite generation diagnostic)**.** Give a rigorous worked analysis of Finite generation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A module is finitely generated if a finite free module surjects onto it. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

12.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 12.11 (Hom modules diagnostic)**.** Give a rigorous worked analysis of Hom modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For A-modules M, N, $\operatorname{Hom}_A(M, N)$ is an A-module under pointwise operations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 12.12 (Contravariance and covariance diagnostic)**.** Give a rigorous worked analysis of Contravariance and covariance. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Hom is contravariant in its first argument and covariant in its second. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 12.13 (Left exactness diagnostic)**.** Give a rigorous worked analysis of Left exactness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\mathrm{Hom}_A(M, -)$ preserves kernels but not generally surjections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 12.14 (Finite generation diagnostic)**.** Give a rigorous worked analysis of Finite generation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is finitely generated if a finite free module surjects onto it. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 12.15 (Finite presentation diagnostic)**.** Give a rigorous worked analysis of Finite presentation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is finitely presented if it is the cokernel of a map between finite free modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 12.16 (Generators and relations diagnostic)**.** Give a rigorous worked analysis of Generators and relations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A finite presentation packages finitely many generators together with finitely many module relations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 12.17 (Hom and filtered colimits diagnostic)**.** Give a rigorous worked analysis of Hom and filtered colimits. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite presentation is characterized by commuting of Hom out of the module with filtered colimits. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 12.18 (Base change of presentations diagnostic)**.** Give a rigorous worked analysis of Base change of presentations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite presentations remain finite after scalar extension. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 12.19 (Finite presentation criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: M is finitely presented iff there exists an exact sequence $A^m \to A^n \to M \to 0$ with finite m, n.

Solution. This is the definition translated into generators and relations: the second map chooses finitely many generators and the image of the first supplies finitely many generators for all relations. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 12.20 (Hom is left exact proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $0 \to N' \to N \to N''$, the sequence $0 \to \mathrm{Hom}(M, N') \to \mathrm{Hom}(M, N) \to \mathrm{Hom}(M, N'')$ is exact.

Solution. A map into N lands in N' exactly when its composite with $N \to N''$ is zero. Injectivity follows because $N' \to N$ is injective. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 12.21 (Localization of Hom for finitely presented modules proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If M is finitely presented, then $S^{-1} \mathrm{Hom}_A(M, N) \cong \mathrm{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N)$.

Solution. Choose a finite presentation of M, apply Hom to it, localize the resulting kernel description, and use exactness of localization together with the evident formula for finite free modules. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 12.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Hom and finite presentation control maps out of modules and the stability of algebraic data under limits and base change. Finitely presented modules are those described by finitely many generators and finitely many relations. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

12.14 Exercises with complete solutions

Exercise 12.23. State and apply the central principle behind Hom modules. Give one concrete algebraic consequence.

Hint. For a free module A^n, identify $\mathrm{Hom}_A(A^n, N)$ with N^n by recording images of basis vectors. □

Solution. For A-modules M, N, $\mathrm{Hom}_A(M, N)$ is an A-module under pointwise operations. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 12.24. State and apply the central principle behind Contravariance and covariance. Give one concrete algebraic consequence.

Hint. For A/I, a homomorphism A/I -> N is determined by an element annihilated by I; translate Hom into an annihilator condition. □

Solution. Hom is contravariant in its first argument and covariant in its second. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 12.25. State and apply the central principle behind Left exactness. Give one concrete algebraic consequence.

Hint. A finite presentation $A^m \to A^n \to M \to 0$ lets you compute Hom(M,N) as a kernel after applying Hom. □

Solution. $\mathrm{Hom}_A(M, -)$ preserves kernels but not generally surjections. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 12.26. State and apply the central principle behind Finite generation. Give one concrete algebraic consequence.

Hint. Finite generation controls generators; finite presentation also finitely controls relations; keep these two finiteness levels separate. □

Solution. A module is finitely generated if a finite free module surjects onto it. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 12.27. State and apply the central principle behind Finite presentation. Give one concrete algebraic consequence.

Hint. To see why finite presentation matters for localization or filtered-colimit statements, inspect where infinitely many relations would prevent a finite denominator choice. □

Solution. A module is finitely presented if it is the cokernel of a map between finite free modules. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 12.28. State and apply the central principle behind Generators and relations. Give one concrete algebraic consequence.

Hint. Construct the presentation matrix from the chosen generators and relation vectors column by column. □

Solution. A finite presentation packages finitely many generators together with finitely many module relations. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 12.29. State and apply the central principle behind Hom and filtered colimits. Give one concrete algebraic consequence.

Hint. Hom is contravariant in the first variable and covariant in the second; track arrow directions before composing induced maps. □

Solution. Finite presentation is characterized by commuting of Hom out of the module with filtered colimits. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 12.30. State and apply the central principle behind Base change of presentations. Give one concrete algebraic consequence.

Hint. Use the universal property of a cokernel/presentation to prove a proposed Hom description is natural. □

Solution. Finite presentations remain finite after scalar extension. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

12.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 12.31 (Hom from ring)**.** Compute $\mathrm{Hom}_R(R, M)$.

Hint. Use Hom and finite presentations. □

Solution. It is M. □

Exercise 12.32 (Hom to zero)**.** Compute $\mathrm{Hom}_R(M, 0)$.

Hint. Use Hom and finite presentations. □

Solution. It is zero. □

Exercise 12.33 (Cyclic Hom)**.** Compute $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/6, \mathbb{Z}/15)$.

Hint. Use Hom and finite presentations. □

Solution. It is isomorphic to $\mathbb{Z}/3$. □

Exercise 12.34 (Cokernel)**.** Find the cokernel of multiplication by f on R.

Hint. Use Hom and finite presentations. □

Solution. It is $R/(f)$. □

Exercise 12.35 (Presentation)**.** Present $R/(f, g)$ using finite free modules.

Hint. Use Hom and finite presentations. □

Solution. Use $R^2 \to R \to R/(f, g) \to 0$. □

Proofs

Exercise 12.36 (Finite presentation criterion proof)**.** Prove: M is finitely presented iff there exists an exact sequence $A^m \to A^n \to M \to 0$ with finite m, n.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. □

Solution. This is the definition translated into generators and relations: the second map chooses finitely many generators and the image of the first supplies finitely many generators for all relations. □

Exercise 12.37 (Hom is left exact proof)**.** Prove: For $0 \to N' \to N \to N''$, the sequence $0 \to \mathrm{Hom}(M, N') \to \mathrm{Hom}(M, N) \to \mathrm{Hom}(M, N'')$ is exact.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. A map into N lands in N' exactly when its composite with $N \to N''$ is zero. Injectivity follows because $N' \to N$ is injective. ☐

Exercise 12.38 (Localization of Hom for finitely presented modules proof)**.** Prove: If M is finitely presented, then $S^{-1}\operatorname{Hom}_A(M,N) \cong \operatorname{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N)$.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Choose a finite presentation of M, apply Hom to it, localize the resulting kernel description, and use exactness of localization together with the evident formula for finite free modules. ☐

Exercise 12.39 (Structural synthesis)**.** Prove a short corollary by combining Finite presentation criterion with Hom is left exact. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes Finite presentation criterion and then applies Hom is left exact; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 12.40 (Finite generation)**.** Test the claim: Every finitely generated module is finitely presented over every ring.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False over non-Noetherian rings. ☐

Exercise 12.41 (Hom exactness)**.** Test the claim: $\operatorname{Hom}(M,-)$ is always right exact.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False unless M has extra projectivity. ☐

Exercise 12.42 (Hom localization)**.** Test the claim: Localization commutes with Hom for every first module.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False in general; finite presentation is a standard sufficient hypothesis. ☐

Applications and investigations

Exercise 12.43 (Linear equations)**.** How does a presentation matrix encode a module?

Hint. Translate the situation into Hom and finite presentations. ☐

Solution. Generators are free coordinates and matrix columns or rows encode relations. ☐

Exercise 12.44 (Stable base change)**.** Why are finitely presented modules convenient under base change?

Hint. Translate the situation into Hom and finite presentations. ☐

Solution. The same finite relation matrix can be transported coefficientwise. ☐

Challenge problems

Exercise 12.45 (Local-global synthesis)**.** Connect the chapter ideas 'Hom modules' and 'Matrix viewpoint' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. □

Solution. The argument begins with For A-modules M, N, $\mathrm{Hom}_A(M, N)$ is an A-module under pointwise operations. It then uses the later viewpoint: Maps between finite free modules are matrices, so finite presentations translate module questions into matrix algebra. The bridge is supplied by the structural theorems proved in the chapter. □

Exercise 12.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Chapter 13

Free and Projective Modules

Free modules have bases, while projective modules are direct summands of free modules. Projectivity is exactly the lifting property that makes Hom out of the module preserve surjections.

Learning goals

After this chapter, the reader should be able to:

- recognize free modules from explicit bases and coordinate decompositions;
- test projectivity using lifting properties, direct summands, or idempotent matrices;
- prove that free modules are projective and that direct summands of projective modules are projective;
- construct projective modules from idempotents and splitting exact sequences;
- distinguish projective modules from free modules by examples where available;
- localize projective modules and analyze finite projectivity through local criteria;

Conceptual roadmap

$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$

13.1 Free modules

A free module is a direct sum of copies of the ring with a distinguished basis.

13.2 Rank

Over suitable rings, finite free modules have a well-defined rank; locally, projective modules acquire ranks prime by prime.

Example 13.1 (Idempotent projective)**.** If $e^2 = e$, then $R = eR \oplus (1-e)R$. Thus eR is a direct summand of the free module R and is projective.

13.3 Projective modules

A module is projective if maps to quotients lift across surjections.

13.4 Direct summands

Projective modules are exactly direct summands of free modules.

13.5 Idempotent matrices

Finite projective modules arise as images of idempotent endomorphisms of finite free modules.

Example 13.2 (Splitting criterion)**.** A module P is projective exactly when every surjection onto P splits. Applying this to a free surjection shows that projectives are precisely direct summands of free modules.

13.6 Splitting exact sequences

A surjection onto a projective module splits.

13.7 Hom exactness

P is projective iff $\mathrm{Hom}(P, -)$ is exact.

Example 13.3 (Local freeness)**.** A finitely generated projective module becomes free after localization at a prime. Its local rank is the algebraic analogue of vector-bundle rank.

13.8 Localization

Localization preserves projectivity, and finite projective modules are locally free.

13.9 Examples

Free modules, ideals generated by idempotents, and vector bundles over affine rings illustrate projectivity.

13.10 Core structural results

Theorem 13.4 (Projective summand criterion)**.** *An A-module P is projective iff it is a direct summand of a free module.*

Proof. Choose a free module F surjecting onto P. If P is projective, the surjection splits, so $F \cong P \oplus K$. Conversely direct summands of free modules inherit the lifting property. □

Theorem 13.5 (Splitting criterion)**.** *A module P is projective iff every short exact sequence $0 \to K \to M \to P \to 0$ splits.*

Proof. Projectivity lifts the identity map of P through the surjection $M \to P$, producing a section. Conversely apply the splitting property to pullbacks of arbitrary surjections. □

Theorem 13.6 (Hom exactness criterion)**.** *P is projective iff $\mathrm{Hom}_A(P, -)$ is an exact functor.*

Proof. Hom is always left exact. Exactness on surjections is precisely the lifting property defining projectivity. □

13.11 Worked examples

Example 13.7 (Free modules diagnostic)**.** Give a rigorous worked analysis of Free modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A free module is a direct sum of copies of the ring with a distinguished basis. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 13.8 (Rank diagnostic)**.** Give a rigorous worked analysis of Rank. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Over suitable rings, finite free modules have a well-defined rank; locally, projective modules acquire ranks prime by prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 13.9 (Projective modules diagnostic)**.** Give a rigorous worked analysis of Projective modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A module is projective if maps to quotients lift across surjections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 13.10 (Direct summands diagnostic)**.** Give a rigorous worked analysis of Direct summands. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Projective modules are exactly direct summands of free modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

13.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 13.11 (Free modules diagnostic)**.** Give a rigorous worked analysis of Free modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A free module is a direct sum of copies of the ring with a distinguished basis. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 13.12 (Rank diagnostic)**.** Give a rigorous worked analysis of Rank. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over suitable rings, finite free modules have a well-defined rank; locally, projective modules acquire ranks prime by prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 13.13 (Projective modules diagnostic)**.** Give a rigorous worked analysis of Projective modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is projective if maps to quotients lift across surjections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 13.14 (Direct summands diagnostic)**.** Give a rigorous worked analysis of Direct summands. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Projective modules are exactly direct summands of free modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 13.15 (Idempotent matrices diagnostic)**.** Give a rigorous worked analysis of Idempotent matrices. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite projective modules arise as images of idempotent endomorphisms of finite free modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 13.16 (Splitting exact sequences diagnostic)**.** Give a rigorous worked analysis of Splitting exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A surjection onto a projective module splits. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 13.17 (Hom exactness diagnostic)**.** Give a rigorous worked analysis of Hom exactness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. P is projective iff $\mathrm{Hom}(P, -)$ is exact. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 13.18 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization preserves projectivity, and finite projective modules are locally free. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 13.19 (Projective summand criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: An A-module P is projective iff it is a direct summand of a free module.

Solution. Choose a free module F surjecting onto P. If P is projective, the surjection splits, so $F \cong P \oplus K$. Conversely direct summands of free modules inherit the lifting property. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 13.20 (Splitting criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A module P is projective iff every short exact sequence $0 \to K \to M \to P \to 0$ splits.

Solution. Projectivity lifts the identity map of P through the surjection $M \to P$, producing a section. Conversely apply the splitting property to pullbacks of arbitrary surjections. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 13.21 (Hom exactness criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: P is projective iff $\mathrm{Hom}_A(P, -)$ is an exact functor.

Solution. Hom is always left exact. Exactness on surjections is precisely the lifting property defining projectivity. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 13.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Free modules have bases, while projective modules are direct summands of free modules. Projectivity is exactly the lifting property that makes Hom out of the module preserve surjections. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. ☐

13.13 Exercises with complete solutions

Exercise 13.23. State and apply the central principle behind Free modules. Give one concrete algebraic consequence.

Hint. A free basis gives a direct coordinate isomorphism with A^n; verify both spanning and uniqueness of coordinates. ☐

Solution. A free module is a direct sum of copies of the ring with a distinguished basis. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 13.24. State and apply the central principle behind Rank. Give one concrete algebraic consequence.

Hint. Projectivity means lifting maps across surjections; draw the lifting square and identify the map that must exist. ☐

Solution. Over suitable rings, finite free modules have a well-defined rank; locally, projective modules acquire ranks prime by prime. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 13.25. State and apply the central principle behind Projective modules. Give one concrete algebraic consequence.

Hint. To show a direct summand is projective, combine a lifting for the ambient projective module with the projection onto the summand. ☐

Solution. A module is projective if maps to quotients lift across surjections. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 13.26. State and apply the central principle behind Direct summands. Give one concrete algebraic consequence.

Hint. A splitting of 0 -> K -> F -> P -> 0 exhibits P as a direct summand of a free module. ☐

Solution. Projective modules are exactly direct summands of free modules. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 13.27. State and apply the central principle behind Idempotent matrices. Give one concrete algebraic consequence.

Hint. An idempotent matrix e with $e^2 = e$ gives the projective module im(e); use $A^n = \mathrm{im}(e) \oplus \ker(e)$. □

Solution. Finite projective modules arise as images of idempotent endomorphisms of finite free modules. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 13.28. State and apply the central principle behind Splitting exact sequences. Give one concrete algebraic consequence.

Hint. Localizing a direct-summand decomposition proves localization of a projective module is projective. □

Solution. A surjection onto a projective module splits. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 13.29. State and apply the central principle behind Hom exactness. Give one concrete algebraic consequence.

Hint. For finitely generated projectives, test local freeness at primes only in settings where the local criterion has been established. □

Solution. P is projective iff $\mathrm{Hom}(P, -)$ is exact. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 13.30. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. Do not assume projective implies free globally; search for nontrivial idempotents or geometric examples when the ring is not local/PID-like. □

Solution. Localization preserves projectivity, and finite projective modules are locally free. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

13.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 13.31 (Free projective)**.** Is R^5 projective?

Hint. Use free modules, projective splittings, and idempotents. ☐

Solution. Yes. ☐

Exercise 13.32 (Direct summand)**.** If $R^4 = P \oplus Q$, is P projective?

Hint. Use free modules, projective splittings, and idempotents. ☐

Solution. Yes. ☐

Exercise 13.33 (Idempotent complement)**.** What complements eR when $e^2 = e$?

Hint. Use free modules, projective splittings, and idempotents. ☐

Solution. $(1-e)R$. ☐

Exercise 13.34 (Localized free)**.** Compute $(R^n)_P$.

Hint. Use free modules, projective splittings, and idempotents. ☐

Solution. It is R_P^n. ☐

Exercise 13.35 (Split quotient)**.** What does a section of $F \to P$ imply?

Hint. Use free modules, projective splittings, and idempotents. ☐

Solution. It makes P a direct summand of F. ☐

Proofs

Exercise 13.36 (Projective summand criterion proof)**.** Prove: An A-module P is projective iff it is a direct summand of a free module.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Choose a free module F surjecting onto P. If P is projective, the surjection splits, so $F \cong P \oplus K$. Conversely direct summands of free modules inherit the lifting property. ☐

Exercise 13.37 (Splitting criterion proof)**.** Prove: A module P is projective iff every short exact sequence $0 \to K \to M \to P \to 0$ splits.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Projectivity lifts the identity map of P through the surjection $M \to P$, producing a section. Conversely apply the splitting property to pullbacks of arbitrary surjections. ☐

Exercise 13.38 (Hom exactness criterion proof)**.** Prove: P is projective iff $\mathrm{Hom}_A(P, -)$ is an exact functor.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Hom is always left exact. Exactness on surjections is precisely the lifting property defining projectivity. ☐

Exercise 13.39 (Structural synthesis)**.** Prove a short corollary by combining Projective summand criterion with Splitting criterion. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes Projective summand criterion and then applies Splitting criterion; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 13.40 (Projective free)**.** Test the claim: Every projective module is free.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False over general rings; idempotent summands give examples. ☐

Exercise 13.41 (Submodule free)**.** Test the claim: Every submodule of a free module is projective over every ring.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False in general. ☐

Exercise 13.42 (Finite generation)**.** Test the claim: Every projective module is finitely generated.

Hint. Try a smallest standard ring or module from this chapter. ☐

Solution. False: infinite-rank free modules are projective. ☐

Applications and investigations

Exercise 13.43 (Vector bundles)**.** Why do finite projectives model algebraic vector bundles?

Hint. Translate the situation into free modules, projective splittings, and idempotents. ☐

Solution. They are locally free of finite rank. ☐

Exercise 13.44 (Resolutions)**.** Why are projectives useful in resolutions?

Hint. Translate the situation into free modules, projective splittings, and idempotents. ☐

Solution. Surjections onto them split, eliminating lifting obstructions. ☐

Challenge problems

Exercise 13.45 (Local-global synthesis)**.** Connect the chapter ideas 'Free modules' and 'Examples' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. □

Solution. The argument begins with A free module is a direct sum of copies of the ring with a distinguished basis. It then uses the later viewpoint: Free modules, ideals generated by idempotents, and vector bundles over affine rings illustrate projectivity. The bridge is supplied by the structural theorems proved in the chapter. □

Exercise 13.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Chapter 14

Flat Modules

Flat modules are those whose tensor product functor preserves injections and hence all finite exact sequences. Flatness is the homological condition behind well-behaved base change.

Learning goals

After this chapter, the reader should be able to:

- test flatness using preservation of injective maps or exact sequences under tensor product;
- prove that free and projective modules are flat;
- verify that localization is flat as a module over the original ring;
- use ideal criteria for flatness in finitely generated situations where developed;
- construct examples of nonflat modules from torsion or failed injectivity after tensoring;
- distinguish flatness from projectivity and freeness by structural consequences and examples;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

14.1 Definition of flatness

An A-module F is flat if $- \otimes_A F$ is exact.

14.2 Injection test

Because tensor is always right exact, flatness is equivalent to preserving injections.

Example 14.1 (Localization is flat)**.** For every multiplicative set S, $S^{-1}R$ is flat because $S^{-1}R \otimes_R M \cong S^{-1}M$ and localization is exact.

14.3 Free and projective modules

Free modules are flat, and direct summands of flat modules are flat; hence projectives are flat.

14.4 Localization

Every localization $S^{-1}A$ is flat over A.

14.5 Filtered colimits

Filtered colimits of flat modules are flat.

Example 14.2 (Nonflat cyclic module)**.** Tensor $\mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z}$ with $\mathbb{Z}/2$. The induced map is zero, so injectivity is lost and $\mathbb{Z}/2$ is not flat.

14.6 Ideal criterion

Flatness can be tested by injectivity of $I \otimes F \to F$ for finitely generated ideals under standard hypotheses.

14.7 Tor preview

$\mathrm{Tor}_1^A(-, F)$ measures the failure of flatness.

Example 14.3 (Ideal criterion)**.** Flatness can be tested by requiring $I \otimes_R M \to M$ to be injective for finitely generated ideals I. This turns exactness into a relation test.

14.8 Faithful flatness

A flat module is faithfully flat when tensoring also detects nonzero modules.

14.9 Base change

Flat scalar extension preserves short exact sequences and scheme-theoretic fiber behavior.

14.10 Examples and failures

Quotient modules are generally not flat; localization provides the basic positive example.

14.11 Core structural results

Theorem 14.4 (Free implies flat)**.** *Every free A-module is flat.*

Proof. Tensoring with a direct sum of copies of A gives the corresponding direct sum of the original sequence. Direct sums preserve exactness in modules. □

Theorem 14.5 (Projective implies flat)**.** *Every projective module is flat.*

Proof. A projective module is a direct summand of a free module. Tensoring with the corresponding direct-sum decomposition shows the tensor functor for the projective summand is a direct summand of an exact functor and hence exact. □

Theorem 14.6 (Localization is flat)**.** *$S^{-1}A$ is flat as an A-module.*

Proof. Tensoring with $S^{-1}A$ is localization of modules, and localization is exact. □

14.12 Worked examples

Example 14.7 (Definition of flatness diagnostic)**.** Give a rigorous worked analysis of Definition of flatness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An A-module F is flat if $- \otimes_A F$ is exact. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 14.8 (Injection test diagnostic)**.** Give a rigorous worked analysis of Injection test. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Because tensor is always right exact, flatness is equivalent to preserving injections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 14.9 (Free and projective modules diagnostic)**.** Give a rigorous worked analysis of Free and projective modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Free modules are flat, and direct summands of flat modules are flat; hence projectives are flat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 14.10 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every localization $S^{-1}A$ is flat over A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

14.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 14.11 (Definition of flatness diagnostic)**.** Give a rigorous worked analysis of Definition of flatness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An A-module F is flat if $- \otimes_A F$ is exact. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 14.12 (Injection test diagnostic)**.** Give a rigorous worked analysis of Injection test. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Because tensor is always right exact, flatness is equivalent to preserving injections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 14.13 (Free and projective modules diagnostic)**.** Give a rigorous worked analysis of Free and projective modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Free modules are flat, and direct summands of flat modules are flat; hence projectives are flat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 14.14 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every localization $S^{-1}A$ is flat over A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 14.15 (Filtered colimits diagnostic)**.** Give a rigorous worked analysis of Filtered colimits. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Filtered colimits of flat modules are flat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 14.16 (Ideal criterion diagnostic)**.** Give a rigorous worked analysis of Ideal criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Flatness can be tested by injectivity of $I \otimes F \to F$ for finitely generated ideals under standard hypotheses. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 14.17 (Tor preview diagnostic)**.** Give a rigorous worked analysis of Tor preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\mathrm{Tor}_1^A(-, F)$ measures the failure of flatness. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 14.18 (Faithful flatness diagnostic)**.** Give a rigorous worked analysis of Faithful flatness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A flat module is faithfully flat when tensoring also detects nonzero modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 14.19 (Free implies flat proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Every free A-module is flat.

Solution. Tensoring with a direct sum of copies of A gives the corresponding direct sum of the original sequence. Direct sums preserve exactness in modules. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 14.20 (Projective implies flat proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Every projective module is flat.

Solution. A projective module is a direct summand of a free module. Tensoring with the corresponding direct-sum decomposition shows the tensor functor for the projective summand is a direct summand of an exact functor and hence exact. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 14.21 (Localization is flat proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: $S^{-1}A$ is flat as an A-module.

Solution. Tensoring with $S^{-1}A$ is localization of modules, and localization is exact. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 14.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Flat modules are those whose tensor product functor preserves injections and hence all finite exact sequences. Flatness is the homological condition behind well-behaved base change. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. ☐

14.14 Exercises with complete solutions

Exercise 14.23. State and apply the central principle behind Definition of flatness. Give one concrete algebraic consequence.

Hint. To test flatness, tensor an injective map N' -> N with M and check whether injectivity survives. ☐

Solution. An A-module F is flat if $- \otimes_A F$ is exact. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 14.24. State and apply the central principle behind Injection test. Give one concrete algebraic consequence.

Hint. Free modules are flat because tensoring with A^I gives direct sums of the original sequence. ☐

Solution. Because tensor is always right exact, flatness is equivalent to preserving injections. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 14.25. State and apply the central principle behind Free and projective modules. Give one concrete algebraic consequence.

Hint. Projective modules are flat because direct summands of flat modules remain flat. ☐

Solution. Free modules are flat, and direct summands of flat modules are flat; hence projectives are flat. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 14.26. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. Localization is flat because localized exact sequences remain exact; connect the two viewpoints explicitly. ☐

Solution. Every localization $S^{-1}A$ is flat over A. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 14.27. State and apply the central principle behind Filtered colimits. Give one concrete algebraic consequence.

Hint. Use the ideal criterion by testing I tensor M -> M for finitely generated ideals only when the theorem's hypotheses permit it. □

Solution. Filtered colimits of flat modules are flat. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 14.28. State and apply the central principle behind Ideal criterion. Give one concrete algebraic consequence.

Hint. For a nonflat quotient A/I, tensor a short exact sequence involving multiplication by an element and look for a new kernel. □

Solution. Flatness can be tested by injectivity of $I \otimes F \to F$ for finitely generated ideals under standard hypotheses. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 14.29. State and apply the central principle behind Tor preview. Give one concrete algebraic consequence.

Hint. Tor_1 detects failure of flatness once Tor is available: nonzero Tor_1 is an obstruction. □

Solution. $\mathrm{Tor}_1^A(-, F)$ measures the failure of flatness. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 14.30. State and apply the central principle behind Faithful flatness. Give one concrete algebraic consequence.

Hint. Separate flat from projective: flatness preserves finite equations/exactness, while projectivity is a lifting/direct-summand condition. □

Solution. A flat module is faithfully flat when tensoring also detects nonzero modules. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

14.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 14.31 (Free flat)**.** Is R^n flat?

Hint. Use flatness and exactness under tensor product. □

Solution. Yes. □

Exercise 14.32 (Localization flat)**.** Is R_f flat over R?

Hint. Use flatness and exactness under tensor product. □

Solution. Yes. ☐

Exercise 14.33 (Injection preservation)**.** What must a flat tensor functor do to injections?

Hint. Use flatness and exactness under tensor product. ☐

Solution. It must preserve them. ☐

Exercise 14.34 (Integer quotient)**.** Is $\mathbb{Z}/3$ flat over $\mathbb{Z}$?

Hint. Use flatness and exactness under tensor product. ☐

Solution. No. ☐

Exercise 14.35 (Direct sum)**.** Is a direct sum of flat modules flat?

Hint. Use flatness and exactness under tensor product. ☐

Solution. Yes. ☐

Proofs

Exercise 14.36 (Free implies flat proof)**.** Prove: Every free A-module is flat.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Tensoring with a direct sum of copies of A gives the corresponding direct sum of the original sequence. Direct sums preserve exactness in modules. ☐

Exercise 14.37 (Projective implies flat proof)**.** Prove: Every projective module is flat.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. A projective module is a direct summand of a free module. Tensoring with the corresponding direct-sum decomposition shows the tensor functor for the projective summand is a direct summand of an exact functor and hence exact. ☐

Exercise 14.38 (Localization is flat proof)**.** Prove: $S^{-1}A$ is flat as an A-module.

Hint. Start from the chapter theorem hypotheses and identify the decisive algebraic map or containment. ☐

Solution. Tensoring with $S^{-1}A$ is localization of modules, and localization is exact. ☐

Exercise 14.39 (Structural synthesis)**.** Prove a short corollary by combining Free implies flat with Projective implies flat. State every hypothesis you use.

Hint. Apply the first theorem to move to its structural description, then use the second theorem on that description. ☐

Solution. A valid synthesis first invokes Free implies flat and then applies Projective implies flat; the conclusion follows after checking the shared hypotheses. ☐

Counterexamples and hypothesis tests

Exercise 14.40 (Flat projective)**.** Test the claim: Every flat module is projective.

Hint. Try a smallest standard ring or module from this chapter. □

Solution. False: $\mathbb{Q}$ is flat but not projective over $\mathbb{Z}$. □

Exercise 14.41 (Torsion-free)**.** Test the claim: Torsion-free is equivalent to flat over every domain.

Hint. Try a smallest standard ring or module from this chapter. □

Solution. False in general. □

Exercise 14.42 (Tensor exact)**.** Test the claim: Tensoring with any module is exact.

Hint. Try a smallest standard ring or module from this chapter. □

Solution. False: tensor is always right exact, not always left exact. □

Applications and investigations

Exercise 14.43 (Families)**.** Why is flatness associated with well-behaved families?

Hint. Translate the situation into flatness and exactness under tensor product. □

Solution. It prevents new kernel relations from appearing under base change. □

Exercise 14.44 (Local test)**.** Why can flatness be checked locally?

Hint. Translate the situation into flatness and exactness under tensor product. □

Solution. Flatness is local on the base ring. □

Challenge problems

Exercise 14.45 (Local-global synthesis)**.** Connect the chapter ideas 'Definition of flatness' and 'Examples and failures' in one rigorous argument.

Hint. State the relevant map, ideal, module, or universal property before drawing the conclusion. □

Solution. The argument begins with An A-module F is flat if $-\otimes_A F$ is exact. It then uses the later viewpoint: Quotient modules are generally not flat; localization provides the basic positive example. The bridge is supplied by the structural theorems proved in the chapter. □

Exercise 14.46 (Hypothesis audit)**.** Choose one theorem from the chapter and explain precisely what can fail if one key hypothesis is removed.

Hint. Use one of the explicit counterexamples in the graded set and compare it with the theorem statement. □

Solution. The theorem hypotheses are essential because the chapter's quotient, localization, exactness, or finiteness mechanism can fail outside them. A correct answer identifies the dropped hypothesis and exhibits a concrete failure. □

Part IV

Noetherian Algebra

Chapter 15

Noetherian Rings and Modules

Noetherian conditions impose finiteness on ideals and submodules. They are stable under quotients and finite-type polynomial extension and support induction arguments throughout algebraic geometry.

Learning goals

After this chapter, the reader should be able to:

- verify Noetherianity using ascending chains or finite generation of ideals/submodules;
- apply Hilbert basis-type arguments to polynomial extensions where established;
- prove that submodules and quotients of finitely generated modules over Noetherian rings retain finiteness;
- construct stabilization arguments from ascending chains;
- compute finite generating sets extracted from chain conditions;
- distinguish Noetherian hypotheses from merely finitely generated individual ideals or modules;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

15.1 ACC on submodules

A module is Noetherian when every ascending chain of submodules stabilizes.

15.2 Finite-generation criterion

A module is Noetherian iff every submodule is finitely generated.

Example 15.1 (A chain that stabilizes)**.** In $\mathbb{Z}$, every ideal is principal. An ascending chain $(a_1) \subseteq (a_2) \subseteq \cdots$ corresponds to a descending chain of positive divisors, so it cannot strictly increase forever. This is a concrete manifestation of the Noetherian condition.

15.3 Noetherian rings

A ring is Noetherian when it is Noetherian as a module over itself, equivalently every ideal is finitely generated.

15.4 Exact sequences

In a short exact sequence, the middle module is Noetherian iff both ends are Noetherian.

15.5 Finite modules

Finite modules over a Noetherian ring are Noetherian.

15.6 Quotients

Quotients of Noetherian rings and modules are Noetherian.

Example 15.2 (Finite generation in a polynomial ideal)**.** In $k[x, y]$, the ideal $I = (x^2, xy, y^2)$ is generated by three quadratic monomials. Every higher-degree monomial in I is obtained by multiplying one of these generators, illustrating finite generation in a Noetherian polynomial ring.

15.7 Hilbert basis theorem

If A is Noetherian, then $A[x]$ is Noetherian.

15.8 Finite-type algebras

Iterating Hilbert's theorem shows finitely generated A-algebras are Noetherian.

Example 15.3 (Quotient remains Noetherian)**.** Since $k[x, y]$ is Noetherian, the quotient $k[x, y]/(xy)$ is Noetherian. Ideals of the quotient correspond to ideals of $k[x, y]$ containing (xy), so finite generation descends through the quotient.

15.9 Noetherian induction

Properties can be proved by assuming all strictly larger subobjects satisfy them.

15.10 Examples

Fields, PID-like rings, polynomial rings over fields, and many coordinate rings are Noetherian.

15.11 Core structural results

Theorem 15.4 (Finite-generation criterion)**.** *An A-module is Noetherian iff every submodule is finitely generated.*

Proof. If every submodule is finite, the union of an ascending chain is finitely generated and therefore contained in one stage. Conversely, if a submodule were not finitely generated, repeatedly adjoining new elements would produce a strictly increasing chain. $\square$

Theorem 15.5 (Exact-sequence criterion)**.** *For $0 \to M' \to M \to M'' \to 0$, M is Noetherian iff M' and M'' are Noetherian.*

Proof. Submodules of M' and images in M'' control any chain in M. For the converse, submodules and quotients of a Noetherian module are Noetherian. □

Theorem 15.6 (Hilbert basis theorem)**.** *If A is Noetherian, then $A[x]$ is Noetherian.*

Proof. For an ideal $I \subset A[x]$, leading coefficients form an ideal of A and stabilize under finite generation. Choose finitely many polynomials controlling high-degree leading terms and finitely many lower-degree remainders; together they generate I. □

15.12 Worked examples

Example 15.7 (ACC on submodules diagnostic)**.** Give a rigorous worked analysis of ACC on submodules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A module is Noetherian when every ascending chain of submodules stabilizes. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 15.8 (Finite-generation criterion diagnostic)**.** Give a rigorous worked analysis of Finite-generation criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A module is Noetherian iff every submodule is finitely generated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 15.9 (Noetherian rings diagnostic)**.** Give a rigorous worked analysis of Noetherian rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A ring is Noetherian when it is Noetherian as a module over itself, equivalently every ideal is finitely generated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 15.10 (Exact sequences diagnostic)**.** Give a rigorous worked analysis of Exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. In a short exact sequence, the middle module is Noetherian iff both ends are Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

15.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 15.11 (ACC on submodules diagnostic)**.** Give a rigorous worked analysis of ACC on submodules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is Noetherian when every ascending chain of submodules stabilizes. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 15.12 (Finite-generation criterion diagnostic)**.** Give a rigorous worked analysis of Finite-generation criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is Noetherian iff every submodule is finitely generated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 15.13 (Noetherian rings diagnostic)**.** Give a rigorous worked analysis of Noetherian rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A ring is Noetherian when it is Noetherian as a module over itself, equivalently every ideal is finitely generated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 15.14 (Exact sequences diagnostic)**.** Give a rigorous worked analysis of Exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. In a short exact sequence, the middle module is Noetherian iff both ends are Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 15.15 (Finite modules diagnostic)**.** Give a rigorous worked analysis of Finite modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite modules over a Noetherian ring are Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 15.16 (Quotients diagnostic)**.** Give a rigorous worked analysis of Quotients. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Quotients of Noetherian rings and modules are Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 15.17 (Hilbert basis theorem diagnostic)**.** Give a rigorous worked analysis of Hilbert basis theorem. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. If A is Noetherian, then $A[x]$ is Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 15.18 (Finite-type algebras diagnostic)**.** Give a rigorous worked analysis of Finite-type algebras. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Iterating Hilbert's theorem shows finitely generated A-algebras are Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 15.19 (Finite-generation criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: An A-module is Noetherian iff every submodule is finitely generated.

Solution. If every submodule is finite, the union of an ascending chain is finitely generated and therefore contained in one stage. Conversely, if a submodule were not finitely generated, repeatedly adjoining new elements would produce a strictly increasing chain. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 15.20 (Exact-sequence criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $0 \to M' \to M \to M'' \to 0$, M is Noetherian iff M' and M'' are Noetherian.

Solution. Submodules of M' and images in M'' control any chain in M. For the converse, submodules and quotients of a Noetherian module are Noetherian. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 15.21 (Hilbert basis theorem proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If A is Noetherian, then $A[x]$ is Noetherian.

Solution. For an ideal $I \subset A[x]$, leading coefficients form an ideal of A and stabilize under finite generation. Choose finitely many polynomials controlling high-degree leading terms and finitely many lower-degree remainders; together they generate I. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 15.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Noetherian conditions impose finiteness on ideals and submodules. They are stable under quotients and finite-type polynomial extension and support induction arguments throughout algebraic geometry. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

15.14 Exercises with complete solutions

Exercise 15.23. State and apply the central principle behind ACC on submodules. Give one concrete algebraic consequence.

Hint. To prove Noetherianity from ACC, start an ascending chain of ideals and show its union is finitely generated, forcing stabilization. □

Solution. A module is Noetherian when every ascending chain of submodules stabilizes. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 15.24. State and apply the central principle behind Finite-generation criterion. Give one concrete algebraic consequence.

Hint. To prove ACC from finite generation, take the union ideal of a chain and place its finitely many generators in one stage. □

Solution. A module is Noetherian iff every submodule is finitely generated. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 15.25. State and apply the central principle behind Noetherian rings. Give one concrete algebraic consequence.

Hint. Submodules of finitely generated modules over a Noetherian ring are finitely generated; this is the key closure property to use. ☐

Solution. A ring is Noetherian when it is Noetherian as a module over itself, equivalently every ideal is finitely generated. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 15.26. State and apply the central principle behind Exact sequences. Give one concrete algebraic consequence.

Hint. Quotients of finitely generated modules are generated by images of any finite generating set. ☐

Solution. In a short exact sequence, the middle module is Noetherian iff both ends are Noetherian. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 15.27. State and apply the central principle behind Finite modules. Give one concrete algebraic consequence.

Hint. For polynomial extensions, isolate leading coefficients and use induction/Hilbert basis reasoning rather than assuming finite generation termwise. ☐

Solution. Finite modules over a Noetherian ring are Noetherian. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 15.28. State and apply the central principle behind Quotients. Give one concrete algebraic consequence.

Hint. To extract generators from a chain, adjoin one new element whenever possible; Noetherianity says this process must stop. ☐

Solution. Quotients of Noetherian rings and modules are Noetherian. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 15.29. State and apply the central principle behind Hilbert basis theorem. Give one concrete algebraic consequence.

Hint. Check whether the ring is Noetherian before inferring every submodule of a finite module is finitely generated. ☐

Solution. If A is Noetherian, then $A[x]$ is Noetherian. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 15.30. State and apply the central principle behind Finite-type algebras. Give one concrete algebraic consequence.

Hint. Distinguish 'this ideal is finitely generated' from 'every ideal is finitely generated'; only the latter characterizes a Noetherian ring. ☐

Solution. Iterating Hilbert's theorem shows finitely generated A-algebras are Noetherian. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

15.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 15.31 (Chain check)**.** Does $(8) \subset (4) \subset (2) \subset (1)$ stabilize in $\mathbb{Z}$?

Hint. Use ascending chains, finite generation, and Hilbert basis arguments. □

Solution. Yes, after reaching (1) no larger proper stage exists. □

Exercise 15.32 (Finite ideal)**.** Give generators for (x^2, xy, y^2).

Hint. Use ascending chains, finite generation, and Hilbert basis arguments. □

Solution. The displayed three monomials generate it. □

Exercise 15.33 (Quotient)**.** Is $k[x]/(x^5)$ Noetherian?

Hint. Use ascending chains, finite generation, and Hilbert basis arguments. □

Solution. Yes, it is a quotient of the Noetherian ring $k[x]$. □

Exercise 15.34 (Finite module)**.** Is A^3 Noetherian when A is Noetherian?

Hint. Use ascending chains, finite generation, and Hilbert basis arguments. □

Solution. Yes, finite free modules are Noetherian. □

Exercise 15.35 (Polynomial extension)**.** Is $k[x_1, x_2, x_3]$ Noetherian?

Hint. Use ascending chains, finite generation, and Hilbert basis arguments. □

Solution. Yes, by iterating Hilbert's basis theorem. □

Proofs

Exercise 15.36 (Finite-generation criterion proof)**.** Prove: An A-module is Noetherian iff every submodule is finitely generated.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. If every submodule is finite, the union of an ascending chain is finitely generated and therefore contained in one stage. Conversely, if a submodule were not finitely generated, repeatedly adjoining new elements would produce a strictly increasing chain. □

Exercise 15.37 (Exact-sequence criterion proof)**.** Prove: For $0 \to M' \to M \to M'' \to 0$, M is Noetherian iff M' and M'' are Noetherian.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Submodules of M' and images in M'' control any chain in M. For the converse, submodules and quotients of a Noetherian module are Noetherian. ☐

Exercise 15.38 (Hilbert basis theorem proof)**.** Prove: If A is Noetherian, then $A[x]$ is Noetherian.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. For an ideal $I \subset A[x]$, leading coefficients form an ideal of A and stabilize under finite generation. Choose finitely many polynomials controlling high-degree leading terms and finitely many lower-degree remainders; together they generate I. ☐

Exercise 15.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Finite-generation criterion and Exact-sequence criterion.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. ☐

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. ☐

Counterexamples and hypothesis tests

Exercise 15.40 (Infinite polynomial ring)**.** Test the claim: $k[x_1, x_2, \ldots]$ is Noetherian.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False: $(x_1) \subset (x_1, x_2) \subset \cdots$ never stabilizes. ☐

Exercise 15.41 (Submodule)**.** Test the claim: A submodule of a Noetherian module need not be finitely generated.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False: every submodule is finitely generated. ☐

Exercise 15.42 (Infinite direct sum)**.** Test the claim: An arbitrary direct sum of Noetherian modules is Noetherian.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; an infinite direct sum of copies of a field has a strictly increasing chain of finite partial sums. ☐

Applications and investigations

Exercise 15.43 (Finite algebra)**.** Why are coordinate rings of affine algebraic sets over a field Noetherian?

Hint. Translate the question into ascending chains, finite generation, and Hilbert basis arguments. □

Solution. They are quotients of finitely generated polynomial rings. □

Exercise 15.44 (Induction)**.** What does Noetherian induction replace?

Hint. Translate the question into ascending chains, finite generation, and Hilbert basis arguments. □

Solution. It replaces induction on integers by induction over strict enlargement of subobjects. □

Challenge problems

Exercise 15.45 (Local-global synthesis)**.** Connect 'ACC on submodules' with 'Examples' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. □

Solution. The opening idea is: A module is Noetherian when every ascending chain of submodules stabilizes. The later viewpoint is: Fields, PID-like rings, polynomial rings over fields, and many coordinate rings are Noetherian. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. □

Exercise 15.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. □

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. □

Chapter 16

Support

The support of a module records the primes where the module remains nonzero after localization. It turns module-theoretic information into a subset of the prime spectrum and behaves well under exact sequences.

Learning goals

After this chapter, the reader should be able to:

- compute the support of a module from localizations at prime ideals;
- identify support as the set of primes containing the annihilator for finitely generated modules under the chapter's hypotheses;
- compute supports of quotients, direct sums, and localized modules;
- relate support to vanishing of a module after localization;
- prove basic closedness properties of support for finitely generated modules;
- use support to detect where module-theoretic phenomena are nontrivial;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

16.1 Definition

$\operatorname{Supp} M = \{\mathfrak{p} : M_{\mathfrak{p}} \neq 0\}$.

16.2 Annihilator

The annihilator $\operatorname{Ann}(M)$ consists of scalars killing every element of M.

Example 16.1 (Support of an integer module)**.** For $M = \mathbb{Z}/12\mathbb{Z}$, the localization $M_{(p)}$ is nonzero exactly for $p = 2$ or 3. Thus the support consists of the prime ideals (2) and (3) together with primes containing the annihilator (12) in the spectrum description.

16.3 Finite-module support

For finitely generated M, $\operatorname{Supp} M = V(\operatorname{Ann} M)$.

16.4 Cyclic modules

$\operatorname{Supp}(A/I) = V(I)$.

Example 16.2 (Cyclic module support)**.** For $A = k[x, y]$ and $M = A/(xy)$, one has $\operatorname{Supp} M = V(xy) = V(x) \cup V(y)$. The module detects the union of the two coordinate axes.

16.5 Exact sequences

Support of the middle term in a short exact sequence is the union of supports of the ends.

16.6 Direct sums

Support of a direct sum is the union of the supports.

Example 16.3 (Support in an exact sequence)**.** From $0 \to A/(x) \to A/(x^2) \to A/(x) \to 0$, the support of the middle module equals the union of the supports of the ends, namely $V(x)$.

16.7 Tensor products

Support of a tensor product is contained in the intersection of supports and equals it under common finite-generation hypotheses over local rings.

16.8 Localization

Support localizes by restricting to primes contained in the chosen localization domain.

16.9 Vanishing detection

A module is zero iff all localizations at maximal ideals are zero.

16.10 Geometric role

Support marks the locus where a sheaf-like algebraic object is present.

16.11 Core structural results

Theorem 16.4 (Support of a finite module)**.** *If M is finitely generated, then* $\operatorname{Supp} M = V(\operatorname{Ann} M)$*.*

Proof. If an annihilator element becomes a unit at $\mathfrak{p}$, the localized module vanishes. Conversely, if $M_{\mathfrak{p}} = 0$, finitely many generators are each killed by denominators outside $\mathfrak{p}$; their product lies in $\operatorname{Ann} M$ but outside $\mathfrak{p}$. $\square$

Theorem 16.5 (Support in a short exact sequence)**.** *For* $0 \to M' \to M \to M'' \to 0$*,* $\operatorname{Supp} M = \operatorname{Supp} M' \cup \operatorname{Supp} M''$*.*

Proof. Localize the sequence. Exactness of localization shows the middle localization is zero exactly when both end localizations are zero. $\square$

Theorem 16.6 (Maximal-local detection)**.** *An A-module M is zero iff $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$.*

Proof. If $m \neq 0$, its annihilator is proper and lies in some maximal ideal $\mathfrak{m}$. Then $m/1$ cannot vanish in $M_{\mathfrak{m}}$, otherwise an element outside $\mathfrak{m}$ would annihilate m. □

16.12 Worked examples

Example 16.7 (Definition diagnostic)**.** Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\operatorname{Supp} M = \{\mathfrak{p} : M_{\mathfrak{p}} \neq 0\}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 16.8 (Annihilator diagnostic)**.** Give a rigorous worked analysis of Annihilator. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The annihilator $\operatorname{Ann}(M)$ consists of scalars killing every element of M. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 16.9 (Finite-module support diagnostic)**.** Give a rigorous worked analysis of Finite-module support. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For finitely generated M, $\operatorname{Supp} M = V(\operatorname{Ann} M)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 16.10 (Cyclic modules diagnostic)**.** Give a rigorous worked analysis of Cyclic modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\operatorname{Supp}(A/I) = V(I)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

16.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 16.11 (Definition diagnostic)**.** Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\operatorname{Supp} M = \{\mathfrak{p} : M_{\mathfrak{p}} \neq 0\}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 16.12 (Annihilator diagnostic)**.** Give a rigorous worked analysis of Annihilator. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The annihilator $\operatorname{Ann}(M)$ consists of scalars killing every element of M. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 16.13 (Finite-module support diagnostic)**.** Give a rigorous worked analysis of Finite-module support. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For finitely generated M, $\operatorname{Supp} M = V(\operatorname{Ann} M)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 16.14 (Cyclic modules diagnostic)**.** Give a rigorous worked analysis of Cyclic modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\operatorname{Supp}(A/I) = V(I)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 16.15 (Exact sequences diagnostic)**.** Give a rigorous worked analysis of Exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Support of the middle term in a short exact sequence is the union of supports of the ends. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 16.16 (Direct sums diagnostic)**.** Give a rigorous worked analysis of Direct sums. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Support of a direct sum is the union of the supports. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 16.17 (Tensor products diagnostic)**.** Give a rigorous worked analysis of Tensor products. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Support of a tensor product is contained in the intersection of supports and equals it under common finite-generation hypotheses over local rings. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 16.18 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Support localizes by restricting to primes contained in the chosen localization domain. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 16.19 (Support of a finite module proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If M is finitely generated, then $\operatorname{Supp} M = V(\operatorname{Ann} M)$.

Solution. If an annihilator element becomes a unit at $\mathfrak{p}$, the localized module vanishes. Conversely, if $M_{\mathfrak{p}} = 0$, finitely many generators are each killed by denominators outside $\mathfrak{p}$; their product lies in $\operatorname{Ann} M$ but outside $\mathfrak{p}$. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 16.20 (Support in a short exact sequence proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $0 \to M' \to M \to M'' \to 0$, $\operatorname{Supp} M = \operatorname{Supp} M' \cup \operatorname{Supp} M''$.

Solution. Localize the sequence. Exactness of localization shows the middle localization is zero exactly when both end localizations are zero. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 16.21 (Maximal-local detection proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: An A-module M is zero iff $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$.

Solution. If $m \neq 0$, its annihilator is proper and lies in some maximal ideal $\mathfrak{m}$. Then $m/1$ cannot vanish in $M_{\mathfrak{m}}$, otherwise an element outside $\mathfrak{m}$ would annihilate m. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 16.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. The support of a module records the primes where the module remains nonzero after localization. It turns module-theoretic information into a subset of the prime spectrum and behaves well under exact sequences. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

16.14 Exercises with complete solutions

Exercise 16.23. State and apply the central principle behind Definition. Give one concrete algebraic consequence.

Hint. By definition p lies in Supp(M) exactly when M_p is nonzero; localize before drawing geometric conclusions. □

Solution. $\operatorname{Supp} M = \{\mathfrak{p} : M_{\mathfrak{p}} \neq 0\}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 16.24. State and apply the central principle behind Annihilator. Give one concrete algebraic consequence.

Hint. For finitely generated M, use Supp(M)=V(Ann M); prove both directions with a finite set of generators. □

Solution. The annihilator $\operatorname{Ann}(M)$ consists of scalars killing every element of M. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 16.25. State and apply the central principle behind Finite-module support. Give one concrete algebraic consequence.

Hint. Supp(A/I)=V(I); reduce quotient examples to prime containment. □

Solution. For finitely generated M, $\operatorname{Supp} M = V(\operatorname{Ann} M)$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 16.26. State and apply the central principle behind Cyclic modules. Give one concrete algebraic consequence.

Hint. Support of a direct sum is the union of supports; localize the direct sum and test nonvanishing componentwise. ☐

Solution. $\operatorname{Supp}(A/I) = V(I)$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 16.27. State and apply the central principle behind Exact sequences. Give one concrete algebraic consequence.

Hint. Localization restricts support to primes compatible with the multiplicative set; identify the corresponding prime set explicitly. ☐

Solution. Support of the middle term in a short exact sequence is the union of supports of the ends. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 16.28. State and apply the central principle behind Direct sums. Give one concrete algebraic consequence.

Hint. To prove support is closed for finite M, compute the annihilator and use the Zariski-closed set it defines. ☐

Solution. Support of a direct sum is the union of the supports. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 16.29. State and apply the central principle behind Tensor products. Give one concrete algebraic consequence.

Hint. If $M_p = 0$ for every prime p, use finite generation/local-global reasoning when needed to conclude M=0. ☐

Solution. Support of a tensor product is contained in the intersection of supports and equals it under common finite-generation hypotheses over local rings. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 16.30. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. Do not confuse support with associated primes: associated primes are selected annihilator primes inside the generally larger support. ☐

Solution. Support localizes by restricting to primes contained in the chosen localization domain. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

16.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 16.31 (Cyclic support)**.** Find $\mathrm{Supp}(A/I)$.

Hint. Use annihilators, localization, and support. □

Solution. It is $V(I)$. □

Exercise 16.32 (Free support)**.** Find the support of a nonzero finite free A-module.

Hint. Use annihilators, localization, and support. □

Solution. It is all of $\mathrm{Spec}\, A$. □

Exercise 16.33 (Zero support)**.** Find $\mathrm{Supp}(0)$.

Hint. Use annihilators, localization, and support. □

Solution. It is empty. □

Exercise 16.34 (Annihilator)**.** Find $\mathrm{Ann}_{\mathbb{Z}}(\mathbb{Z}/18)$.

Hint. Use annihilators, localization, and support. □

Solution. It is (18). □

Exercise 16.35 (Direct sum)**.** Find $\mathrm{Supp}(M \oplus N)$.

Hint. Use annihilators, localization, and support. □

Solution. It is $\mathrm{Supp}\, M \cup \mathrm{Supp}\, N$. □

Proofs

Exercise 16.36 (Support of a finite module proof)**.** Prove: If M is finitely generated, then $\mathrm{Supp}\, M = V(\mathrm{Ann}\, M)$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. If an annihilator element becomes a unit at $\mathfrak{p}$, the localized module vanishes. Conversely, if $M_{\mathfrak{p}} = 0$, finitely many generators are each killed by denominators outside $\mathfrak{p}$; their product lies in $\mathrm{Ann}\, M$ but outside $\mathfrak{p}$. □

Exercise 16.37 (Support in a short exact sequence proof)**.** Prove: For $0 \to M' \to M \to M'' \to 0$, $\mathrm{Supp}\, M = \mathrm{Supp}\, M' \cup \mathrm{Supp}\, M''$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. Localize the sequence. Exactness of localization shows the middle localization is zero exactly when both end localizations are zero. □

Exercise 16.38 (Maximal-local detection proof)**.** Prove: An A-module M is zero iff $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. If $m \neq 0$, its annihilator is proper and lies in some maximal ideal $\mathfrak{m}$. Then $m/1$ cannot vanish in $M_{\mathfrak{m}}$, otherwise an element outside $\mathfrak{m}$ would annihilate m. □

Exercise 16.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Support of a finite module and Support in a short exact sequence.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. □

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. □

Counterexamples and hypothesis tests

Exercise 16.40 (Nonzero global)**.** Test the claim: A nonzero module can have empty support.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False: some maximal localization is nonzero. □

Exercise 16.41 (Finite formula)**.** Test the claim: $\operatorname{Supp} M = V(\operatorname{Ann} M)$ for every arbitrary module without hypotheses.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False in this simple form; finite generation is the standard hypothesis. □

Exercise 16.42 (Submodule equality)**.** Test the claim: A submodule always has the same support as its ambient module.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False; its support can be strictly smaller. □

Applications and investigations

Exercise 16.43 (Geometric locus)**.** What geometric set does module support record?

Hint. Translate the question into annihilators, localization, and support. □

Solution. It records the primes or points where the module remains locally nonzero. □

Exercise 16.44 (Local vanishing)**.** How can support test whether a finite module vanishes on an open region?

Hint. Translate the question into annihilators, localization, and support. □

Solution. Localize on that region; vanishing is equivalent to absence of support there. □

Challenge problems

Exercise 16.45 (Local-global synthesis)**.** Connect 'Definition' with 'Geometric role' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. ☐

Solution. The opening idea is: $\operatorname{Supp} M = \{\mathfrak{p} : M_{\mathfrak{p}} \neq 0\}$. The later viewpoint is: Support marks the locus where a sheaf-like algebraic object is present. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. ☐

Exercise 16.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. ☐

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. ☐

Chapter 17

Associated Primes

Associated primes identify prime annihilators of individual module elements. They refine support by detecting the primes that occur as embedded or minimal algebraic components.

Learning goals

After this chapter, the reader should be able to:

- identify associated primes from annihilators of module elements;
- compute associated primes in basic quotient and primary-type examples;
- prove that associated primes lie inside the support;
- use zero divisors on a module to detect unions of associated primes under Noetherian hypotheses;
- analyze behavior of associated primes under localization;
- distinguish minimal primes of support from embedded associated primes in representative examples;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

17.1 Definition

A prime $\mathfrak{p}$ is associated to M if $\mathfrak{p} = \operatorname{Ann}(m)$ for some $m \in M$.

17.2 Cyclic submodules

Associated primes arise exactly from embedded copies of $A/\mathfrak{p}$ inside M.

Example 17.1 (Associated primes of a cyclic integer module)**.** For $\mathbb{Z}/12\mathbb{Z}$, the elements of orders 2 and 3 have annihilators (2) and (3) after choosing appropriate representatives. The associated primes are (2) and (3).

17.3 Existence in Noetherian modules

Every nonzero finitely generated module over a Noetherian ring has an associated prime.

17.4 Containment in support

Every associated prime belongs to the support.

17.5 Minimal support primes

Minimal primes in the support of a finite module are associated.

Example 17.2 (A zero-divisor example)**.** Let $A = k[x, y]/(xy)$. The class of x is annihilated by the class of y, and conversely. The minimal primes (x) and (y) appear naturally as annihilators of suitable elements.

17.6 Exact sequence behavior

Associated primes of ends and middle terms satisfy useful inclusions.

17.7 Localization

Associated primes localize by keeping precisely those contained in the localization prime.

Example 17.3 (Associated versus support)**.** For a finitely generated module over a Noetherian ring, every associated prime lies in the support. In $A/(x^2)$, the prime (x) is associated and the support is the full closed set $V(x)$.

17.8 Zero divisors

For finite modules over Noetherian rings, the zero divisors on M are the union of associated primes.

17.9 Primary decomposition preview

Associated primes index components that later appear in primary decompositions.

17.10 Core structural results

Theorem 17.4 (Associated prime existence)**.** *If $M \neq 0$ is finitely generated over a Noetherian ring, then M has an associated prime.*

Proof. Choose an annihilator maximal among annihilators of nonzero elements. If ab lies in it and b does not, the annihilator of bm contains the chosen annihilator and also a; maximality forces a into the original annihilator, proving primality. $\square$

Theorem 17.5 (Associated implies support)**.** *If $\mathfrak{p} \in \operatorname{Ass} M$, then $\mathfrak{p} \in \operatorname{Supp} M$.*

Proof. An embedding $A/\mathfrak{p} \hookrightarrow M$ localizes to a nonzero embedding at $\mathfrak{p}$, since $(A/\mathfrak{p})_{\mathfrak{p}}$ is nonzero. $\square$

Theorem 17.6 (Minimal support primes are associated)**.** *For a finitely generated module over a Noetherian ring, every prime minimal in $\operatorname{Supp} M$ lies in $\operatorname{Ass} M$.*

Proof. Localize at the minimal prime. The localized support has only the maximal prime, so an associated prime of the nonzero localized module must be that maximal prime; contract back. □

17.11 Worked examples

Example 17.7 (Definition diagnostic)**.** Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A prime $\mathfrak{p}$ is associated to M if $\mathfrak{p} = \mathrm{Ann}(m)$ for some $m \in M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 17.8 (Cyclic submodules diagnostic)**.** Give a rigorous worked analysis of Cyclic submodules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Associated primes arise exactly from embedded copies of $A/\mathfrak{p}$ inside M. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 17.9 (Existence in Noetherian modules diagnostic)**.** Give a rigorous worked analysis of Existence in Noetherian modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every nonzero finitely generated module over a Noetherian ring has an associated prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 17.10 (Containment in support diagnostic)**.** Give a rigorous worked analysis of Containment in support. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every associated prime belongs to the support. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

17.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 17.11 (Definition diagnostic)**.** Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A prime $\mathfrak{p}$ is associated to M if $\mathfrak{p} = \mathrm{Ann}(m)$ for some $m \in M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 17.12 (Cyclic submodules diagnostic)**.** Give a rigorous worked analysis of Cyclic submodules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Associated primes arise exactly from embedded copies of $A/\mathfrak{p}$ inside M. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 17.13 (Existence in Noetherian modules diagnostic)**.** Give a rigorous worked analysis of Existence in Noetherian modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every nonzero finitely generated module over a Noetherian ring has an associated prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 17.14 (Containment in support diagnostic)**.** Give a rigorous worked analysis of Containment in support. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every associated prime belongs to the support. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 17.15 (Minimal support primes diagnostic)**.** Give a rigorous worked analysis of Minimal support primes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Minimal primes in the support of a finite module are associated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 17.16 (Exact sequence behavior diagnostic)**.** Give a rigorous worked analysis of Exact sequence behavior. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Associated primes of ends and middle terms satisfy useful inclusions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 17.17 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Associated primes localize by keeping precisely those contained in the localization prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 17.18 (Zero divisors diagnostic)**.** Give a rigorous worked analysis of Zero divisors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For finite modules over Noetherian rings, the zero divisors on M are the union of associated primes. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 17.19 (Associated prime existence proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $M \neq 0$ is finitely generated over a Noetherian ring, then M has an associated prime.

Solution. Choose an annihilator maximal among annihilators of nonzero elements. If ab lies in it and b does not, the annihilator of bm contains the chosen annihilator and also a; maximality forces a into the original annihilator, proving primality. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 17.20 (Associated implies support proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $\mathfrak{p} \in \operatorname{Ass} M$, then $\mathfrak{p} \in \operatorname{Supp} M$.

Solution. An embedding $A/\mathfrak{p} \hookrightarrow M$ localizes to a nonzero embedding at $\mathfrak{p}$, since $(A/\mathfrak{p})_{\mathfrak{p}}$ is nonzero. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 17.21 (Minimal support primes are associated proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For a finitely generated module over a Noetherian ring, every prime minimal in $\operatorname{Supp} M$ lies in $\operatorname{Ass} M$.

Solution. Localize at the minimal prime. The localized support has only the maximal prime, so an associated prime of the nonzero localized module must be that maximal prime; contract back. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 17.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Associated primes identify prime annihilators of individual module elements. They refine support by detecting the primes that occur as embedded or minimal algebraic components. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

17.13 Exercises with complete solutions

Exercise 17.23. State and apply the central principle behind Definition. Give one concrete algebraic consequence.

Hint. p is associated to M when p=Ann(m) for some element m; search for elements with prime annihilator. □

Solution. A prime $\mathfrak{p}$ is associated to M if $\mathfrak{p} = \operatorname{Ann}(m)$ for some $m \in M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 17.24. State and apply the central principle behind Cyclic submodules. Give one concrete algebraic consequence.

Hint. For A/I, annihilators of residue classes can expose associated primes more directly than primary decomposition. □

Solution. Associated primes arise exactly from embedded copies of $A/\mathfrak{p}$ inside M. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 17.25. State and apply the central principle behind Existence in Noetherian modules. Give one concrete algebraic consequence.

Hint. Every associated prime lies in the support because localizing the witnessing element at its annihilator prime keeps it nonzero. □

Solution. Every nonzero finitely generated module over a Noetherian ring has an associated prime. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 17.26. State and apply the central principle behind Containment in support. Give one concrete algebraic consequence.

Hint. Under Noetherian hypotheses, zero divisors on M are detected by associated primes; identify the element killed by a zero divisor. □

Solution. Every associated prime belongs to the support. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 17.27. State and apply the central principle behind Minimal support primes. Give one concrete algebraic consequence.

Hint. Localize the witnessing equation Ann(m)=p to relate associated primes of M to those of M_p. □

Solution. Minimal primes in the support of a finite module are associated. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 17.28. State and apply the central principle behind Exact sequence behavior. Give one concrete algebraic consequence.

Hint. Minimal primes of Supp(M) are associated in standard Noetherian settings; distinguish them from possible embedded primes. □

Solution. Associated primes of ends and middle terms satisfy useful inclusions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 17.29. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. For direct sums, compare annihilators of pairs and use known associated-prime union formulas where established. □

Solution. Associated primes localize by keeping precisely those contained in the localization prime. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 17.30. State and apply the central principle behind Zero divisors. Give one concrete algebraic consequence.

Hint. Check the Noetherian/finite hypotheses before invoking finiteness of the associated-prime set. □

Solution. For finite modules over Noetherian rings, the zero divisors on M are the union of associated primes. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

17.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 17.31 (Integer module)**.** List the associated primes of $\mathbb{Z}/18\mathbb{Z}$.

Hint. Use element annihilators and associated primes. ☐

Solution. They are (2) and (3). ☐

Exercise 17.32 (Domain)**.** Find $\operatorname{Ass}_A(A)$ when A is a domain.

Hint. Use element annihilators and associated primes. ☐

Solution. It contains the zero prime; in the Noetherian domain case it is exactly (0). ☐

Exercise 17.33 (Cyclic prime)**.** Find an associated prime of A/P for prime P.

Hint. Use element annihilators and associated primes. ☐

Solution. P itself. ☐

Exercise 17.34 (Annihilator element)**.** In $k[x, y]/(xy)$, what annihilates the class of x?

Hint. Use element annihilators and associated primes. ☐

Solution. The ideal generated by the class of y. ☐

Exercise 17.35 (Support containment)**.** Where must an associated prime of M lie?

Hint. Use element annihilators and associated primes. ☐

Solution. In $\operatorname{Supp} M$. ☐

Proofs

Exercise 17.36 (Associated prime existence proof)**.** Prove: If $M \neq 0$ is finitely generated over a Noetherian ring, then M has an associated prime.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Choose an annihilator maximal among annihilators of nonzero elements. If ab lies in it and b does not, the annihilator of bm contains the chosen annihilator and also a; maximality forces a into the original annihilator, proving primality. ☐

Exercise 17.37 (Associated implies support proof)**.** Prove: If $\mathfrak{p} \in \operatorname{Ass} M$, then $\mathfrak{p} \in \operatorname{Supp} M$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. An embedding $A/\mathfrak{p} \hookrightarrow M$ localizes to a nonzero embedding at $\mathfrak{p}$, since $(A/\mathfrak{p})_{\mathfrak{p}}$ is nonzero. ☐

Exercise 17.38 (Minimal support primes are associated proof)**.** Prove: For a finitely generated module over a Noetherian ring, every prime minimal in $\operatorname{Supp} M$ lies in $\operatorname{Ass} M$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Localize at the minimal prime. The localized support has only the maximal prime, so an associated prime of the nonzero localized module must be that maximal prime; contract back. ☐

Exercise 17.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Associated prime existence and Associated implies support.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. ☐

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. ☐

Counterexamples and hypothesis tests

Exercise 17.40 (All support primes)**.** Test the claim: Every prime in the support is associated.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; associated primes are a distinguished finite subset under Noetherian finiteness. ☐

Exercise 17.41 (Empty associated set)**.** Test the claim: A nonzero finite module over a Noetherian ring can have no associated prime.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False. ☐

Exercise 17.42 (Annihilator prime)**.** Test the claim: The annihilator of every nonzero element is prime.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; associated primes arise only when such an annihilator happens to be prime. ☐

Applications and investigations

Exercise 17.43 (Zero divisors)**.** How do associated primes detect zero divisors on a Noetherian module?

Hint. Translate the question into element annihilators and associated primes. ☐

Solution. Zero divisors are captured by the union of associated primes. ☐

Exercise 17.44 (Embedded components)**.** Why are nonminimal associated primes important geometrically?

Hint. Translate the question into element annihilators and associated primes. ☐

Solution. They record embedded algebraic components invisible from the underlying support alone. ☐

Challenge problems

Exercise 17.45 (Local-global synthesis)**.** Connect 'Definition' with 'Primary decomposition preview' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. □

Solution. The opening idea is: A prime $\mathfrak{p}$ is associated to M if $\mathfrak{p} = \mathrm{Ann}(m)$ for some $m \in M$. The later viewpoint is: Associated primes index components that later appear in primary decompositions. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. □

Exercise 17.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. □

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. □

Chapter 18

Completion and I-Adic Topology

I-adic topology and completion encode infinite approximation by powers of an ideal. Completion turns compatible residue classes modulo I^n into actual elements and is fundamental for local algebra and formal geometry.

Learning goals

After this chapter, the reader should be able to:

- construct inverse systems of quotients defining I-adic completion;
- compute elementary completions such as power-series or p-adic-type examples;
- verify separatedness by analyzing intersections of powers of an ideal;
- compare a module with its completion through the canonical map;
- use completeness to solve compatible systems of congruences;
- distinguish completion from localization and from ordinary product constructions;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

18.1 I-adic filtration

The powers $M \supset IM \supset I^2M \supset \cdots$ define progressively finer congruences.

18.2 I-adic topology

Cosets of I^nM form neighborhoods of module elements.

Example 18.1 (Successive p-adic quotients)**.** For $A = \mathbb{Z}$ and $I = (p)$, the quotients $A/I^n = \mathbb{Z}/p^n\mathbb{Z}$ form an inverse system. A compatible sequence of residues is a p-adic integer in the completion.

18.3 Separatedness

The topology is separated when $\bigcap_n I^nM = 0$.

18.4 Completion

$\widehat{M} = \varprojlim_n M/I^n M$ consists of compatible systems of residues.

18.5 Ring completion

$\widehat{A}$ is a ring and $\widehat{M}$ naturally a module over it.

Example 18.2 (Power-series completion)**.** The (x)-adic completion of $k[x]_{(x)}$ is $k[[x]]$. Modulo x^n, a formal power series is determined by its polynomial truncation of degree less than n.

18.6 Universal approximation

A completion element determines all finite-order approximations simultaneously.

18.7 Examples

p-adic integers and formal power-series rings arise as completions.

Example 18.3 (Separatedness test)**.** In a separated I-adic topology, $\bigcap_{n\geq 1} I^n = 0$. For $k[x]$ with $I = (x)$, a polynomial divisible by every power of x must be zero.

18.8 Krull intersection preview

Under Noetherian local hypotheses, intersections of powers often vanish.

18.9 Exactness issues

Completion is not always exact, but for finite modules over Noetherian rings it behaves much better.

18.10 Comparison with tensor

For finitely generated modules in the Noetherian setting, completion often agrees with tensoring by $\widehat{A}$.

18.11 Core structural results

Theorem 18.4 (Inverse-limit description)**.** *The I-adic completion of M is the inverse limit $\varprojlim M/I^n M$.*

Proof. A Cauchy-compatible system is exactly a sequence of residue classes whose class modulo $I^n M$ agrees with every later approximation. This is the defining inverse-limit condition. □

Theorem 18.5 (Completion map kernel)**.** *The kernel of the natural map $M \to \widehat{M}$ is $\bigcap_{n\geq 1} I^n M$.*

Proof. An element maps to zero in the inverse limit exactly when it is zero in every quotient $M/I^n M$, equivalently when it lies in every $I^n M$. □

Theorem 18.6 (Power-series completion)**.** *The (x)-adic completion of $k[x]$ is canonically $k[[x]]$.*

Proof. A compatible sequence modulo x^n uniquely specifies all coefficients of a formal power series, and truncating a power series gives a compatible residue system. $\square$

18.12 Worked examples

Example 18.7 (I-adic filtration diagnostic)**.** Give a rigorous worked analysis of I-adic filtration. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The powers $M \supset IM \supset I^2M \supset \cdots$ define progressively finer congruences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 18.8 (I-adic topology diagnostic)**.** Give a rigorous worked analysis of I-adic topology. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Cosets of I^nM form neighborhoods of module elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 18.9 (Separatedness diagnostic)**.** Give a rigorous worked analysis of Separatedness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The topology is separated when $\bigcap_n I^nM = 0$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 18.10 (Completion diagnostic)**.** Give a rigorous worked analysis of Completion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\widehat{M} = \varprojlim_n M/I^nM$ consists of compatible systems of residues. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

18.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 18.11 (I-adic filtration diagnostic)**.** Give a rigorous worked analysis of I-adic filtration. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The powers $M \supset IM \supset I^2M \supset \cdots$ define progressively finer congruences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 18.12 (I-adic topology diagnostic)**.** Give a rigorous worked analysis of I-adic topology. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Cosets of I^nM form neighborhoods of module elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 18.13 (Separatedness diagnostic)**.** Give a rigorous worked analysis of Separatedness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The topology is separated when $\bigcap_n I^n M = 0$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 18.14 (Completion diagnostic)**.** Give a rigorous worked analysis of Completion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\widehat{M} = \varprojlim_n M/I^n M$ consists of compatible systems of residues. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 18.15 (Ring completion diagnostic)**.** Give a rigorous worked analysis of Ring completion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\widehat{A}$ is a ring and $\widehat{M}$ naturally a module over it. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 18.16 (Universal approximation diagnostic)**.** Give a rigorous worked analysis of Universal approximation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A completion element determines all finite-order approximations simultaneously. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 18.17 (Examples diagnostic)**.** Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. p-adic integers and formal power-series rings arise as completions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 18.18 (Krull intersection preview diagnostic)**.** Give a rigorous worked analysis of Krull intersection preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Under Noetherian local hypotheses, intersections of powers often vanish. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 18.19 (Inverse-limit description proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: The I-adic completion of M is the inverse limit $\varprojlim M/I^n M$.

Solution. A Cauchy-compatible system is exactly a sequence of residue classes whose class modulo $I^n M$ agrees with every later approximation. This is the defining inverse-limit condition. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 18.20 (Completion map kernel proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: The kernel of the natural map $M \to \widehat{M}$ is $\bigcap_{n\geq 1} I^n M$.

Solution. An element maps to zero in the inverse limit exactly when it is zero in every quotient $M/I^n M$, equivalently when it lies in every $I^n M$. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 18.21 (Power-series completion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: The (x)-adic completion of $k[x]$ is canonically $k[[x]]$.

Solution. A compatible sequence modulo x^n uniquely specifies all coefficients of a formal power series, and truncating a power series gives a compatible residue system. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 18.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. I-adic topology and completion encode infinite approximation by powers of an ideal. Completion turns compatible residue classes modulo I^n into actual elements and is fundamental for local algebra and formal geometry. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

18.14 Exercises with complete solutions

Exercise 18.23. State and apply the central principle behind I-adic filtration. Give one concrete algebraic consequence.

Hint. Write the completion as $\varprojlim A/I^n$ or $\varprojlim M/I^n M$ and keep the transition maps explicit. □

Solution. The powers $M \supset IM \supset I^2M \supset \cdots$ define progressively finer congruences. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 18.24. State and apply the central principle behind I-adic topology. Give one concrete algebraic consequence.

Hint. A compatible sequence of residue classes must agree modulo every lower power; verify this before declaring it a point of the inverse limit. □

Solution. Cosets of $I^n M$ form neighborhoods of module elements. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 18.25. State and apply the central principle behind Separatedness. Give one concrete algebraic consequence.

Hint. For k[x] completed at (x), identify coefficients successively modulo x^n to recover a formal power series. □

Solution. The topology is separated when $\bigcap_n I^n M = 0$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 18.26. State and apply the central principle behind Completion. Give one concrete algebraic consequence.

Hint. Separatedness asks whether the intersection of all $I^n M$ is zero; test an element lying in every power. ☐

Solution. $\widehat{M} = \varprojlim_n M/I^n M$ consists of compatible systems of residues. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 18.27. State and apply the central principle behind Ring completion. Give one concrete algebraic consequence.

Hint. The canonical map sends m to its compatible residue classes; its kernel is exactly the I-adic intersection. ☐

Solution. $\widehat{A}$ is a ring and $\widehat{M}$ naturally a module over it. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 18.28. State and apply the central principle behind Universal approximation. Give one concrete algebraic consequence.

Hint. Completeness means every compatible congruence system comes from an element; formulate the reconstruction one stage at a time. ☐

Solution. A completion element determines all finite-order approximations simultaneously. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 18.29. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. Do not confuse product of quotients with inverse limit: compatibility conditions cut out a proper subset of the product. ☐

Solution. p-adic integers and formal power-series rings arise as completions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 18.30. State and apply the central principle behind Krull intersection preview. Give one concrete algebraic consequence.

Hint. Compare localization and completion carefully; one inverts elements while the other records increasingly fine congruence data. ☐

Solution. Under Noetherian local hypotheses, intersections of powers often vanish. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

18.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 18.31 (Adic quotient)**.** Compute $\mathbb{Z}/(5^3)$.

Hint. Use inverse limits, powers of ideals, and adic topology. ☐

Solution. It is $\mathbb{Z}/125\mathbb{Z}$. ☐

Exercise 18.32 (Polynomial truncation)**.** Describe $k[x]/(x^4)$.

Hint. Use inverse limits, powers of ideals, and adic topology. ☐

Solution. Every class has a representative $a_0 + a_1x + a_2x^2 + a_3x^3$. ☐

Exercise 18.33 (Completion model)**.** What is the (x)-adic completion of $k[x]_{(x)}$?

Hint. Use inverse limits, powers of ideals, and adic topology. ☐

Solution. $k[[x]]$. ☐

Exercise 18.34 (Inverse map)**.** What is the transition map $A/I^{n+1} \to A/I^n$?

Hint. Use inverse limits, powers of ideals, and adic topology. ☐

Solution. Natural reduction modulo I^n. ☐

Exercise 18.35 (Intersection)**.** Compute $\bigcap_n (x^n)$ in $k[x]$.

Hint. Use inverse limits, powers of ideals, and adic topology. ☐

Solution. It is zero. ☐

Proofs

Exercise 18.36 (Inverse-limit description proof)**.** Prove: The I-adic completion of M is the inverse limit $\varprojlim M/I^nM$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. A Cauchy-compatible system is exactly a sequence of residue classes whose class modulo I^nM agrees with every later approximation. This is the defining inverse-limit condition. ☐

Exercise 18.37 (Completion map kernel proof)**.** Prove: The kernel of the natural map $M \to \widehat{M}$ is $\bigcap_{n\geq 1} I^nM$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. An element maps to zero in the inverse limit exactly when it is zero in every quotient M/I^nM, equivalently when it lies in every I^nM. ☐

Exercise 18.38 (Power-series completion proof)**.** Prove: The (x)-adic completion of $k[x]$ is canonically $k[[x]]$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. A compatible sequence modulo x^n uniquely specifies all coefficients of a formal power series, and truncating a power series gives a compatible residue system. □

Exercise 18.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Inverse-limit description and Completion map kernel.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. □

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. □

Counterexamples and hypothesis tests

Exercise 18.40 (Direct limit)**.** Test the claim: Adic completion is a direct limit of A/I^n.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False: it is an inverse limit. □

Exercise 18.41 (Polynomial equality)**.** Test the claim: $k[x]$ is already complete in the (x)-adic topology.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False; the completion adds infinite formal power series. □

Exercise 18.42 (Separated automatic)**.** Test the claim: Every I-adic topology is automatically separated without hypotheses.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False; separatedness requires the intersection of powers to vanish. □

Applications and investigations

Exercise 18.43 (Approximation)**.** How does completion encode increasingly accurate congruence data?

Hint. Translate the question into inverse limits, powers of ideals, and adic topology. □

Solution. An element is a compatible choice modulo every power I^n. □

Exercise 18.44 (Local algebra)**.** Why is completion useful near a maximal ideal?

Hint. Translate the question into inverse limits, powers of ideals, and adic topology. □

Solution. It replaces a local ring by a limit in which formal expansions and infinitesimal neighborhoods are easier to analyze. □

Challenge problems

Exercise 18.45 (Local-global synthesis)**.** Connect 'I-adic filtration' with 'Comparison with tensor' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. ☐

Solution. The opening idea is: The powers $M \supset IM \supset I^2M \supset \cdots$ define progressively finer congruences. The later viewpoint is: For finitely generated modules in the Noetherian setting, completion often agrees with tensoring by $\widehat{A}$. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. ☐

Exercise 18.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. ☐

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. ☐

Part V

Integral and Valuation Theory

Chapter 19

Integral Dependence

Integral dependence generalizes algebraicity from field extensions to ring extensions. An element is integral when it satisfies a monic polynomial, and integrality is stable under finite algebraic constructions.

Learning goals

After this chapter, the reader should be able to:

- test whether an element is integral over a subring by finding a monic polynomial relation;
- prove closure of integral elements under addition and multiplication using finite-module criteria;
- apply the determinant trick or finite-generation criterion for integral dependence;
- analyze integral extensions under quotients and localization;
- show how integrality constrains prime ideals through lying-over-type statements developed in the chapter;
- distinguish integral dependence from algebraic dependence or arbitrary polynomial relations;

Conceptual roadmap

$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$

19.1 Integral elements

An element b of an A-algebra B is integral over A if it satisfies a monic polynomial with coefficients in A.

19.2 Integral extensions

An extension $A \subset B$ is integral when every element of B is integral over A.

Example 19.1 (Integral element over a subring)**.** Let $A = k[t^2, t^3] \subset k[t]$. The element t is integral over A because it satisfies the monic equation $X^2 - t^2 = 0$ with coefficient $t^2 \in A$.

19.3 Finite module criterion

An element is integral iff the subalgebra it generates is finite as an A-module.

19.4 Finite algebras

Every finite A-algebra is integral over A.

19.5 Transitivity

Integrality is transitive through towers of ring extensions.

Example 19.2 (Gaussian integers)**.** The element i is integral over $\mathbb{Z}$ because it satisfies $X^2 + 1 = 0$. Consequently every element of $\mathbb{Z}[i]$ is integral over $\mathbb{Z}$.

19.6 Integral elements form a ring

The set of elements of B integral over A is a subring.

19.7 Determinant trick

Finite-generation plus stability under multiplication produces monic equations by a determinant argument.

Example 19.3 (Finite-module criterion)**.** If b is integral over A, then $A[b]$ is generated as an A-module by $1, b, \dots, b^{n-1}$ for a monic equation of degree n. Integrality converts an algebra extension into finite module data.

19.8 Prime behavior preview

Integral extensions satisfy lying-over and contraction properties for prime ideals.

19.9 Examples

Algebraic integers, roots of monic equations, and normalization rings illustrate integral dependence.

19.10 Core structural results

Theorem 19.4 (Finite module criterion)**.** *For $b \in B$, b is integral over A iff $A[b]$ is finitely generated as an A-module.*

Proof. A monic equation expresses every high power of b in terms of lower powers. Conversely, if finitely many powers generate an A-module stable under multiplication by b, the determinant trick gives a monic polynomial annihilating b. $\square$

Theorem 19.5 (Finite algebra implies integral)**.** *If B is a finite A-module and an A-algebra, then every $b \in B$ is integral over A.*

Proof. Multiplication by b is an A-linear endomorphism of the finite module B. Apply the determinant trick or Cayley–Hamilton to obtain a monic polynomial with coefficients in A annihilating b. □

Theorem 19.6 (Transitivity of integrality)**.** *If B is integral over A and C integral over B, then C is integral over A.*

Proof. For $c \in C$, its monic equation uses finitely many coefficients from B. Those coefficients generate a finite A-algebra because they are integral; adjoining c is finite over that finite algebra, hence finite over A, so c is integral over A. □

19.11 Worked examples

Example 19.7 (Integral elements diagnostic)**.** Give a rigorous worked analysis of Integral elements. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An element b of an A-algebra B is integral over A if it satisfies a monic polynomial with coefficients in A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 19.8 (Integral extensions diagnostic)**.** Give a rigorous worked analysis of Integral extensions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An extension $A \subset B$ is integral when every element of B is integral over A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 19.9 (Finite module criterion diagnostic)**.** Give a rigorous worked analysis of Finite module criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An element is integral iff the subalgebra it generates is finite as an A-module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 19.10 (Finite algebras diagnostic)**.** Give a rigorous worked analysis of Finite algebras. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every finite A-algebra is integral over A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

19.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 19.11 (Integral elements diagnostic)**.** Give a rigorous worked analysis of Integral elements. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element b of an A-algebra B is integral over A if it satisfies a monic polynomial with coefficients in A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 19.12 (Integral extensions diagnostic)**.** Give a rigorous worked analysis of Integral extensions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An extension $A \subset B$ is integral when every element of B is integral over A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 19.13 (Finite module criterion diagnostic)**.** Give a rigorous worked analysis of Finite module criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element is integral iff the subalgebra it generates is finite as an A-module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 19.14 (Finite algebras diagnostic)**.** Give a rigorous worked analysis of Finite algebras. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every finite A-algebra is integral over A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 19.15 (Transitivity diagnostic)**.** Give a rigorous worked analysis of Transitivity. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Integrality is transitive through towers of ring extensions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 19.16 (Integral elements form a ring diagnostic)**.** Give a rigorous worked analysis of Integral elements form a ring. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The set of elements of B integral over A is a subring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 19.17 (Determinant trick diagnostic)**.** Give a rigorous worked analysis of Determinant trick. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite-generation plus stability under multiplication produces monic equations by a determinant argument. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 19.18 (Prime behavior preview diagnostic)**.** Give a rigorous worked analysis of Prime behavior preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Integral extensions satisfy lying-over and contraction properties for prime ideals. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 19.19 (Finite module criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $b \in B$, b is integral over A iff $A[b]$ is finitely generated as an A-module.

Solution. A monic equation expresses every high power of b in terms of lower powers. Conversely, if finitely many powers generate an A-module stable under multiplication by b, the determinant trick gives a monic polynomial annihilating b. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 19.20 (Finite algebra implies integral proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If B is a finite A-module and an A-algebra, then every $b \in B$ is integral over A.

Solution. Multiplication by b is an A-linear endomorphism of the finite module B. Apply the determinant trick or Cayley–Hamilton to obtain a monic polynomial with coefficients in A annihilating b. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 19.21 (Transitivity of integrality proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If B is integral over A and C integral over B, then C is integral over A.

Solution. For $c \in C$, its monic equation uses finitely many coefficients from B. Those coefficients generate a finite A-algebra because they are integral; adjoining c is finite over that finite algebra, hence finite over A, so c is integral over A. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 19.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Integral dependence generalizes algebraicity from field extensions to ring extensions. An element is integral when it satisfies a monic polynomial, and integrality is stable under finite algebraic constructions. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

19.13 Exercises with complete solutions

Exercise 19.23. State and apply the central principle behind Integral elements. Give one concrete algebraic consequence.

Hint. Integral means satisfying a monic polynomial over A; monicity is essential, so do not use an arbitrary algebraic relation. □

Solution. An element b of an A-algebra B is integral over A if it satisfies a monic polynomial with coefficients in A. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 19.24. State and apply the central principle behind Integral extensions. Give one concrete algebraic consequence.

Hint. To prove closure under addition/multiplication, place the elements inside a common finite A-module algebra and use the finite-module criterion. □

Solution. An extension $A \subset B$ is integral when every element of B is integral over A. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 19.25. State and apply the central principle behind Finite module criterion. Give one concrete algebraic consequence.

Hint. The determinant trick starts from a finite module stable under multiplication by x and produces a monic characteristic relation. □

Solution. An element is integral iff the subalgebra it generates is finite as an A-module. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 19.26. State and apply the central principle behind Finite algebras. Give one concrete algebraic consequence.

Hint. Integrality descends to quotients by reducing the monic polynomial modulo the ideal. □

Solution. Every finite A-algebra is integral over A. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 19.27. State and apply the central principle behind Transitivity. Give one concrete algebraic consequence.

Hint. Integrality localizes because the same monic relation survives after localizing coefficients. □

Solution. Integrality is transitive through towers of ring extensions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 19.28. State and apply the central principle behind Integral elements form a ring. Give one concrete algebraic consequence.

Hint. For prime behavior, extend/contraction arguments use integrality together with the lying-over mechanism; identify the exact theorem before applying it. □

Solution. The set of elements of B integral over A is a subring. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 19.29. State and apply the central principle behind Determinant trick. Give one concrete algebraic consequence.

Hint. Every element of a finite A-algebra is integral over A; use Cayley–Hamilton on multiplication by that element. □

Solution. Finite-generation plus stability under multiplication produces monic equations by a determinant argument. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 19.30. State and apply the central principle behind Prime behavior preview. Give one concrete algebraic consequence.

Hint. Separate 'algebraic over the fraction field' from 'integral over A': denominators in a polynomial relation matter. □

Solution. Integral extensions satisfy lying-over and contraction properties for prime ideals. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

19.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 19.31 (Square root)**.** Is $\sqrt{2}$ integral over $\mathbb{Z}$?

Hint. Use monic equations and finite module criteria. ☐

Solution. Yes, it satisfies $X^2 - 2 = 0$. ☐

Exercise 19.32 (Rational)**.** Is $1/2$ integral over $\mathbb{Z}$?

Hint. Use monic equations and finite module criteria. ☐

Solution. No. ☐

Exercise 19.33 (Cusp parameter)**.** Is t integral over $k[t^2, t^3]$?

Hint. Use monic equations and finite module criteria. ☐

Solution. Yes. ☐

Exercise 19.34 (Finite algebra)**.** If $b^3 + a_2 b^2 + a_1 b + a_0 = 0$, which powers generate $A[b]$?

Hint. Use monic equations and finite module criteria. ☐

Solution. $1, b, b^2$. ☐

Exercise 19.35 (Transitivity)**.** If C is integral over B and B over A, what follows?

Hint. Use monic equations and finite module criteria. ☐

Solution. C is integral over A. ☐

Proofs

Exercise 19.36 (Finite module criterion proof)**.** Prove: For $b \in B$, b is integral over A iff $A[b]$ is finitely generated as an A-module.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. A monic equation expresses every high power of b in terms of lower powers. Conversely, if finitely many powers generate an A-module stable under multiplication by b, the determinant trick gives a monic polynomial annihilating b. ☐

Exercise 19.37 (Finite algebra implies integral proof)**.** Prove: If B is a finite A-module and an A-algebra, then every $b \in B$ is integral over A.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Multiplication by b is an A-linear endomorphism of the finite module B. Apply the determinant trick or Cayley–Hamilton to obtain a monic polynomial with coefficients in A annihilating b. ☐

Exercise 19.38 (Transitivity of integrality proof)**.** Prove: If B is integral over A and C integral over B, then C is integral over A.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. For $c \in C$, its monic equation uses finitely many coefficients from B. Those coefficients generate a finite A-algebra because they are integral; adjoining c is finite over that finite algebra, hence finite over A, so c is integral over A. ☐

Exercise 19.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Finite module criterion and Finite algebra implies integral.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. ☐

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. ☐

Counterexamples and hypothesis tests

Exercise 19.40 (Arbitrary fraction)**.** Test the claim: Every element of the fraction field of an integral domain is integral over the domain.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False: $1/2$ over $\mathbb{Z}$. ☐

Exercise 19.41 (Monic condition)**.** Test the claim: A root of any polynomial with coefficients in A is automatically integral.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; the polynomial must be monic or an equivalent criterion must apply. ☐

Exercise 19.42 (Sum product)**.** Test the claim: The sum and product of integral elements need not be integral.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False: integral elements form a subring. ☐

Applications and investigations

Exercise 19.43 (Finite maps)**.** Why does integrality correspond to finite algebra behavior?

Hint. Translate the question into monic equations and finite module criteria. ☐

Solution. An integral finite-type algebra is finite as a module. ☐

Exercise 19.44 (Eliminating parameters)**.** Why is $k[t]$ integral over the cusp ring $k[t^2, t^3]$?

Hint. Translate the question into monic equations and finite module criteria. ☐

Solution. The missing parameter satisfies a monic relation over the cusp ring. ☐

Challenge problems

Exercise 19.45 (Local-global synthesis)**.** Connect 'Integral elements' with 'Examples' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. □

Solution. The opening idea is: An element b of an A-algebra B is integral over A if it satisfies a monic polynomial with coefficients in A. The later viewpoint is: Algebraic integers, roots of monic equations, and normalization rings illustrate integral dependence. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. □

Exercise 19.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. □

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. □

Chapter 20

Integral Closure and Normalization

Integral closure collects all elements of a fraction field integral over the base ring. Normalization repairs non-normal domains by adjoining these integral elements.

Learning goals

After this chapter, the reader should be able to:

- compute integral closures in elementary domains and fraction fields;
- verify that a domain is integrally closed using monic-equation criteria;
- construct normalization maps from a domain to its integral closure;
- analyze normalization under localization in standard finite-type examples;
- identify singular algebraic presentations that become simpler after normalization;
- distinguish normality of a domain from reducedness, factoriality, or smoothness;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

20.1 Integrally closed domains

A domain is integrally closed when every element of its fraction field integral over the ring already lies in the ring.

20.2 Integral closure

The integral closure of A in an extension field consists of all elements integral over A.

Example 20.1 (Normalization of a cusp)**.** The cusp ring $A = k[t^2, t^3]$ is not integrally closed in $k(t)$ because t is integral over A but $t \notin A$. Its normalization is $k[t]$.

20.3 Normalization

For a domain, normalization is its integral closure in its fraction field.

20.4 UFDs and normality

Unique factorization domains are integrally closed.

20.5 Localization

Integral closure behaves well under localization.

Example 20.2 (Integers in the rationals)**.** The integral closure of $\mathbb{Z}$ in $\mathbb{Q}$ is $\mathbb{Z}$. A rational number integral over $\mathbb{Z}$ must have denominator 1 by the rational-root argument for a monic integer polynomial.

20.6 Finite normalization

For many finitely generated algebras over fields, normalization is finite, though this requires hypotheses.

20.7 Singular examples

Cusp-like rings such as $k[t^2, t^3]$ normalize to $k[t]$.

Example 20.3 (Normal domain check)**.** A unique factorization domain is integrally closed. Thus $k[x, y]$ is normal, while the singular cusp subring $k[t^2, t^3]$ provides a standard nonnormal contrast.

20.8 Prime spectra

Normalization induces an integral map on spectra and separates certain singular algebraic behaviors.

20.9 Conductor

The conductor measures the largest ideal shared by a ring and its normalization.

20.10 Core structural results

Theorem 20.4 (UFDs are integrally closed)**.** *Every unique factorization domain is integrally closed in its fraction field.*

Proof. Write an integral element as a/b in lowest terms. A monic equation implies a^n is divisible by b; coprimality and unique factorization force b to be a unit. ☐

Theorem 20.5 (Localization of integral closure)**.** *If B is the integral closure of a domain A in a field extension and $S \subset A$ is multiplicative, then $S^{-1}B$ is the integral closure of $S^{-1}A$ in the same localized field.*

Proof. Localization preserves monic equations, giving one inclusion. For the reverse, clear finitely many denominators in a monic equation to show a suitable multiple of the element is integral over A, hence lies in B. ☐

Theorem 20.6 (Normalization of a cusp ring)**.** *The normalization of $k[t^2, t^3]$ in $k(t)$ is $k[t]$.*

Proof. The element t is integral because it satisfies $X^2 - t^2 = 0$ with coefficient t^2 in the subring. Since $k[t]$ is a PID and therefore integrally closed, no larger integral subring inside $k(t)$ is needed. □

20.11 Worked examples

Example 20.7 (Integrally closed domains diagnostic)**.** Give a rigorous worked analysis of Integrally closed domains. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A domain is integrally closed when every element of its fraction field integral over the ring already lies in the ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 20.8 (Integral closure diagnostic)**.** Give a rigorous worked analysis of Integral closure. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The integral closure of A in an extension field consists of all elements integral over A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 20.9 (Normalization diagnostic)**.** Give a rigorous worked analysis of Normalization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For a domain, normalization is its integral closure in its fraction field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 20.10 (UFDs and normality diagnostic)**.** Give a rigorous worked analysis of UFDs and normality. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Unique factorization domains are integrally closed. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

20.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 20.11 (Integrally closed domains diagnostic)**.** Give a rigorous worked analysis of Integrally closed domains. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A domain is integrally closed when every element of its fraction field integral over the ring already lies in the ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 20.12 (Integral closure diagnostic)**.** Give a rigorous worked analysis of Integral closure. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The integral closure of A in an extension field consists of all elements integral over A. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 20.13 (Normalization diagnostic)**.** Give a rigorous worked analysis of Normalization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For a domain, normalization is its integral closure in its fraction field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 20.14 (UFDs and normality diagnostic)**.** Give a rigorous worked analysis of UFDs and normality. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Unique factorization domains are integrally closed. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 20.15 (Localization diagnostic)**.** Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Integral closure behaves well under localization. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 20.16 (Finite normalization diagnostic)**.** Give a rigorous worked analysis of Finite normalization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For many finitely generated algebras over fields, normalization is finite, though this requires hypotheses. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 20.17 (Singular examples diagnostic)**.** Give a rigorous worked analysis of Singular examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Cusp-like rings such as $k[t^2, t^3]$ normalize to $k[t]$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 20.18 (Prime spectra diagnostic)**.** Give a rigorous worked analysis of Prime spectra. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Normalization induces an integral map on spectra and separates certain singular algebraic behaviors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 20.19 (UFDs are integrally closed proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Every unique factorization domain is integrally closed in its fraction field.

Solution. Write an integral element as a/b in lowest terms. A monic equation implies a^n is divisible by b; coprimality and unique factorization force b to be a unit. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 20.20 (Localization of integral closure proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If B is the integral closure of a domain A in a field extension and $S \subset A$ is multiplicative, then $S^{-1}B$ is the integral closure of $S^{-1}A$ in the same localized field.

Solution. Localization preserves monic equations, giving one inclusion. For the reverse, clear finitely many denominators in a monic equation to show a suitable multiple of the element is integral over A, hence lies in B. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 20.21 (Normalization of a cusp ring proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: The normalization of $k[t^2, t^3]$ in $k(t)$ is $k[t]$.

Solution. The element t is integral because it satisfies $X^2 - t^2 = 0$ with coefficient t^2 in the subring. Since $k[t]$ is a PID and therefore integrally closed, no larger integral subring inside $k(t)$ is needed. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 20.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Integral closure collects all elements of a fraction field integral over the base ring. Normalization repairs non-normal domains by adjoining these integral elements. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

20.13 Exercises with complete solutions

Exercise 20.23. State and apply the central principle behind Integrally closed domains. Give one concrete algebraic consequence.

Hint. To compute integral closure, start with fractions x and test whether a monic equation over A can hold. □

Solution. A domain is integrally closed when every element of its fraction field integral over the ring already lies in the ring. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 20.24. State and apply the central principle behind Integral closure. Give one concrete algebraic consequence.

Hint. An integrally closed domain contains every fraction integral over it; phrase the test inside the fraction field. □

Solution. The integral closure of A in an extension field consists of all elements integral over A. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 20.25. State and apply the central principle behind Normalization. Give one concrete algebraic consequence.

Hint. Normalization is the inclusion A -> Abar in the fraction field or total ring used by the chapter; identify that ambient field first. □

Solution. For a domain, normalization is its integral closure in its fraction field. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 20.26. State and apply the central principle behind UFDs and normality. Give one concrete algebraic consequence.

Hint. Localization of an integrally closed domain remains integrally closed; clear denominators in a monic relation after localization. ☐

Solution. Unique factorization domains are integrally closed. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 20.27. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. For a singular subring such as $k[t^2, t^3]$, test whether t is integral and adjoin it to reach the normalization. ☐

Solution. Integral closure behaves well under localization. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 20.28. State and apply the central principle behind Finite normalization. Give one concrete algebraic consequence.

Hint. Normal does not mean UFD automatically; keep integrally closed and factorial conditions distinct. ☐

Solution. For many finitely generated algebras over fields, normalization is finite, though this requires hypotheses. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 20.29. State and apply the central principle behind Singular examples. Give one concrete algebraic consequence.

Hint. Reducedness only excludes nilpotents; it does not imply integral closure in the fraction field. ☐

Solution. Cusp-like rings such as $k[t^2, t^3]$ normalize to $k[t]$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 20.30. State and apply the central principle behind Prime spectra. Give one concrete algebraic consequence.

Hint. Check finite-generation hypotheses before asserting normalization is finite as an A-module. ☐

Solution. Normalization induces an integral map on spectra and separates certain singular algebraic behaviors. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

20.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 20.31 (Integer closure)**.** Find the integral closure of $\mathbb{Z}$ in $\mathbb{Q}$.

Hint. Use integral closure and normalization. □

Solution. It is $\mathbb{Z}$. □

Exercise 20.32 (Cusp closure)**.** Find the normalization of $k[t^2, t^3]$ in $k(t)$.

Hint. Use integral closure and normalization. □

Solution. It is $k[t]$. □

Exercise 20.33 (Polynomial ring)**.** Is $k[x]$ integrally closed?

Hint. Use integral closure and normalization. □

Solution. Yes. □

Exercise 20.34 (UFD)**.** Is every UFD integrally closed?

Hint. Use integral closure and normalization. □

Solution. Yes. □

Exercise 20.35 (Witness)**.** Give an element witnessing nonnormality of $k[t^2, t^3]$.

Hint. Use integral closure and normalization. □

Solution. The element t. □

Proofs

Exercise 20.36 (UFDs are integrally closed proof)**.** Prove: Every unique factorization domain is integrally closed in its fraction field.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. Write an integral element as a/b in lowest terms. A monic equation implies a^n is divisible by b; coprimality and unique factorization force b to be a unit. □

Exercise 20.37 (Localization of integral closure proof)**.** Prove: If B is the integral closure of a domain A in a field extension and $S \subset A$ is multiplicative, then $S^{-1}B$ is the integral closure of $S^{-1}A$ in the same localized field.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. Localization preserves monic equations, giving one inclusion. For the reverse, clear finitely many denominators in a monic equation to show a suitable multiple of the element is integral over A, hence lies in B. □

Exercise 20.38 (Normalization of a cusp ring proof)**.** Prove: The normalization of $k[t^2, t^3]$ in $k(t)$ is $k[t]$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. The element t is integral because it satisfies $X^2 - t^2 = 0$ with coefficient t^2 in the subring. Since $k[t]$ is a PID and therefore integrally closed, no larger integral subring inside $k(t)$ is needed. ☐

Exercise 20.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both UFDs are integrally closed and Localization of integral closure.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. ☐

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. ☐

Counterexamples and hypothesis tests

Exercise 20.40 (Domain normality)**.** Test the claim: Every integral domain is integrally closed.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False. ☐

Exercise 20.41 (Integral closure field)**.** Test the claim: The integral closure of a domain in its fraction field is always the whole fraction field.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False. ☐

Exercise 20.42 (Normalization trivial)**.** Test the claim: Normalization never changes a ring.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; it changes the cusp ring to $k[t]$. ☐

Applications and investigations

Exercise 20.43 (Singularity repair)**.** What does normalization do to the cusp parametrization?

Hint. Translate the question into integral closure and normalization. ☐

Solution. It replaces the nonnormal cusp coordinate ring by the regular parameter ring $k[t]$. ☐

Exercise 20.44 (Arithmetic closure)**.** Why study integral closure in number fields?

Hint. Translate the question into integral closure and normalization. ☐

Solution. It identifies the natural ring of algebraic integers inside the field. ☐

Challenge problems

Exercise 20.45 (Local-global synthesis)**.** Connect 'Integrally closed domains' with 'Conductor' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. ☐

Solution. The opening idea is: A domain is integrally closed when every element of its fraction field integral over the ring already lies in the ring. The later viewpoint is: The conductor measures the largest ideal shared by a ring and its normalization. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. ☐

Exercise 20.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. ☐

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. ☐

Chapter 21

Valuation Rings

Valuation rings impose a total divisibility order on a field. They are integrally closed local domains and provide a powerful way to test integrality and dominate local rings.

Learning goals

After this chapter, the reader should be able to:

- verify the valuation-ring criterion that for each fraction either it or its inverse lies in the ring;
- construct valuation rings from valuations on fraction fields;
- identify prime ideals and units in elementary valuation rings;
- compare divisibility of principal ideals through the valuation ordering;
- use valuation rings to test integral closure in fraction fields where the chapter establishes the criterion;
- distinguish discrete valuation rings from general valuation rings by value-group behavior;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

21.1 Valuations

A valuation assigns ordered-group values compatible with products and satisfying a triangle inequality for sums.

21.2 Valuation rings

A subring V of a field K is a valuation ring when for every nonzero $x \in K$, either x or x^{-1} lies in V.

Example 21.1 (The p-adic valuation)**.** For a prime p, write a nonzero rational number as $p^n u/v$ with $p \nmid uv$. Then v_p equals n. The valuation ring consists of rationals with nonnegative p-adic valuation.

21.3 Locality

The nonunits of a valuation ring form its unique maximal ideal.

21.4 Divisibility order

Principal ideals are totally ordered by inclusion in a valuation ring.

21.5 Discrete valuation rings

A DVR corresponds to a discrete rank-one valuation and has a principal maximal ideal.

Example 21.2 (The t-adic valuation)**.** On $k(t)$, the order of vanishing at $t = 0$ defines a valuation. The local ring $k[t]_{(t)}$ is its valuation ring: a rational function belongs exactly when it has no pole at 0.

21.6 Integrally closed property

Every valuation ring is integrally closed in its fraction field.

21.7 Dominating local rings

Valuation rings can contain a local domain while having maximal ideal contracting to the local maximal ideal.

Example 21.3 (Either x or x^{-1})**.** A subring V of a field K is a valuation ring exactly when for every nonzero $x \in K$, either $x \in V$ or $x^{-1} \in V$. This total comparability forces ideals of V to be linearly ordered.

21.8 Prime ideals

Prime ideals of a valuation ring are linearly ordered.

21.9 Integral closure criterion

Integral closure of a domain can be recovered as an intersection of valuation rings containing it.

21.10 Core structural results

Theorem 21.4 (Valuation rings are local)**.** *If V is a valuation ring of a field, its nonunits form an ideal and hence the unique maximal ideal.*

Proof. For nonunits a, b, compare a/b or b/a when both are nonzero; one divides the other, so their sum is a multiple of one nonunit. Multiplication by arbitrary ring elements preserves nonunits. □

Theorem 21.5 (Principal ideals are totally ordered)**.** *For nonzero $a, b \in V$, either $(a) \subseteq (b)$ or $(b) \subseteq (a)$.*

Proof. Apply the valuation-ring condition to a/b: if $a/b \in V$, then $a \in (b)$; otherwise $b/a \in V$, so $b \in (a)$. □

Theorem 21.6 (Valuation rings are integrally closed)**.** *A valuation ring V is integrally closed in its fraction field.*

Proof. If x is integral over V but $x \notin V$, then $x^{-1} \in V$ and is a nonunit. Divide a monic equation for x by the highest power of x to express 1 as an element of the maximal ideal generated by powers of x^{-1}, contradiction. □

21.11 Worked examples

Example 21.7 (Valuations diagnostic)**.** Give a rigorous worked analysis of Valuations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A valuation assigns ordered-group values compatible with products and satisfying a triangle inequality for sums. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 21.8 (Valuation rings diagnostic)**.** Give a rigorous worked analysis of Valuation rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A subring V of a field K is a valuation ring when for every nonzero $x \in K$, either x or x^{-1} lies in V. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 21.9 (Locality diagnostic)**.** Give a rigorous worked analysis of Locality. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The nonunits of a valuation ring form its unique maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 21.10 (Divisibility order diagnostic)**.** Give a rigorous worked analysis of Divisibility order. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Principal ideals are totally ordered by inclusion in a valuation ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

21.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 21.11 (Valuations diagnostic)**.** Give a rigorous worked analysis of Valuations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A valuation assigns ordered-group values compatible with products and satisfying a triangle inequality for sums. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 21.12 (Valuation rings diagnostic)**.** Give a rigorous worked analysis of Valuation rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A subring V of a field K is a valuation ring when for every nonzero $x \in K$, either x or x^{-1} lies in V. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 21.13 (Locality diagnostic)**.** Give a rigorous worked analysis of Locality. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The nonunits of a valuation ring form its unique maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 21.14 (Divisibility order diagnostic)**.** Give a rigorous worked analysis of Divisibility order. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Principal ideals are totally ordered by inclusion in a valuation ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 21.15 (Discrete valuation rings diagnostic)**.** Give a rigorous worked analysis of Discrete valuation rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A DVR corresponds to a discrete rank-one valuation and has a principal maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 21.16 (Integrally closed property diagnostic)**.** Give a rigorous worked analysis of Integrally closed property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every valuation ring is integrally closed in its fraction field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 21.17 (Dominating local rings diagnostic)**.** Give a rigorous worked analysis of Dominating local rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Valuation rings can contain a local domain while having maximal ideal contracting to the local maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 21.18 (Prime ideals diagnostic)**.** Give a rigorous worked analysis of Prime ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Prime ideals of a valuation ring are linearly ordered. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 21.19 (Valuation rings are local proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If V is a valuation ring of a field, its nonunits form an ideal and hence the unique maximal ideal.

Solution. For nonunits a, b, compare a/b or b/a when both are nonzero; one divides the other, so their sum is a multiple of one nonunit. Multiplication by arbitrary ring elements preserves nonunits. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 21.20 (Principal ideals are totally ordered proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For nonzero $a, b \in V$, either $(a) \subseteq (b)$ or $(b) \subseteq (a)$.

Solution. Apply the valuation-ring condition to a/b: if $a/b \in V$, then $a \in (b)$; otherwise $b/a \in V$, so $b \in (a)$. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 21.21 (Valuation rings are integrally closed proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A valuation ring V is integrally closed in its fraction field.

Solution. If x is integral over V but $x \notin V$, then $x^{-1} \in V$ and is a nonunit. Divide a monic equation for x by the highest power of x to express 1 as an element of the maximal ideal generated by powers of x^{-1}, contradiction. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 21.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Valuation rings impose a total divisibility order on a field. They are integrally closed local domains and provide a powerful way to test integrality and dominate local rings. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

21.13 Exercises with complete solutions

Exercise 21.23. State and apply the central principle behind Valuations. Give one concrete algebraic consequence.

Hint. For each nonzero fraction x in K, test whether x or x^{-1} lies in V; this total divisibility criterion characterizes valuation rings. □

Solution. A valuation assigns ordered-group values compatible with products and satisfying a triangle inequality for sums. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 21.24. State and apply the central principle behind Valuation rings. Give one concrete algebraic consequence.

Hint. From a valuation $v : K^* \to \Gamma$, define $V = \{x : v(x) \geq 0\} \cup \{0\}$ and verify closure under ring operations. □

Solution. A subring V of a field K is a valuation ring when for every nonzero $x \in K$, either x or x^{-1} lies in V. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 21.25. State and apply the central principle behind Locality. Give one concrete algebraic consequence.

Hint. Units are exactly elements of valuation zero; positive valuation elements form the maximal ideal in the standard valuation-domain construction. □

Solution. The nonunits of a valuation ring form its unique maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 21.26. State and apply the central principle behind Divisibility order. Give one concrete algebraic consequence.

Hint. Principal ideals are ordered by reverse valuation: larger value means smaller principal ideal. □

Solution. Principal ideals are totally ordered by inclusion in a valuation ring. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 21.27. State and apply the central principle behind Discrete valuation rings. Give one concrete algebraic consequence.

Hint. A discrete valuation ring has value group isomorphic to Z and a uniformizer of least positive value; identify that element. □

Solution. A DVR corresponds to a discrete rank-one valuation and has a principal maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 21.28. State and apply the central principle behind Integrally closed property. Give one concrete algebraic consequence.

Hint. To show valuation rings are integrally closed, compare valuations in a monic integral equation and derive a contradiction for negative valuation. □

Solution. Every valuation ring is integrally closed in its fraction field. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 21.29. State and apply the central principle behind Dominating local rings. Give one concrete algebraic consequence.

Hint. Prime ideals correspond to convex/order data in the value group in general; do not force the DVR picture onto every valuation ring. □

Solution. Valuation rings can contain a local domain while having maximal ideal contracting to the local maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 21.30. State and apply the central principle behind Prime ideals. Give one concrete algebraic consequence.

Hint. Use valuation-overring criteria for integral closure only after fixing the common fraction field. □

Solution. Prime ideals of a valuation ring are linearly ordered. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

21.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 21.31 (Valuation)**.** Compute $v_2(24/5)$.

Hint. Use valuations, units, and ordered ideals. □

Solution. It is 3. □

Exercise 21.32 (Negative valuation)**.** Compute $v_3(2/27)$.

Hint. Use valuations, units, and ordered ideals. □

Solution. It is -3. □

Exercise 21.33 (Unit criterion)**.** When is an element of a valuation ring a unit?

Hint. Use valuations, units, and ordered ideals. □

Solution. When its valuation is zero. □

Exercise 21.34 (Maximal ideal)**.** Which elements lie in the maximal ideal for a discrete valuation?

Hint. Use valuations, units, and ordered ideals. □

Solution. Those with positive valuation. □

Exercise 21.35 (Parameter valuation)**.** What is the t-adic valuation of $t^4(1+t)/(1-t)$?

Hint. Use valuations, units, and ordered ideals. □

Solution. It is 4. □

Proofs

Exercise 21.36 (Valuation rings are local proof)**.** Prove: If V is a valuation ring of a field, its nonunits form an ideal and hence the unique maximal ideal.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. For nonunits a, b, compare a/b or b/a when both are nonzero; one divides the other, so their sum is a multiple of one nonunit. Multiplication by arbitrary ring elements preserves nonunits. □

Exercise 21.37 (Principal ideals are totally ordered proof)**.** Prove: For nonzero $a, b \in V$, either $(a) \subseteq (b)$ or $(b) \subseteq (a)$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. Apply the valuation-ring condition to a/b: if $a/b \in V$, then $a \in (b)$; otherwise $b/a \in V$, so $b \in (a)$. □

Exercise 21.38 (Valuation rings are integrally closed proof)**.** Prove: A valuation ring V is integrally closed in its fraction field.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. If x is integral over V but $x \notin V$, then $x^{-1} \in V$ and is a nonunit. Divide a monic equation for x by the highest power of x to express 1 as an element of the maximal ideal generated by powers of x^{-1}, contradiction. □

Exercise 21.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Valuation rings are local and Principal ideals are totally ordered.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. □

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. □

Counterexamples and hypothesis tests

Exercise 21.40 (Two-variable local ring)**.** Test the claim: Every local domain is a valuation ring.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False; ideals in a valuation ring must be totally ordered. □

Exercise 21.41 (Negative member)**.** Test the claim: An element of negative valuation lies in the valuation ring.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False. □

Exercise 21.42 (Ideal ordering)**.** Test the claim: Ideals of a valuation ring can be incomparable.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False: they are linearly ordered. □

Applications and investigations

Exercise 21.43 (Order of vanishing)**.** What geometric information does a discrete valuation record?

Hint. Translate the question into valuations, units, and ordered ideals. □

Solution. It records order of zero or pole along a local parameter. □

Exercise 21.44 (Divisibility)**.** Why are valuation rings useful for divisibility?

Hint. Translate the question into valuations, units, and ordered ideals. □

Solution. The valuation totally orders divisibility by comparing numerical values. □

Challenge problems

Exercise 21.45 (Local-global synthesis)**.** Connect 'Valuations' with 'Integral closure criterion' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. □

Solution. The opening idea is: A valuation assigns ordered-group values compatible with products and satisfying a triangle inequality for sums. The later viewpoint is: Integral closure of a domain can be recovered as an intersection of valuation rings containing it. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. □

Exercise 21.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. □

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. □

Part VI

Homological Algebra

Chapter 22

Chain Complexes

Chain complexes organize sequences of modules with consecutive compositions zero. Homology measures the failure of exactness and is the central invariant of homological algebra.

Learning goals

After this chapter, the reader should be able to:

- verify that a sequence of modules and differentials forms a chain complex by checking consecutive composites vanish;
- compute cycles, boundaries, and homology groups in explicit finite complexes;
- construct chain maps and verify compatibility with differentials;
- identify exact complexes by vanishing homology;
- compute mapping behavior induced on homology by chain maps;
- distinguish chain complexes, exact sequences, and graded modules by their additional differential structure;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

22.1 Chain complexes

A chain complex has modules C_n and differentials $d_n : C_n \to C_{n-1}$ with $d_{n-1}d_n = 0$.

22.2 Cycles and boundaries

Cycles are kernels of differentials; boundaries are images of the next differentials.

Example 22.1 (A two-term complex)**.** The sequence $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to 0$ is a chain complex. Its degree-zero homology is $\mathbb{Z}/2\mathbb{Z}$, while the preceding homology vanishes because multiplication by 2 is injective.

22.3 Homology

$H_n(C) = \ker d_n / \operatorname{im} d_{n+1}$.

22.4 Exact complexes

A complex is exact exactly when all homology modules vanish.

22.5 Chain maps

A chain map commutes with differentials and therefore sends cycles to cycles and boundaries to boundaries.

Example 22.2 (Boundary squared is zero)**.** In any chain complex, $d_{n-1}d_n = 0$, so every boundary is automatically a cycle. Homology measures the quotient of cycles by those cycles already explained as boundaries.

22.6 Induced maps

Every chain map induces maps on homology.

22.7 Short exact sequences of complexes

Degreewise short exact sequences produce long exact sequences in homology.

Example 22.3 (Exact complex)**.** For a short exact sequence $0 \to A \to B \to C \to 0$ regarded as a complex, every homology group vanishes. Exactness is precisely acyclicity for this finite complex.

22.8 Homotopies

Chain-homotopic maps induce the same map on homology.

22.9 Examples

Two-term complexes, resolutions, Koszul-style complexes, and singular-like algebraic complexes illustrate the formalism.

22.10 Core structural results

Theorem 22.4 (Chain maps induce homology maps)**.** *A chain map $f : C \to D$ induces $H_n(f) : H_n(C) \to H_n(D)$.*

Proof. Commutation with differentials sends cycles to cycles. It also sends boundaries to boundaries because $f_n d_{n+1} = d_{n+1} f_{n+1}$, so the map descends to the quotient. □

Theorem 22.5 (Exactness equals zero homology)**.** *A chain complex is exact at every degree iff $H_n(C) = 0$ for all n.*

Proof. By definition, zero homology at degree n means the cycle module equals the boundary module, which is exactly exactness at that term. □

Theorem 22.6 (Chain homotopic maps agree on homology)**.** *If $f - g = dh + hd$, then f and g induce the same homology maps.*

Proof. For a cycle z, $(f - g)(z) = dh(z)$ because $d(z) = 0$, so the difference of images is a boundary. □

22.11 Worked examples

Example 22.7 (Chain complexes diagnostic)**.** Give a rigorous worked analysis of Chain complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A chain complex has modules C_n and differentials $d_n : C_n \to C_{n-1}$ with $d_{n-1}d_n = 0$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 22.8 (Cycles and boundaries diagnostic)**.** Give a rigorous worked analysis of Cycles and boundaries. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Cycles are kernels of differentials; boundaries are images of the next differentials. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 22.9 (Homology diagnostic)**.** Give a rigorous worked analysis of Homology. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $H_n(C) = \ker d_n / \operatorname{im} d_{n+1}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 22.10 (Exact complexes diagnostic)**.** Give a rigorous worked analysis of Exact complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A complex is exact exactly when all homology modules vanish. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

22.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 22.11 (Chain complexes diagnostic)**.** Give a rigorous worked analysis of Chain complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A chain complex has modules C_n and differentials $d_n : C_n \to C_{n-1}$ with $d_{n-1}d_n = 0$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 22.12 (Cycles and boundaries diagnostic)**.** Give a rigorous worked analysis of Cycles and boundaries. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Cycles are kernels of differentials; boundaries are images of the next differentials. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 22.13 (Homology diagnostic)**.** Give a rigorous worked analysis of Homology. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $H_n(C) = \ker d_n / \operatorname{im} d_{n+1}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 22.14 (Exact complexes diagnostic)**.** Give a rigorous worked analysis of Exact complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A complex is exact exactly when all homology modules vanish. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 22.15 (Chain maps diagnostic)**.** Give a rigorous worked analysis of Chain maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A chain map commutes with differentials and therefore sends cycles to cycles and boundaries to boundaries. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 22.16 (Induced maps diagnostic)**.** Give a rigorous worked analysis of Induced maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every chain map induces maps on homology. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 22.17 (Short exact sequences of complexes diagnostic)**.** Give a rigorous worked analysis of Short exact sequences of complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Degreewise short exact sequences produce long exact sequences in homology. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 22.18 (Homotopies diagnostic)**.** Give a rigorous worked analysis of Homotopies. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Chain-homotopic maps induce the same map on homology. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 22.19 (Chain maps induce homology maps proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A chain map $f : C \to D$ induces $H_n(f) : H_n(C) \to H_n(D)$.

Solution. Commutation with differentials sends cycles to cycles. It also sends boundaries to boundaries because $f_n d_{n+1} = d_{n+1} f_{n+1}$, so the map descends to the quotient. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 22.20 (Exactness equals zero homology proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A chain complex is exact at every degree iff $H_n(C) = 0$ for all n.

Solution. By definition, zero homology at degree n means the cycle module equals the boundary module, which is exactly exactness at that term. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 22.21 (Chain homotopic maps agree on homology proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $f - g = dh + hd$, then f and g induce the same homology maps.

Solution. For a cycle z, $(f - g)(z) = dh(z)$ because $d(z) = 0$, so the difference of images is a boundary. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 22.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Chain complexes organize sequences of modules with consecutive compositions zero. Homology measures the failure of exactness and is the central invariant of homological algebra. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. ☐

22.13 Exercises with complete solutions

Exercise 22.23. State and apply the central principle behind Chain complexes. Give one concrete algebraic consequence.

Hint. Check $d_n d_{n+1} = 0$ at every degree before calling the sequence a chain complex. ☐

Solution. A chain complex has modules C_n and differentials $d_n : C_n \to C_{n-1}$ with $d_{n-1} d_n = 0$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 22.24. State and apply the central principle behind Cycles and boundaries. Give one concrete algebraic consequence.

Hint. Cycles are $\ker d_n$ and boundaries are $\operatorname{im} d_{n+1}$; homology is Z_n/B_n. ☐

Solution. Cycles are kernels of differentials; boundaries are images of the next differentials. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 22.25. State and apply the central principle behind Homology. Give one concrete algebraic consequence.

Hint. To prove exactness at degree n, show every cycle is already a boundary. ☐

Solution. $H_n(C) = \ker d_n / \operatorname{im} d_{n+1}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 22.26. State and apply the central principle behind Exact complexes. Give one concrete algebraic consequence.

Hint. A chain map f satisfies d f = f d degreewise; write the commuting square at the degree in question. □

Solution. A complex is exact exactly when all homology modules vanish. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 22.27. State and apply the central principle behind Chain maps. Give one concrete algebraic consequence.

Hint. An induced homology map is well-defined because chain maps send cycles to cycles and boundaries to boundaries. □

Solution. A chain map commutes with differentials and therefore sends cycles to cycles and boundaries to boundaries. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 22.28. State and apply the central principle behind Induced maps. Give one concrete algebraic consequence.

Hint. For a short complex, compute kernels and images explicitly rather than using homological terminology as a substitute for calculation. □

Solution. Every chain map induces maps on homology. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 22.29. State and apply the central principle behind Short exact sequences of complexes. Give one concrete algebraic consequence.

Hint. Degree shifts and sign conventions matter when comparing chain and cochain complexes; state the convention first. □

Solution. Degreewise short exact sequences produce long exact sequences in homology. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 22.30. State and apply the central principle behind Homotopies. Give one concrete algebraic consequence.

Hint. Zero homology at every degree means exactness, but an exact complex need not be split or contractible. □

Solution. Chain-homotopic maps induce the same map on homology. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

22.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 22.31 (Homology)**.** Compute the cokernel homology of $\mathbb{Z} \xrightarrow{\cdot 3} \mathbb{Z}$.

Hint. Use cycles, boundaries, and homology. □

Solution. It is $\mathbb{Z}/3$. □

Exercise 22.32 (Kernel homology)**.** Compute the kernel of multiplication by 3 on $\mathbb{Z}$.

Hint. Use cycles, boundaries, and homology. □

Solution. It is zero. □

Exercise 22.33 (Zero differential)**.** If all differentials are zero, what is homology?

Hint. Use cycles, boundaries, and homology. □

Solution. It equals the chain modules themselves. □

Exercise 22.34 (Exact complex)**.** What is the homology of an exact complex?

Hint. Use cycles, boundaries, and homology. □

Solution. It is zero in every degree. □

Exercise 22.35 (Boundary inclusion)**.** Why is every boundary a cycle?

Hint. Use cycles, boundaries, and homology. □

Solution. Because consecutive differentials compose to zero. □

Proofs

Exercise 22.36 (Chain maps induce homology maps proof)**.** Prove: A chain map $f : C \to D$ induces $H_n(f) : H_n(C) \to H_n(D)$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. Commutation with differentials sends cycles to cycles. It also sends boundaries to boundaries because $f_n d_{n+1} = d_{n+1} f_{n+1}$, so the map descends to the quotient. □

Exercise 22.37 (Exactness equals zero homology proof)**.** Prove: A chain complex is exact at every degree iff $H_n(C) = 0$ for all n.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. By definition, zero homology at degree n means the cycle module equals the boundary module, which is exactly exactness at that term. □

Exercise 22.38 (Chain homotopic maps agree on homology proof)**.** Prove: If $f - g = dh + hd$, then f and g induce the same homology maps.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. For a cycle z, $(f-g)(z) = dh(z)$ because $d(z) = 0$, so the difference of images is a boundary. ☐

Exercise 22.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Chain maps induce homology maps and Exactness equals zero homology.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. ☐

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. ☐

Counterexamples and hypothesis tests

Exercise 22.40 (Arbitrary maps)**.** Test the claim: Any sequence of module maps is a chain complex.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False: consecutive maps must compose to zero. ☐

Exercise 22.41 (Cycles boundaries)**.** Test the claim: Every cycle is a boundary in every complex.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; the quotient measures homology. ☐

Exercise 22.42 (Zero homology)**.** Test the claim: Vanishing homology at one degree implies the whole complex is exact.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; exactness must hold at every degree. ☐

Applications and investigations

Exercise 22.43 (Obstruction measure)**.** What does homology measure?

Hint. Translate the question into cycles, boundaries, and homology. ☐

Solution. It measures failure of exactness at each degree. ☐

Exercise 22.44 (Topological bridge)**.** Why are chain complexes useful beyond algebra?

Hint. Translate the question into cycles, boundaries, and homology. ☐

Solution. They encode boundary operators whose homology captures invariants of spaces and resolutions. ☐

Challenge problems

Exercise 22.45 (Local-global synthesis)**.** Connect 'Chain complexes' with 'Examples' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. □

Solution. The opening idea is: A chain complex has modules C_n and differentials $d_n : C_n \to C_{n-1}$ with $d_{n-1}d_n = 0$. The later viewpoint is: Two-term complexes, resolutions, Koszul-style complexes, and singular-like algebraic complexes illustrate the formalism. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. □

Exercise 22.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. □

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. □

Chapter 23

Free Resolutions

Free resolutions replace a module by an exact complex of free modules. They allow non-free modules to be studied using explicit matrices and are the computational input for Tor and Ext.

Learning goals

After this chapter, the reader should be able to:

- construct free resolutions of elementary modules from generators and relations;
- verify exactness of a proposed resolution at each stage;
- compare two free resolutions through chain maps lifting the identity;
- use resolutions to compute invariants that are independent of the chosen resolution;
- truncate or extend resolutions while preserving exactness where justified;
- distinguish a presentation from a full resolution by identifying higher syzygies;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

23.1 Augmented complexes

A resolution ends in a surjection $F_0 \to M$ and is exact before and at M.

23.2 Free resolutions

All modules F_i in a free resolution are free.

Example 23.1 (Resolution of a cyclic group)**.** A free resolution of $\mathbb{Z}/n\mathbb{Z}$ begins $0 \to \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$. It is exact and both nonzero resolving modules are free.

23.3 Construction by kernels

Choose free generators for M, then free generators for the kernel, and repeat.

23.4 Projective resolutions

Projective resolutions generalize free resolutions and are often sufficient for derived functors.

23.5 Comparison maps

A module homomorphism lifts to a chain map between projective resolutions.

Example 23.2 (Resolution of a quotient ring)**.** For a nonzerodivisor $f \in R$, the sequence $0 \to R \xrightarrow{\cdot f} R \to R/(f) \to 0$ is a free resolution of length one.

23.6 Homotopy uniqueness

Two such lifts are chain homotopic.

23.7 Length

Finite projective dimension means a resolution terminates after finitely many nonzero projectives.

Example 23.3 (Computing after tensoring)**.** Applying a right-exact functor to a free resolution converts derived questions into homology calculations. Tensoring the resolution of $\mathbb{Z}/n$ with $\mathbb{Z}/m$ already reveals the kernel that becomes Tor_1.

23.8 Tensoring resolutions

Tensoring a free resolution computes Tor homology.

23.9 Applying Hom

Applying Hom to a projective resolution computes Ext cohomology.

23.10 Examples

Cyclic quotients and regular sequences yield explicit short or structured resolutions.

23.11 Core structural results

Theorem 23.4 (Existence of free resolutions)**.** *Every module admits a free resolution.*

Proof. Choose a free module surjecting onto the module. Take its kernel, choose a free module surjecting onto that kernel, and iterate indefinitely. □

Theorem 23.5 (Comparison theorem)**.** *A homomorphism $M \to N$ lifts to a chain map from any projective resolution of M to any resolution of N ending in a surjection.*

Proof. Lift degree zero using projectivity. Inductively, the image of the previous differential lands in the next kernel by the chain condition, and projectivity lifts through the corresponding surjection. □

Theorem 23.6 (Homotopy uniqueness of comparison maps)**.** *Two chain maps between projective resolutions inducing the same module map are chain homotopic.*

Proof. Construct the homotopy inductively. At each degree, the difference of the two maps has image in a kernel already identified with the image of the next differential, and projectivity provides a lift. □

23.12 Worked examples

Example 23.7 (Augmented complexes diagnostic)**.** Give a rigorous worked analysis of Augmented complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A resolution ends in a surjection $F_0 \to M$ and is exact before and at M. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 23.8 (Free resolutions diagnostic)**.** Give a rigorous worked analysis of Free resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. All modules F_i in a free resolution are free. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 23.9 (Construction by kernels diagnostic)**.** Give a rigorous worked analysis of Construction by kernels. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Choose free generators for M, then free generators for the kernel, and repeat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 23.10 (Projective resolutions diagnostic)**.** Give a rigorous worked analysis of Projective resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Projective resolutions generalize free resolutions and are often sufficient for derived functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

23.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 23.11 (Augmented complexes diagnostic)**.** Give a rigorous worked analysis of Augmented complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A resolution ends in a surjection $F_0 \to M$ and is exact before and at M. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 23.12 (Free resolutions diagnostic)**.** Give a rigorous worked analysis of Free resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. All modules F_i in a free resolution are free. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 23.13 (Construction by kernels diagnostic)**.** Give a rigorous worked analysis of Construction by kernels. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Choose free generators for M, then free generators for the kernel, and repeat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 23.14 (Projective resolutions diagnostic)**.** Give a rigorous worked analysis of Projective resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Projective resolutions generalize free resolutions and are often sufficient for derived functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 23.15 (Comparison maps diagnostic)**.** Give a rigorous worked analysis of Comparison maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module homomorphism lifts to a chain map between projective resolutions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 23.16 (Homotopy uniqueness diagnostic)**.** Give a rigorous worked analysis of Homotopy uniqueness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Two such lifts are chain homotopic. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 23.17 (Length diagnostic)**.** Give a rigorous worked analysis of Length. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite projective dimension means a resolution terminates after finitely many nonzero projectives. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 23.18 (Tensoring resolutions diagnostic)**.** Give a rigorous worked analysis of Tensoring resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Tensoring a free resolution computes Tor homology. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 23.19 (Existence of free resolutions proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Every module admits a free resolution.

Solution. Choose a free module surjecting onto the module. Take its kernel, choose a free module surjecting onto that kernel, and iterate indefinitely. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 23.20 (Comparison theorem proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A homomorphism $M \to N$ lifts to a chain map from any projective resolution of M to any resolution of N ending in a surjection.

Solution. Lift degree zero using projectivity. Inductively, the image of the previous differential lands in the next kernel by the chain condition, and projectivity lifts through the corresponding surjection. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 23.21 (Homotopy uniqueness of comparison maps proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Two chain maps between projective resolutions inducing the same module map are chain homotopic.

Solution. Construct the homotopy inductively. At each degree, the difference of the two maps has image in a kernel already identified with the image of the next differential, and projectivity provides a lift. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 23.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Free resolutions replace a module by an exact complex of free modules. They allow non-free modules to be studied using explicit matrices and are the computational input for Tor and Ext. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

23.14 Exercises with complete solutions

Exercise 23.23. State and apply the central principle behind Augmented complexes. Give one concrete algebraic consequence.

Hint. Start from a surjection $F_0 \to M$ from a free module on chosen generators; the first syzygy is its kernel. □

Solution. A resolution ends in a surjection $F_0 \to M$ and is exact before and at M. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 23.24. State and apply the central principle behind Free resolutions. Give one concrete algebraic consequence.

Hint. Choose a free module F_1 surjecting onto that kernel, then repeat recursively to build a free resolution. □

Solution. All modules F_i in a free resolution are free. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 23.25. State and apply the central principle behind Construction by kernels. Give one concrete algebraic consequence.

Hint. At each degree, exactness requires image of the new differential to equal the previous kernel. □

Solution. Choose free generators for M, then free generators for the kernel, and repeat. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 23.26. State and apply the central principle behind Projective resolutions. Give one concrete algebraic consequence.

Hint. A presentation stops after $F_1 \to F_0 \to M$; a resolution continues by resolving the relation modules. □

Solution. Projective resolutions generalize free resolutions and are often sufficient for derived functors. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 23.27. State and apply the central principle behind Comparison maps. Give one concrete algebraic consequence.

Hint. Comparison maps between resolutions are built inductively using projectivity of the free modules. □

Solution. A module homomorphism lifts to a chain map between projective resolutions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 23.28. State and apply the central principle behind Homotopy uniqueness. Give one concrete algebraic consequence.

Hint. Different resolutions may look different, so only use invariants proved independent up to chain homotopy. □

Solution. Two such lifts are chain homotopic. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 23.29. State and apply the central principle behind Length. Give one concrete algebraic consequence.

Hint. Truncating a resolution changes exactness at the cut unless the remaining kernel is projective/free as required. □

Solution. Finite projective dimension means a resolution terminates after finitely many nonzero projectives. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 23.30. State and apply the central principle behind Tensoring resolutions. Give one concrete algebraic consequence.

Hint. For explicit modules, derive the first differential from a presentation matrix before inventing higher maps. □

Solution. Tensoring a free resolution computes Tor homology. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

23.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 23.31 (Cyclic resolution)**.** Write a free resolution of $\mathbb{Z}/5$ of length one.

Hint. Use free resolutions and exact augmentations. ☐

Solution. $0 \to \mathbb{Z} \xrightarrow{\cdot 5} \mathbb{Z} \to \mathbb{Z}/5 \to 0$. ☐

Exercise 23.32 (Quotient resolution)**.** Resolve $R/(f)$ when f is a nonzerodivisor.

Hint. Use free resolutions and exact augmentations. ☐

Solution. $0 \to R \xrightarrow{\cdot f} R \to R/(f) \to 0$. ☐

Exercise 23.33 (Projective dimension)**.** What upper bound follows from a length-one free resolution?

Hint. Use free resolutions and exact augmentations. ☐

Solution. Projective dimension at most one. ☐

Exercise 23.34 (Free module)**.** What resolution does a free module need?

Hint. Use free resolutions and exact augmentations. ☐

Solution. Length zero. ☐

Exercise 23.35 (Augmentation)**.** What is the last map in a resolution called?

Hint. Use free resolutions and exact augmentations. ☐

Solution. The augmentation onto the resolved module. ☐

Proofs

Exercise 23.36 (Existence of free resolutions proof)**.** Prove: Every module admits a free resolution.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Choose a free module surjecting onto the module. Take its kernel, choose a free module surjecting onto that kernel, and iterate indefinitely. ☐

Exercise 23.37 (Comparison theorem proof)**.** Prove: A homomorphism $M \to N$ lifts to a chain map from any projective resolution of M to any resolution of N ending in a surjection.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Lift degree zero using projectivity. Inductively, the image of the previous differential lands in the next kernel by the chain condition, and projectivity lifts through the corresponding surjection. ☐

Exercise 23.38 (Homotopy uniqueness of comparison maps proof)**.** Prove: Two chain maps between projective resolutions inducing the same module map are chain homotopic.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Construct the homotopy inductively. At each degree, the difference of the two maps has image in a kernel already identified with the image of the next differential, and projectivity provides a lift. ☐

Exercise 23.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Existence of free resolutions and Comparison theorem.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. ☐

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. ☐

Counterexamples and hypothesis tests

Exercise 23.40 (Unique resolution)**.** Test the claim: A module has a unique free resolution.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; resolutions are highly nonunique. ☐

Exercise 23.41 (Finite resolution)**.** Test the claim: Every module has a finite free resolution.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False in general. ☐

Exercise 23.42 (Resolution exactness)**.** Test the claim: A resolution may have nonzero homology in positive degrees.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; it is exact there. ☐

Applications and investigations

Exercise 23.43 (Derived computation)**.** Why resolve by free or projective modules?

Hint. Translate the question into free resolutions and exact augmentations. ☐

Solution. Such modules make Hom and tensor computations tractable and define derived functors. ☐

Exercise 23.44 (Complexity)**.** What does minimal resolution length measure?

Hint. Translate the question into free resolutions and exact augmentations. ☐

Solution. It contributes to projective dimension and homological complexity. ☐

Challenge problems

Exercise 23.45 (Local-global synthesis)**.** Connect 'Augmented complexes' with 'Examples' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. □

Solution. The opening idea is: A resolution ends in a surjection $F_0 \to M$ and is exact before and at M. The later viewpoint is: Cyclic quotients and regular sequences yield explicit short or structured resolutions. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. □

Exercise 23.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. □

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. □

Chapter 24

Syzygies

Syzygies are relations among generators, then relations among relations, and so on. Iterating syzygies turns presentations into resolutions and exposes hidden structure in modules.

Learning goals

After this chapter, the reader should be able to:

- compute first syzygies as relations among a chosen generating set;
- construct higher syzygies recursively from kernels in a free resolution;
- translate presentation matrices into syzygy modules;
- analyze how changing generators changes a presentation but not the underlying module;
- use syzygies to continue a free resolution;
- distinguish minimal relations from redundant generators or redundant syzygies;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

24.1 First syzygy

Given a surjection $F_0 \to M$, the first syzygy module is its kernel.

24.2 Higher syzygies

The nth syzygy is the kernel at stage $n-1$ of a chosen free or projective resolution.

Example 24.1 (A first syzygy)**.** For the ideal $(x, y) \subset k[x, y]$, the map $R^2 \to (x, y)$ sends (a, b) to $ax + by$. The vector $(-y, x)$ lies in the kernel, giving the basic syzygy $-yx + xy = 0$.

24.3 Dependence on presentation

Syzygies depend on choices, but stable projective information is controlled up to adding free summands.

24.4 Presentations

A finite presentation gives a finitely generated first syzygy.

24.5 Noetherian setting

Over a Noetherian ring, syzygies of finite modules in finite free presentations remain finitely generated.

Example 24.2 (Syzygy from a presentation matrix)**.** If $R^m \xrightarrow{A} R^n \to M \to 0$ presents M, then the first syzygy module is $\ker A$. A second syzygy is obtained by presenting that kernel and taking the next kernel.

24.6 Dimension shifting

Higher Tor and Ext groups can be expressed through lower groups of syzygies.

24.7 Regular sequences

Koszul-type relations provide canonical syzygies for complete-intersection situations.

Example 24.3 (Koszul relation)**.** For a regular pair x, y, the relation $(-y, x)$ generates the first relation among the two generators of (x, y). This is the opening step of the Koszul resolution.

24.8 Hilbert syzygy preview

Over polynomial rings over fields, finite modules have bounded projective resolution length.

24.9 Computational matrices

Syzygies are kernels of matrices and can be studied algorithmically.

24.10 Core structural results

Theorem 24.4 (Syzygies over Noetherian rings are finite)**.** *If A is Noetherian and M is finitely generated, then the kernel of a map $A^n \to M$ is finitely generated.*

Proof. The finite free module A^n is Noetherian, so every submodule, including the kernel, is finitely generated. □

Theorem 24.5 (Schanuel lemma)**.** *If $0 \to K \to P \to M \to 0$ and $0 \to K' \to P' \to M \to 0$ have projective P, P', then $K \oplus P' \cong K' \oplus P$.*

Proof. Form the pullback of $P \to M$ and $P' \to M$. Its two projections fit into short exact sequences that split because P and P' are projective, yielding the two decompositions. □

Theorem 24.6 (Dimension shifting for Tor)**.** *If $0 \to \Omega M \to P \to M \to 0$ with P projective, then $\mathrm{Tor}_n^A(M, N) \cong \mathrm{Tor}_{n-1}^A(\Omega M, N)$ for $n \geq 2$.*

Proof. Apply the long exact Tor sequence. Higher Tor of the projective module P vanishes, so adjacent terms give the stated isomorphism. □

24.11 Worked examples

Example 24.7 (First syzygy diagnostic). Give a rigorous worked analysis of First syzygy. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Given a surjection $F_0 \to M$, the first syzygy module is its kernel. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 24.8 (Higher syzygies diagnostic). Give a rigorous worked analysis of Higher syzygies. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The nth syzygy is the kernel at stage $n-1$ of a chosen free or projective resolution. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 24.9 (Dependence on presentation diagnostic). Give a rigorous worked analysis of Dependence on presentation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Syzygies depend on choices, but stable projective information is controlled up to adding free summands. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 24.10 (Presentations diagnostic). Give a rigorous worked analysis of Presentations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A finite presentation gives a finitely generated first syzygy. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

24.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 24.11 (First syzygy diagnostic). Give a rigorous worked analysis of First syzygy. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Given a surjection $F_0 \to M$, the first syzygy module is its kernel. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 24.12 (Higher syzygies diagnostic). Give a rigorous worked analysis of Higher syzygies. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The nth syzygy is the kernel at stage $n-1$ of a chosen free or projective resolution. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 24.13 (Dependence on presentation diagnostic). Give a rigorous worked analysis of Dependence on presentation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Syzygies depend on choices, but stable projective information is controlled up to adding free summands. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 24.14 (Presentations diagnostic)**.** Give a rigorous worked analysis of Presentations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A finite presentation gives a finitely generated first syzygy. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 24.15 (Noetherian setting diagnostic)**.** Give a rigorous worked analysis of Noetherian setting. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over a Noetherian ring, syzygies of finite modules in finite free presentations remain finitely generated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 24.16 (Dimension shifting diagnostic)**.** Give a rigorous worked analysis of Dimension shifting. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Higher Tor and Ext groups can be expressed through lower groups of syzygies. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 24.17 (Regular sequences diagnostic)**.** Give a rigorous worked analysis of Regular sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Koszul-type relations provide canonical syzygies for complete-intersection situations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 24.18 (Hilbert syzygy preview diagnostic)**.** Give a rigorous worked analysis of Hilbert syzygy preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over polynomial rings over fields, finite modules have bounded projective resolution length. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 24.19 (Syzygies over Noetherian rings are finite proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If A is Noetherian and M is finitely generated, then the kernel of a map $A^n \to M$ is finitely generated.

Solution. The finite free module A^n is Noetherian, so every submodule, including the kernel, is finitely generated. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 24.20 (Schanuel lemma proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $0 \to K \to P \to M \to 0$ and $0 \to K' \to P' \to M \to 0$ have projective P, P', then $K \oplus P' \cong K' \oplus P$.

Solution. Form the pullback of $P \to M$ and $P' \to M$. Its two projections fit into short exact sequences that split because P and P' are projective, yielding the two decompositions. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 24.21 (Dimension shifting for Tor proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $0 \to \Omega M \to P \to M \to 0$ with P projective, then $\mathrm{Tor}_n^A(M,N) \cong \mathrm{Tor}_{n-1}^A(\Omega M, N)$ for $n \geq 2$.

Solution. Apply the long exact Tor sequence. Higher Tor of the projective module P vanishes, so adjacent terms give the stated isomorphism. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 24.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Syzygies are relations among generators, then relations among relations, and so on. Iterating syzygies turns presentations into resolutions and exposes hidden structure in modules. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. ☐

24.13 Exercises with complete solutions

Exercise 24.23. State and apply the central principle behind First syzygy. Give one concrete algebraic consequence.

Hint. Given generators of M, place them in a surjection $F_0 \to M$; the kernel consists exactly of the first syzygies. ☐

Solution. Given a surjection $F_0 \to M$, the first syzygy module is its kernel. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 24.24. State and apply the central principle behind Higher syzygies. Give one concrete algebraic consequence.

Hint. Write each relation as a coordinate vector in F_0; those vectors generate the syzygy module. ☐

Solution. The nth syzygy is the kernel at stage $n-1$ of a chosen free or projective resolution. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 24.25. State and apply the central principle behind Dependence on presentation. Give one concrete algebraic consequence.

Hint. To compute higher syzygies, repeat the kernel computation for the map from a free module onto the previous syzygy module. ☐

Solution. Syzygies depend on choices, but stable projective information is controlled up to adding free summands. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 24.26. State and apply the central principle behind Presentations. Give one concrete algebraic consequence.

Hint. A presentation matrix encodes relations in its columns or rows according to convention; fix that convention before computing kernels. ☐

Solution. A finite presentation gives a finitely generated first syzygy. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 24.27. State and apply the central principle behind Noetherian setting. Give one concrete algebraic consequence.

Hint. Changing generators changes the presentation matrix by free-module operations but not the isomorphism class of M. ☐

Solution. Over a Noetherian ring, syzygies of finite modules in finite free presentations remain finitely generated. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 24.28. State and apply the central principle behind Dimension shifting. Give one concrete algebraic consequence.

Hint. Remove a redundant relation only after proving it lies in the submodule generated by the others. ☐

Solution. Higher Tor and Ext groups can be expressed through lower groups of syzygies. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 24.29. State and apply the central principle behind Regular sequences. Give one concrete algebraic consequence.

Hint. Syzygies are the data needed for the next differential in a free resolution; verify consecutive maps compose to zero. ☐

Solution. Koszul-type relations provide canonical syzygies for complete-intersection situations. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 24.30. State and apply the central principle behind Hilbert syzygy preview. Give one concrete algebraic consequence.

Hint. In graded settings, track degrees of generators and relations so the syzygy maps are homogeneous. ☐

Solution. Over polynomial rings over fields, finite modules have bounded projective resolution length. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

24.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 24.31 (Basic relation)**.** Give a syzygy between x and y.

Hint. Use kernels of presentation maps and relations among generators. ☐

Solution. $(-y, x)$. ☐

Exercise 24.32 (Kernel definition)**.** What is the first syzygy module in a presentation $F_1 \to F_0 \to M$?

Hint. Use kernels of presentation maps and relations among generators. ☐

Solution. The kernel of $F_0 \to M$, equivalently the image from F_1 in an exact resolution. ☐

Exercise 24.33 (Free module)**.** What are the positive syzygies of a free module in a length-zero resolution?

Hint. Use kernels of presentation maps and relations among generators. ☐

Solution. They vanish. ☐

Exercise 24.34 (Cyclic quotient)**.** What is the first syzygy of $R/(f)$ in the standard presentation?

Hint. Use kernels of presentation maps and relations among generators. ☐

Solution. The principal submodule $fR \cong R$ when f is regular. ☐

Exercise 24.35 (Iterated kernel)**.** How is the second syzygy obtained?

Hint. Use kernels of presentation maps and relations among generators. ☐

Solution. As the kernel of a free map presenting the first syzygy. ☐

Proofs

Exercise 24.36 (Syzygies over Noetherian rings are finite proof)**.** Prove: If A is Noetherian and M is finitely generated, then the kernel of a map $A^n \to M$ is finitely generated.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. The finite free module A^n is Noetherian, so every submodule, including the kernel, is finitely generated. ☐

Exercise 24.37 (Schanuel lemma proof)**.** Prove: If $0 \to K \to P \to M \to 0$ and $0 \to K' \to P' \to M \to 0$ have projective P, P', then $K \oplus P' \cong K' \oplus P$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Form the pullback of $P \to M$ and $P' \to M$. Its two projections fit into short exact sequences that split because P and P' are projective, yielding the two decompositions. ☐

Exercise 24.38 (Dimension shifting for Tor proof)**.** Prove: If $0 \to \Omega M \to P \to M \to 0$ with P projective, then $\mathrm{Tor}_n^A(M, N) \cong \mathrm{Tor}_{n-1}^A(\Omega M, N)$ for $n \geq 2$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Apply the long exact Tor sequence. Higher Tor of the projective module P vanishes, so adjacent terms give the stated isomorphism. ☐

Exercise 24.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Syzygies over Noetherian rings are finite and Schanuel lemma.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. ☐

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. ☐

Counterexamples and hypothesis tests

Exercise 24.40 (No relations)**.** Test the claim: Every generating set has zero syzygy module.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; relations are usually nontrivial. ☐

Exercise 24.41 (Generator dependence)**.** Test the claim: The concrete syzygy module is independent of the chosen presentation on the nose.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; stable isomorphism phenomena replace literal equality. ☐

Exercise 24.42 (Finite syzygy)**.** Test the claim: Every syzygy over every ring is finitely generated.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False without Noetherian or finiteness hypotheses. ☐

Applications and investigations

Exercise 24.43 (Equation relations)**.** What does a syzygy encode computationally?

Hint. Translate the question into kernels of presentation maps and relations among generators. ☐

Solution. It records relations among chosen generators. ☐

Exercise 24.44 (Resolution building)**.** How do syzygies build a free resolution?

Hint. Translate the question into kernels of presentation maps and relations among generators. ☐

Solution. Resolve the relation module repeatedly by new free modules. ☐

Challenge problems

Exercise 24.45 (Local-global synthesis)**.** Connect 'First syzygy' with 'Computational matrices' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. □

Solution. The opening idea is: Given a surjection $F_0 \to M$, the first syzygy module is its kernel. The later viewpoint is: Syzygies are kernels of matrices and can be studied algorithmically. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. □

Exercise 24.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. □

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. □

Chapter 25

Minimal Resolutions

Minimal resolutions remove redundant free summands. Over local rings, minimality is detected by differentials whose matrix entries lie in the maximal ideal, making Betti numbers intrinsic.

Learning goals

After this chapter, the reader should be able to:

- identify minimal free resolutions over local or graded rings using the chapter's minimality criterion;
- compute Betti numbers from ranks of modules in a minimal resolution;
- detect and remove contractible or unit-entry summands from nonminimal resolutions;
- compare minimal resolutions up to isomorphism in settings where uniqueness holds;
- use residue-field tensoring to read off minimal generators or Betti numbers;
- distinguish homological length, projective dimension, and resolution length;

Conceptual roadmap

$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$

25.1 Minimal free resolutions

A free resolution is minimal when no contractible free summand can be removed.

25.2 Local criterion

Over a local ring $(A, \mathfrak{m})$, a finite free resolution is minimal iff each differential has image contained in $\mathfrak{m}$ times the previous free module.

Example 25.1 (Minimality over a local ring)**.** Over a local ring $(R, \mathfrak{m})$, a finite free resolution is minimal when every differential matrix has entries in $\mathfrak{m}$. Reducing modulo $\mathfrak{m}$ then makes every differential zero.

25.3 Reduction modulo the maximal ideal

Minimal differentials become zero after tensoring with the residue field.

25.4 Betti numbers

Ranks of the free modules in a minimal resolution are invariants of the module.

25.5 Nakayama mechanism

Minimal generating sets are detected after reduction modulo $\mathfrak{m}$.

Example 25.2 (Cancellation of a unit)**.** If a differential matrix contains a unit entry, suitable row and column operations split off a contractible summand $R \xrightarrow{\cong} R$. Removing it shortens the resolution, so a minimal resolution cannot contain such a unit.

25.6 Uniqueness

Minimal free resolutions over local rings are unique up to chain isomorphism.

25.7 Projective dimension

The last nonzero module in a minimal finite resolution detects projective dimension.

Example 25.3 (Residue field over a regular local ring)**.** For $R = k[x, y]_{(x,y)}$, the Koszul complex on x, y gives a minimal free resolution of the residue field k. Its Betti numbers are $1, 2, 1$.

25.8 Graded analogue

Over standard graded rings, minimality is similarly controlled by positive-degree entries.

25.9 Computational significance

Minimal resolutions expose generators, first relations, and higher relations without algebraic cancellation.

25.10 Core structural results

Theorem 25.4 (Local minimality criterion)**.** *A free resolution $F_\bullet$ over a local ring is minimal iff $d(F_n) \subseteq \mathfrak{m}F_{n-1}$ for all n.*

Proof. If a matrix entry is a unit, elementary basis changes split off an identity map between rank-one free summands, producing a contractible piece. Conversely, if all entries lie in the maximal ideal, no such unit cancellation is possible. □

Theorem 25.5 (Betti numbers from Tor)**.** *For a minimal free resolution of a finite module over a local ring with residue field k, $\operatorname{Tor}_n^A(M, k) \cong F_n \otimes_A k$.*

Proof. Tensor the minimal resolution with k. Every differential becomes zero because its entries lie in $\mathfrak{m}$, so homology in degree n is the entire vector space $F_n \otimes k$. ☐

Theorem 25.6 (Uniqueness of minimal resolutions)**.** *Minimal free resolutions of a finite module over a local ring are isomorphic as complexes.*

Proof. Comparison maps in both directions lift the identity of the module. Modulo the maximal ideal they are inverse isomorphisms on Tor, hence degreewise isomorphisms by Nakayama's lemma. ☐

25.11 Worked examples

Example 25.7 (Minimal free resolutions diagnostic)**.** Give a rigorous worked analysis of Minimal free resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A free resolution is minimal when no contractible free summand can be removed. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 25.8 (Local criterion diagnostic)**.** Give a rigorous worked analysis of Local criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Over a local ring $(A, \mathfrak{m})$, a finite free resolution is minimal iff each differential has image contained in $\mathfrak{m}$ times the previous free module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 25.9 (Reduction modulo the maximal ideal diagnostic)**.** Give a rigorous worked analysis of Reduction modulo the maximal ideal. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Minimal differentials become zero after tensoring with the residue field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 25.10 (Betti numbers diagnostic)**.** Give a rigorous worked analysis of Betti numbers. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Ranks of the free modules in a minimal resolution are invariants of the module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

25.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 25.11 (Minimal free resolutions diagnostic)**.** Give a rigorous worked analysis of Minimal free resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A free resolution is minimal when no contractible free summand can be removed. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 25.12 (Local criterion diagnostic)**.** Give a rigorous worked analysis of Local criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over a local ring $(A, \mathfrak{m})$, a finite free resolution is minimal iff each differential has image contained in $\mathfrak{m}$ times the previous free module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 25.13 (Reduction modulo the maximal ideal diagnostic)**.** Give a rigorous worked analysis of Reduction modulo the maximal ideal. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Minimal differentials become zero after tensoring with the residue field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 25.14 (Betti numbers diagnostic)**.** Give a rigorous worked analysis of Betti numbers. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Ranks of the free modules in a minimal resolution are invariants of the module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 25.15 (Nakayama mechanism diagnostic)**.** Give a rigorous worked analysis of Nakayama mechanism. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Minimal generating sets are detected after reduction modulo $\mathfrak{m}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 25.16 (Uniqueness diagnostic)**.** Give a rigorous worked analysis of Uniqueness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Minimal free resolutions over local rings are unique up to chain isomorphism. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 25.17 (Projective dimension diagnostic)**.** Give a rigorous worked analysis of Projective dimension. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The last nonzero module in a minimal finite resolution detects projective dimension. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 25.18 (Graded analogue diagnostic)**.** Give a rigorous worked analysis of Graded analogue. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over standard graded rings, minimality is similarly controlled by positive-degree entries. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 25.19 (Local minimality criterion proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A free resolution $F_\bullet$ over a local ring is minimal iff $d(F_n) \subseteq \mathfrak{m}F_{n-1}$ for all n.

Solution. If a matrix entry is a unit, elementary basis changes split off an identity map between rank-one free summands, producing a contractible piece. Conversely, if all entries lie in the maximal ideal, no such unit cancellation is possible. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 25.20 (Betti numbers from Tor proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For a minimal free resolution of a finite module over a local ring with residue field k, $\mathrm{Tor}_n^A(M,k) \cong F_n \otimes_A k$.

Solution. Tensor the minimal resolution with k. Every differential becomes zero because its entries lie in $\mathfrak{m}$, so homology in degree n is the entire vector space $F_n \otimes k$. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 25.21 (Uniqueness of minimal resolutions proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Minimal free resolutions of a finite module over a local ring are isomorphic as complexes.

Solution. Comparison maps in both directions lift the identity of the module. Modulo the maximal ideal they are inverse isomorphisms on Tor, hence degreewise isomorphisms by Nakayama's lemma. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 25.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Minimal resolutions remove redundant free summands. Over local rings, minimality is detected by differentials whose matrix entries lie in the maximal ideal, making Betti numbers intrinsic. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

25.13 Exercises with complete solutions

Exercise 25.23. State and apply the central principle behind Minimal free resolutions. Give one concrete algebraic consequence.

Hint. Over a local ring, minimality is detected by differential matrices having entries in the maximal ideal; check this degree by degree. □

Solution. A free resolution is minimal when no contractible free summand can be removed. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 25.24. State and apply the central principle behind Local criterion. Give one concrete algebraic consequence.

Hint. Over a graded ring, use positive-degree entries in the irrelevant maximal ideal for the analogous minimality test. □

Solution. Over a local ring $(A, \mathfrak{m})$, a finite free resolution is minimal iff each differential has image contained in $\mathfrak{m}$ times the previous free module. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 25.25. State and apply the central principle behind Reduction modulo the maximal ideal. Give one concrete algebraic consequence.

Hint. Tensor a minimal resolution with the residue field; all differentials vanish, so the graded vector-space dimensions reveal Betti numbers. ☐

Solution. Minimal differentials become zero after tensoring with the residue field. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 25.26. State and apply the central principle behind Betti numbers. Give one concrete algebraic consequence.

Hint. If a differential matrix contains a unit entry, split off the corresponding contractible free summand to reduce the resolution. ☐

Solution. Ranks of the free modules in a minimal resolution are invariants of the module. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 25.27. State and apply the central principle behind Nakayama mechanism. Give one concrete algebraic consequence.

Hint. Minimal resolutions are unique up to isomorphism in the standard local/graded setting; compare ranks degree by degree. ☐

Solution. Minimal generating sets are detected after reduction modulo $\mathfrak{m}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 25.28. State and apply the central principle behind Uniqueness. Give one concrete algebraic consequence.

Hint. Projective dimension is the least possible resolution length by projectives, not simply the length of an arbitrary resolution. ☐

Solution. Minimal free resolutions over local rings are unique up to chain isomorphism. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 25.29. State and apply the central principle behind Projective dimension. Give one concrete algebraic consequence.

Hint. Betti numbers are ranks in a minimal resolution; nonminimal resolutions can inflate them artificially. ☐

Solution. The last nonzero module in a minimal finite resolution detects projective dimension. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 25.30. State and apply the central principle behind Graded analogue. Give one concrete algebraic consequence.

Hint. Check finite-generation/local hypotheses before using minimal-resolution uniqueness or residue-field criteria. □

Solution. Over standard graded rings, minimality is similarly controlled by positive-degree entries. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

25.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 25.31 (Minimal entry)**.** Where must entries of a minimal differential lie over a local ring?

Hint. Use local minimality, maximal-ideal entries, and cancellation. □

Solution. In the maximal ideal. □

Exercise 25.32 (Reduction)**.** What happens to minimal differentials modulo the maximal ideal?

Hint. Use local minimality, maximal-ideal entries, and cancellation. □

Solution. They become zero. □

Exercise 25.33 (Betti numbers)**.** What are the Koszul Betti numbers for two generators?

Hint. Use local minimality, maximal-ideal entries, and cancellation. □

Solution. $1, 2, 1$. □

Exercise 25.34 (Unit entry)**.** What does a unit entry signal?

Hint. Use local minimality, maximal-ideal entries, and cancellation. □

Solution. A cancellable free summand. □

Exercise 25.35 (Free module)**.** What is the minimal resolution of a free module?

Hint. Use local minimality, maximal-ideal entries, and cancellation. □

Solution. Length zero. □

Proofs

Exercise 25.36 (Local minimality criterion proof)**.** Prove: A free resolution $F_\bullet$ over a local ring is minimal iff $d(F_n) \subseteq \mathfrak{m}F_{n-1}$ for all n.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. If a matrix entry is a unit, elementary basis changes split off an identity map between rank-one free summands, producing a contractible piece. Conversely, if all entries lie in the maximal ideal, no such unit cancellation is possible. □

Exercise 25.37 (Betti numbers from Tor proof)**.** Prove: For a minimal free resolution of a finite module over a local ring with residue field k, $\mathrm{Tor}_n^A(M, k) \cong F_n \otimes_A k$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. Tensor the minimal resolution with k. Every differential becomes zero because its entries lie in $\mathfrak{m}$, so homology in degree n is the entire vector space $F_n \otimes k$. □

Exercise 25.38 (Uniqueness of minimal resolutions proof)**.** Prove: Minimal free resolutions of a finite module over a local ring are isomorphic as complexes.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. □

Solution. Comparison maps in both directions lift the identity of the module. Modulo the maximal ideal they are inverse isomorphisms on Tor, hence degreewise isomorphisms by Nakayama's lemma. □

Exercise 25.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Local minimality criterion and Betti numbers from Tor.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. □

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. □

Counterexamples and hypothesis tests

Exercise 25.40 (Any field reduction)**.** Test the claim: A resolution is minimal exactly when every differential is zero before reduction.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False; entries need only lie in the maximal ideal. □

Exercise 25.41 (Uniqueness matrices)**.** Test the claim: Minimal resolutions have literally identical matrices for every basis choice.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False; they are unique only up to suitable chain isomorphism. □

Exercise 25.42 (Unit allowed)**.** Test the claim: A minimal local resolution may contain a unit entry in a differential.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. □

Solution. False. □

Applications and investigations

Exercise 25.43 (Betti invariants)**.** Why are ranks in a minimal resolution meaningful invariants?

Hint. Translate the question into local minimality, maximal-ideal entries, and cancellation. ☐

Solution. Minimality prevents trivial cancellations, so the free ranks are intrinsic. ☐

Exercise 25.44 (Complexity)**.** What does growth of Betti numbers indicate?

Hint. Translate the question into local minimality, maximal-ideal entries, and cancellation. ☐

Solution. It measures increasing homological complexity of the module. ☐

Challenge problems

Exercise 25.45 (Local-global synthesis)**.** Connect 'Minimal free resolutions' with 'Computational significance' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. ☐

Solution. The opening idea is: A free resolution is minimal when no contractible free summand can be removed. The later viewpoint is: Minimal resolutions expose generators, first relations, and higher relations without algebraic cancellation. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. ☐

Exercise 25.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. ☐

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. ☐

Chapter 26

The Tor Functor

The Tor functor is the homology left behind when tensor product fails to be exact. It is computed from projective resolutions and detects torsion, non-flatness, and derived intersections.

Learning goals

After this chapter, the reader should be able to:

- compute Tor groups from a projective or free resolution and tensor product;
- verify that Tor_0 recovers ordinary tensor product;
- use long exact Tor sequences arising from short exact sequences;
- detect nonflatness through nonvanishing Tor_1;
- compute Tor for cyclic modules and quotient rings in elementary examples;
- track functoriality and independence of the chosen resolution;

Conceptual roadmap

$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$

26.1 Definition

Choose a projective resolution $P_\bullet \to M$ and set $\mathrm{Tor}_n^A(M,N) = H_n(P_\bullet \otimes_A N)$.

26.2 Independence of resolution

Comparison and homotopy theorems make Tor well-defined up to canonical isomorphism.

Example 26.1 (Tor of cyclic groups)**.** Resolve $\mathbb{Z}/m$ by $0 \to \mathbb{Z} \xrightarrow{\cdot m} \mathbb{Z} \to \mathbb{Z}/m \to 0$ and tensor with $\mathbb{Z}/n$. The kernel of multiplication by m on $\mathbb{Z}/n$ is $\mathbb{Z}/\gcd(m,n)$, so $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/m, \mathbb{Z}/n)$ has that form.

26.3 Degree zero

$\mathrm{Tor}_0^A(M,N) \cong M \otimes_A N$.

26.4 Long exact sequence

Short exact sequences in either variable yield long exact Tor sequences.

26.5 Projective vanishing

Higher Tor with a projective module vanishes.

Example 26.2 (Flatness kills Tor)**.** If F is flat, tensoring a projective resolution stays exact in positive degrees. Therefore $\mathrm{Tor}_i^R(M, F) = 0$ for every $i > 0$.

26.6 Flatness criterion

A module is flat iff $\mathrm{Tor}_1^A(-, F) = 0$ for all modules.

26.7 Symmetry

Over a commutative ring, Tor is symmetric in its two module arguments.

Example 26.3 (Tor detects lost injectivity)**.** Tensoring $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$ with $\mathbb{Z}/2$ destroys injectivity. The resulting kernel is exactly the nonzero Tor_1 term.

26.8 Cyclic calculation

For integers or quotient modules, short resolutions compute Tor explicitly.

26.9 Dimension shifting

Syzygies reduce higher Tor computations to lower degrees.

26.10 Core structural results

Theorem 26.4 (Well-definedness of Tor)**.** *The modules $H_n(P_\bullet \otimes N)$ are independent of the chosen projective resolution of M up to canonical isomorphism.*

Proof. Comparison maps between projective resolutions are unique up to chain homotopy. Tensor preserves chain homotopies, and chain-homotopic maps induce the same homology maps, yielding canonical isomorphisms. □

Theorem 26.5 (Projective vanishing)**.** *If P is projective, then $\mathrm{Tor}_n^A(P, N) = 0$ for $n > 0$.*

Proof. Use the resolution concentrated in degree zero at P. The tensor complex has no positive-degree terms, so its positive homology vanishes. □

Theorem 26.6 (Flatness via Tor)**.** *An A-module F is flat iff $\mathrm{Tor}_1^A(A/I, F) = 0$ for every finitely generated ideal I under the standard finite-ideal criterion, and in particular iff $\mathrm{Tor}_1^A(M, F) = 0$ for all M.*

Proof. Apply Tor to $0 \to I \to A \to A/I \to 0$. Vanishing of the connecting kernel is exactly injectivity of $I \otimes F \to F$; the general criterion follows from the flatness tests and long exact sequences. □

26.11 Worked examples

Example 26.7 (Definition diagnostic)**.** Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Choose a projective resolution $P_\bullet \to M$ and set $\mathrm{Tor}_n^A(M, N) = H_n(P_\bullet \otimes_A N)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 26.8 (Independence of resolution diagnostic)**.** Give a rigorous worked analysis of Independence of resolution. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Comparison and homotopy theorems make Tor well-defined up to canonical isomorphism. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 26.9 (Degree zero diagnostic)**.** Give a rigorous worked analysis of Degree zero. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\mathrm{Tor}_0^A(M, N) \cong M \otimes_A N$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 26.10 (Long exact sequence diagnostic)**.** Give a rigorous worked analysis of Long exact sequence. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Short exact sequences in either variable yield long exact Tor sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

26.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 26.11 (Definition diagnostic)**.** Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Choose a projective resolution $P_\bullet \to M$ and set $\mathrm{Tor}_n^A(M, N) = H_n(P_\bullet \otimes_A N)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 26.12 (Independence of resolution diagnostic)**.** Give a rigorous worked analysis of Independence of resolution. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Comparison and homotopy theorems make Tor well-defined up to canonical isomorphism. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 26.13 (Degree zero diagnostic)**.** Give a rigorous worked analysis of Degree zero. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\mathrm{Tor}_0^A(M, N) \cong M \otimes_A N$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 26.14 (Long exact sequence diagnostic)**.** Give a rigorous worked analysis of Long exact sequence. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Short exact sequences in either variable yield long exact Tor sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 26.15 (Projective vanishing diagnostic)**.** Give a rigorous worked analysis of Projective vanishing. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Higher Tor with a projective module vanishes. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 26.16 (Flatness criterion diagnostic)**.** Give a rigorous worked analysis of Flatness criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is flat iff $\mathrm{Tor}_1^A(-,F) = 0$ for all modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 26.17 (Symmetry diagnostic)**.** Give a rigorous worked analysis of Symmetry. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over a commutative ring, Tor is symmetric in its two module arguments. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 26.18 (Cyclic calculation diagnostic)**.** Give a rigorous worked analysis of Cyclic calculation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For integers or quotient modules, short resolutions compute Tor explicitly. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 26.19 (Well-definedness of Tor proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: The modules $H_n(P_\bullet \otimes N)$ are independent of the chosen projective resolution of M up to canonical isomorphism.

Solution. Comparison maps between projective resolutions are unique up to chain homotopy. Tensor preserves chain homotopies, and chain-homotopic maps induce the same homology maps, yielding canonical isomorphisms. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 26.20 (Projective vanishing proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If P is projective, then $\mathrm{Tor}_n^A(P,N) = 0$ for $n > 0$.

Solution. Use the resolution concentrated in degree zero at P. The tensor complex has no positive-degree terms, so its positive homology vanishes. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 26.21 (Flatness via Tor proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: An A-module F is flat iff $\mathrm{Tor}_1^A(A/I, F) = 0$ for every finitely generated ideal I under the standard finite-ideal criterion, and in particular iff $\mathrm{Tor}_1^A(M, F) = 0$ for all M.

Solution. Apply Tor to $0 \to I \to A \to A/I \to 0$. Vanishing of the connecting kernel is exactly injectivity of $I \otimes F \to F$; the general criterion follows from the flatness tests and long exact sequences. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 26.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. The Tor functor is the homology left behind when tensor product fails to be exact. It is computed from projective resolutions and detects torsion, non-flatness, and derived intersections. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

26.13 Exercises with complete solutions

Exercise 26.23. State and apply the central principle behind Definition. Give one concrete algebraic consequence.

Hint. Choose a free/projective resolution $P_\bullet \to M$, tensor it with N, then compute homology of the resulting complex. □

Solution. Choose a projective resolution $P_\bullet \to M$ and set $\mathrm{Tor}_n^A(M, N) = H_n(P_\bullet \otimes_A N)$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 26.24. State and apply the central principle behind Independence of resolution. Give one concrete algebraic consequence.

Hint. Degree zero homology is $\mathrm{coker}(P_1 \otimes N \to P_0 \otimes N)$, naturally $M \otimes N$. □

Solution. Comparison and homotopy theorems make Tor well-defined up to canonical isomorphism. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 26.25. State and apply the central principle behind Degree zero. Give one concrete algebraic consequence.

Hint. For cyclic quotients, use a short resolution by multiplication and compute kernel/cokernel after tensoring. □

Solution. $\mathrm{Tor}_0^A(M, N) \cong M \otimes_A N$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 26.26. State and apply the central principle behind Long exact sequence. Give one concrete algebraic consequence.

Hint. A short exact sequence in one variable gives a long exact Tor sequence; write the neighboring terms before solving for the unknown group. □

Solution. Short exact sequences in either variable yield long exact Tor sequences. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 26.27. State and apply the central principle behind Projective vanishing. Give one concrete algebraic consequence.

Hint. Nonzero Tor_1 records failure of tensoring to preserve injectivity; compare with the flatness criterion. □

Solution. Higher Tor with a projective module vanishes. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 26.28. State and apply the central principle behind Flatness criterion. Give one concrete algebraic consequence.

Hint. Projective modules have vanishing higher Tor because they admit a resolution concentrated in degree zero. □

Solution. A module is flat iff $\mathrm{Tor}_1^A(-, F) = 0$ for all modules. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 26.29. State and apply the central principle behind Symmetry. Give one concrete algebraic consequence.

Hint. Independence of resolution comes from comparison maps unique up to homotopy; do not compare chain groups literally. □

Solution. Over a commutative ring, Tor is symmetric in its two module arguments. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 26.30. State and apply the central principle behind Cyclic calculation. Give one concrete algebraic consequence.

Hint. Track which argument is fixed and how ring/module structures act before writing Tor over A. □

Solution. For integers or quotient modules, short resolutions compute Tor explicitly. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

26.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 26.31 (Cyclic Tor)**.** Compute $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/6, \mathbb{Z}/15)$.

Hint. Use tensoring resolutions and Tor homology. ☐

Solution. It is $\mathbb{Z}/3$. ☐

Exercise 26.32 (Coprime Tor)**.** Compute $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/4, \mathbb{Z}/9)$.

Hint. Use tensoring resolutions and Tor homology. ☐

Solution. It is zero. ☐

Exercise 26.33 (Free argument)**.** Compute $\mathrm{Tor}_1^{R}(R^n, M)$.

Hint. Use tensoring resolutions and Tor homology. ☐

Solution. It is zero. ☐

Exercise 26.34 (Flat argument)**.** Compute positive Tor with a flat second module.

Hint. Use tensoring resolutions and Tor homology. ☐

Solution. It vanishes. ☐

Exercise 26.35 (Degree zero)**.** What is $\mathrm{Tor}_0^{R}(M, N)$?

Hint. Use tensoring resolutions and Tor homology. ☐

Solution. It is $M \otimes_R N$. ☐

Proofs

Exercise 26.36 (Well-definedness of Tor proof)**.** Prove: The modules $H_n(P_\bullet \otimes N)$ are independent of the chosen projective resolution of M up to canonical isomorphism.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Comparison maps between projective resolutions are unique up to chain homotopy. Tensor preserves chain homotopies, and chain-homotopic maps induce the same homology maps, yielding canonical isomorphisms. ☐

Exercise 26.37 (Projective vanishing proof)**.** Prove: If P is projective, then $\mathrm{Tor}_n^{A}(P, N) = 0$ for $n > 0$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Use the resolution concentrated in degree zero at P. The tensor complex has no positive-degree terms, so its positive homology vanishes. ☐

Exercise 26.38 (Flatness via Tor proof)**.** Prove: An A-module F is flat iff $\mathrm{Tor}_1^{A}(A/I, F) = 0$ for every finitely generated ideal I under the standard finite-ideal criterion, and in particular iff $\mathrm{Tor}_1^{A}(M, F) = 0$ for all M.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Apply Tor to $0 \to I \to A \to A/I \to 0$. Vanishing of the connecting kernel is exactly injectivity of $I \otimes F \to F$; the general criterion follows from the flatness tests and long exact sequences. ☐

Exercise 26.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Well-definedness of Tor and Projective vanishing.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. ☐

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. ☐

Counterexamples and hypothesis tests

Exercise 26.40 (Always zero)**.** Test the claim: Tor_1 always vanishes.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False. ☐

Exercise 26.41 (Flat criterion)**.** Test the claim: A flat module can have nonzero positive Tor with some module.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False. ☐

Exercise 26.42 (Resolution dependence)**.** Test the claim: Tor depends on the chosen projective resolution.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False up to canonical isomorphism. ☐

Applications and investigations

Exercise 26.43 (Flatness test)**.** How does Tor measure flatness?

Hint. Translate the question into tensoring resolutions and Tor homology. ☐

Solution. Nonzero Tor_1 records failure of tensoring to preserve injections. ☐

Exercise 26.44 (Intersection excess)**.** Why does Tor appear in geometric intersections?

Hint. Translate the question into tensoring resolutions and Tor homology. ☐

Solution. It measures higher failure of naive tensor products to capture nontransverse intersections. ☐

Challenge problems

Exercise 26.45 (Local-global synthesis)**.** Connect 'Definition' with 'Dimension shifting' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. ☐

Solution. The opening idea is: Choose a projective resolution $P_\bullet \to M$ and set $\mathrm{Tor}_n^A(M,N) = H_n(P_\bullet \otimes_A N)$. The later viewpoint is: Syzygies reduce higher Tor computations to lower degrees. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. ☐

Exercise 26.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. □

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. □

Chapter 27

The Ext Functor

The Ext functor is the derived correction to the left exact Hom functor. It classifies extensions in degree one and measures higher obstructions to lifting maps.

Learning goals

After this chapter, the reader should be able to:

- compute Ext groups from projective resolutions and Hom complexes;
- verify that Ext^0 recovers ordinary Hom;
- use long exact Ext sequences and connecting morphisms in concrete examples;
- interpret Ext^1 in terms of extensions where developed;
- compute Ext for cyclic modules or quotients from short free resolutions;
- distinguish variance in the first and second arguments and track induced maps correctly;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \longrightarrow \text{exactness/localization} \longrightarrow \text{structural consequence.}}$$

27.1 Definition from projectives

Choose a projective resolution $P_\bullet \to M$ and set $\mathrm{Ext}^n_A(M, N) = H^n(\mathrm{Hom}_A(P_\bullet, N))$.

27.2 Degree zero

$\mathrm{Ext}^0_A(M, N) \cong \mathrm{Hom}_A(M, N)$.

Example 27.1 (Ext of cyclic groups)**.** Applying $\mathrm{Hom}_{\mathbb{Z}}(-, \mathbb{Z}/n)$ to the standard resolution of $\mathbb{Z}/m$ gives a cokernel of multiplication by m. Hence $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n)$.

27.3 Long exact sequence

Short exact sequences yield long exact Ext sequences.

27.4 Projective vanishing

Higher Ext out of a projective module is zero.

27.5 Injective viewpoint

Ext can also be computed using injective resolutions in the second variable.

Example 27.2 (Projectives kill Ext)**.** If P is projective, it has a length-zero projective resolution. Therefore $\mathrm{Ext}^i_R(P, N) = 0$ for every $i > 0$.

27.6 Extension classes

$\mathrm{Ext}^1(M, N)$ classifies short exact sequences $0 \to N \to E \to M \to 0$ up to the appropriate equivalence.

27.7 Dimension shifting

Syzygies reduce higher Ext to lower Ext.

Example 27.3 (Extensions classified by Ext)**.** Elements of $\mathrm{Ext}^1_R(M, N)$ correspond to equivalence classes of short exact sequences $0 \to N \to E \to M \to 0$. The zero class is represented by the split extension.

27.8 Local and graded variants

Localization and graded structures refine Ext in common Noetherian settings.

27.9 Duality preview

Ext groups appear in duality, canonical modules, and deformation theory.

27.10 Core structural results

Theorem 27.4 (Well-definedness of Ext)**.** *$H^n(\mathrm{Hom}(P_\bullet, N))$ is independent of the chosen projective resolution of M.*

Proof. Comparison maps between resolutions are unique up to chain homotopy. Applying Hom turns chain homotopies into cochain homotopies, which induce identical cohomology maps. □

Theorem 27.5 (Projective vanishing)**.** *If P is projective, then $\mathrm{Ext}^n_A(P, N) = 0$ for $n > 0$.*

Proof. Use the projective resolution concentrated in degree zero. The resulting Hom cochain complex has no positive cohomology. □

Theorem 27.6 (Ext one classifies extensions)**.** *Equivalence classes of short exact sequences $0 \to N \to E \to M \to 0$ form a group naturally isomorphic to $\mathrm{Ext}^1_A(M, N)$.*

Proof. Pull an extension back along a projective presentation of M and choose a splitting over the projective term. The resulting map from the first syzygy to N defines a degree-one cohomology class; changing the splitting changes it by a coboundary, and the construction is reversible via pushout. □

27.11 Worked examples

Example 27.7 (Definition from projectives diagnostic)**.** Give a rigorous worked analysis of Definition from projectives. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Choose a projective resolution $P_\bullet \to M$ and set $\mathrm{Ext}^n_A(M, N) = H^n(\mathrm{Hom}_A(P_\bullet, N))$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 27.8 (Degree zero diagnostic)**.** Give a rigorous worked analysis of Degree zero. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\mathrm{Ext}^0_A(M, N) \cong \mathrm{Hom}_A(M, N)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 27.9 (Long exact sequence diagnostic)**.** Give a rigorous worked analysis of Long exact sequence. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Short exact sequences yield long exact Ext sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 27.10 (Projective vanishing diagnostic)**.** Give a rigorous worked analysis of Projective vanishing. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Higher Ext out of a projective module is zero. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

27.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 27.11 (Definition from projectives diagnostic)**.** Give a rigorous worked analysis of Definition from projectives. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Choose a projective resolution $P_\bullet \to M$ and set $\mathrm{Ext}^n_A(M, N) = H^n(\mathrm{Hom}_A(P_\bullet, N))$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 27.12 (Degree zero diagnostic)**.** Give a rigorous worked analysis of Degree zero. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\mathrm{Ext}^0_A(M, N) \cong \mathrm{Hom}_A(M, N)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 27.13 (Long exact sequence diagnostic)**.** Give a rigorous worked analysis of Long exact sequence. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Short exact sequences yield long exact Ext sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 27.14 (Projective vanishing diagnostic)**.** Give a rigorous worked analysis of Projective vanishing. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Higher Ext out of a projective module is zero. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 27.15 (Injective viewpoint diagnostic)**.** Give a rigorous worked analysis of Injective viewpoint. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Ext can also be computed using injective resolutions in the second variable. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 27.16 (Extension classes diagnostic)**.** Give a rigorous worked analysis of Extension classes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\mathrm{Ext}^1(M, N)$ classifies short exact sequences $0 \to N \to E \to M \to 0$ up to the appropriate equivalence. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 27.17 (Dimension shifting diagnostic)**.** Give a rigorous worked analysis of Dimension shifting. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Syzygies reduce higher Ext to lower Ext. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 27.18 (Local and graded variants diagnostic)**.** Give a rigorous worked analysis of Local and graded variants. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization and graded structures refine Ext in common Noetherian settings. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. ☐

Problem 27.19 (Well-definedness of Ext proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: $H^n(\mathrm{Hom}(P_\bullet, N))$ is independent of the chosen projective resolution of M.

Solution. Comparison maps between resolutions are unique up to chain homotopy. Applying Hom turns chain homotopies into cochain homotopies, which induce identical cohomology maps. This proof isolates the reusable algebraic mechanism behind the result. ☐

Problem 27.20 (Projective vanishing proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: If P is projective, then $\mathrm{Ext}^n_A(P, N) = 0$ for $n > 0$.

Solution. Use the projective resolution concentrated in degree zero. The resulting Hom cochain complex has no positive cohomology. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 27.21 (Ext one classifies extensions proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: Equivalence classes of short exact sequences $0 \to N \to E \to M \to 0$ form a group naturally isomorphic to $\mathrm{Ext}^1_A(M, N)$.

Solution. Pull an extension back along a projective presentation of M and choose a splitting over the projective term. The resulting map from the first syzygy to N defines a degree-one cohomology class; changing the splitting changes it by a coboundary, and the construction is reversible via pushout. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 27.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. The Ext functor is the derived correction to the left exact Hom functor. It classifies extensions in degree one and measures higher obstructions to lifting maps. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

27.13 Exercises with complete solutions

Exercise 27.23. State and apply the central principle behind Definition from projectives. Give one concrete algebraic consequence.

Hint. Choose a projective resolution $P_\bullet \to M$ and apply $\mathrm{Hom}_A(P_\bullet, N)$; Ext is the cohomology of that cochain complex. □

Solution. Choose a projective resolution $P_\bullet \to M$ and set $\mathrm{Ext}^n_A(M, N) = H^n(\mathrm{Hom}_A(P_\bullet, N))$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 27.24. State and apply the central principle behind Degree zero. Give one concrete algebraic consequence.

Hint. Degree zero cohomology recovers $\mathrm{Hom}_A(M, N)$; verify this from the first two terms of the resolution. □

Solution. $\mathrm{Ext}^0_A(M, N) \cong \mathrm{Hom}_A(M, N)$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. □

Exercise 27.25. State and apply the central principle behind Long exact sequence. Give one concrete algebraic consequence.

Hint. For a cyclic quotient, apply Hom to a short free resolution and compute kernel/cokernel of multiplication. □

Solution. Short exact sequences yield long exact Ext sequences. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 27.26. State and apply the central principle behind Projective vanishing. Give one concrete algebraic consequence.

Hint. A short exact sequence yields a long exact Ext sequence; mark variance so arrows go in the correct direction. ☐

Solution. Higher Ext out of a projective module is zero. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 27.27. State and apply the central principle behind Injective viewpoint. Give one concrete algebraic consequence.

Hint. Ext^1 classifies extensions in the chapter's setting; identify the split extension as the zero class. ☐

Solution. Ext can also be computed using injective resolutions in the second variable. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 27.28. State and apply the central principle behind Extension classes. Give one concrete algebraic consequence.

Hint. Projective first arguments have vanishing higher Ext when computed projectively. ☐

Solution. $\mathrm{Ext}^1(M, N)$ classifies short exact sequences $0 \to N \to E \to M \to 0$ up to the appropriate equivalence. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 27.29. State and apply the central principle behind Dimension shifting. Give one concrete algebraic consequence.

Hint. Changing the second argument is covariant while changing the first is contravariant; write induced maps before composing them. ☐

Solution. Syzygies reduce higher Ext to lower Ext. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 27.30. State and apply the central principle behind Local and graded variants. Give one concrete algebraic consequence.

Hint. Independence of the chosen resolution is a homotopy-invariance statement, not equality of the resulting cochain complexes. ☐

Solution. Localization and graded structures refine Ext in common Noetherian settings. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

27.14 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 27.31 (Cyclic Ext)**.** Compute $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/6, \mathbb{Z}/15)$.

Hint. Use Hom on projective resolutions and extension classes. ☐

Solution. It is $\mathbb{Z}/3$. ☐

Exercise 27.32 (Coprime Ext)**.** Compute $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/4, \mathbb{Z}/9)$.

Hint. Use Hom on projective resolutions and extension classes. ☐

Solution. It is zero. ☐

Exercise 27.33 (Projective)**.** Compute positive Ext from a free module.

Hint. Use Hom on projective resolutions and extension classes. ☐

Solution. It vanishes. ☐

Exercise 27.34 (Degree zero)**.** What is $\mathrm{Ext}^0_R(M, N)$?

Hint. Use Hom on projective resolutions and extension classes. ☐

Solution. It is $\mathrm{Hom}_R(M, N)$. ☐

Exercise 27.35 (Split class)**.** Which Ext class corresponds to a split short exact sequence?

Hint. Use Hom on projective resolutions and extension classes. ☐

Solution. The zero class. ☐

Proofs

Exercise 27.36 (Well-definedness of Ext proof)**.** Prove: $H^n(\mathrm{Hom}(P_\bullet, N))$ is independent of the chosen projective resolution of M.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Comparison maps between resolutions are unique up to chain homotopy. Applying Hom turns chain homotopies into cochain homotopies, which induce identical cohomology maps. ☐

Exercise 27.37 (Projective vanishing proof)**.** Prove: If P is projective, then $\mathrm{Ext}^n_A(P, N) = 0$ for $n > 0$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Use the projective resolution concentrated in degree zero. The resulting Hom cochain complex has no positive cohomology. ☐

Exercise 27.38 (Ext one classifies extensions proof)**.** Prove: Equivalence classes of short exact sequences $0 \to N \to E \to M \to 0$ form a group naturally isomorphic to $\mathrm{Ext}^1_A(M, N)$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Pull an extension back along a projective presentation of M and choose a splitting over the projective term. The resulting map from the first syzygy to N defines a degree-one cohomology class; changing the splitting changes it by a coboundary, and the construction is reversible via pushout. ☐

Exercise 27.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Well-definedness of Ext and Projective vanishing.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. ☐

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. ☐

Counterexamples and hypothesis tests

Exercise 27.40 (Always split)**.** Test the claim: Every short exact sequence of modules splits.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; nonzero Ext classes obstruct splitting. ☐

Exercise 27.41 (Projective first variable)**.** Test the claim: A projective first argument can have nonzero positive Ext.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False. ☐

Exercise 27.42 (Resolution dependence)**.** Test the claim: Ext depends on arbitrary choices in a projective resolution.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False up to canonical isomorphism. ☐

Applications and investigations

Exercise 27.43 (Extension obstruction)**.** What does Ext^1 measure?

Hint. Translate the question into Hom on projective resolutions and extension classes. ☐

Solution. It measures equivalence classes of extensions and obstruction to splitting. ☐

Exercise 27.44 (Deformation flavor)**.** Why does Ext occur in deformation theory?

Hint. Translate the question into Hom on projective resolutions and extension classes. ☐

Solution. Extension groups encode first-order gluing and obstruction data in many algebraic settings. ☐

Challenge problems

Exercise 27.45 (Local-global synthesis)**.** Connect 'Definition from projectives' with 'Duality preview' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. ☐

Solution. The opening idea is: Choose a projective resolution $P_\bullet \to M$ and set $\mathrm{Ext}^n_A(M, N) = H^n(\mathrm{Hom}_A(P_\bullet, N))$. The later viewpoint is: Ext groups appear in duality, canonical modules, and deformation theory. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. ☐

Exercise 27.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. ☐

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. ☐

Chapter 28

Derived-Functor Viewpoint

Derived functors organize Tor, Ext, and similar constructions as systematic corrections to nonexact functors. Resolutions make these corrections computable, while long exact sequences expose their functorial structure.

Learning goals

After this chapter, the reader should be able to:

- identify which failure of exactness a derived functor is measuring;
- construct left derived functors from projective or flat resolutions and right derived functors from injective resolutions;
- explain Tor as derived tensor and Ext as derived Hom;
- derive long exact sequences from short exact sequences of complexes or resolved objects;
- compare different resolutions using chain-homotopy and universality principles;
- connect derived-functor computations to later geometric uses such as sheaf cohomology and intersection theory;

Conceptual roadmap

$$\boxed{\text{presentation or universal property} \;\longrightarrow\; \text{exactness/localization} \;\longrightarrow\; \text{structural consequence.}}$$

28.1 Exact and derived behavior

Left or right exact functors lose information at kernels or cokernels; derived functors measure that failure in higher degrees.

28.2 Left derived functors

Projective or flat resolutions produce left derived functors of right exact functors.

Example 28.1 (Deriving a right-exact functor)**.** Tensor product is right exact but may fail on kernels. Replacing an input by a projective resolution and taking homology produces its left derived functors, the Tor groups.

28.3 Right derived functors

Injective resolutions produce right derived functors of left exact functors.

28.4 Tor as derived tensor

Tor_n is the left derived functor of tensor product.

28.5 Ext as derived Hom

Ext^n is the right derived functor of Hom in the second variable or equivalently computed projectively in the first.

Example 28.2 (Deriving a left-exact functor)**.** Hom in the second variable is left exact. Applying it to a projective resolution in the first variable and taking cohomology produces Ext, which records the missing higher exactness.

28.6 Universality

Derived functors are characterized by effaceability and universal delta-functor properties.

28.7 Long exact sequences

Connecting morphisms package the failure of exactness into functorial long exact sequences.

Example 28.3 (Long exact sequence)**.** A short exact sequence of modules induces long exact sequences in Tor or Ext. The connecting morphisms propagate the failure of exactness into neighboring derived degrees.

28.8 Acyclic resolutions

One may compute derived functors using suitable classes of acyclic objects, not only projectives or injectives.

28.9 Derived-category preview

Complexes and quasi-isomorphisms provide a setting in which derived functors become ordinary functors after localization.

28.10 Algebra-geometry bridge

Derived tensor and derived Hom underpin intersection theory, deformation theory, and sheaf cohomology.

28.11 Core structural results

Theorem 28.4 (Tor is the left derived tensor functor)**.** *For fixed N, the homology of a projective resolution tensor N gives the left derived functors of $- \otimes_A N$.*

Proof. Projective resolutions are acyclic for tensor in positive resolution degrees and comparison maps make the construction functorial and independent of resolution; degree zero recovers ordinary tensor. ☐

Theorem 28.5 (Ext is a derived Hom functor)**.** *For fixed N, cohomology of* $\mathrm{Hom}(P_\bullet, N)$ *gives derived functors extending* $\mathrm{Hom}(-, N)$ *contravariantly in the first variable.*

Proof. Projective resolutions replace the first variable by Hom-acyclic objects. Comparison and homotopy yield functorial well-defined cohomology, and degree zero recovers Hom. ☐

Theorem 28.6 (Derived functors produce long exact sequences)**.** *A short exact sequence in the appropriate variable induces a natural long exact sequence of derived functors.*

Proof. Resolve compatibly or apply the functor to a short exact sequence of suitable resolutions. The resulting short exact sequence of complexes yields the standard long exact sequence in homology or cohomology via connecting morphisms. ☐

28.12 Worked examples

Example 28.7 (Exact and derived behavior diagnostic)**.** Give a rigorous worked analysis of Exact and derived behavior. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Left or right exact functors lose information at kernels or cokernels; derived functors measure that failure in higher degrees. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 28.8 (Left derived functors diagnostic)**.** Give a rigorous worked analysis of Left derived functors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Projective or flat resolutions produce left derived functors of right exact functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 28.9 (Right derived functors diagnostic)**.** Give a rigorous worked analysis of Right derived functors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Injective resolutions produce right derived functors of left exact functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 28.10 (Tor as derived tensor diagnostic)**.** Give a rigorous worked analysis of Tor as derived tensor. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Tor_n is the left derived functor of tensor product. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

28.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 28.11 (Exact and derived behavior diagnostic)**.** Give a rigorous worked analysis of Exact and derived behavior. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Left or right exact functors lose information at kernels or cokernels; derived functors measure that failure in higher degrees. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 28.12 (Left derived functors diagnostic)**.** Give a rigorous worked analysis of Left derived functors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Projective or flat resolutions produce left derived functors of right exact functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 28.13 (Right derived functors diagnostic)**.** Give a rigorous worked analysis of Right derived functors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Injective resolutions produce right derived functors of left exact functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 28.14 (Tor as derived tensor diagnostic)**.** Give a rigorous worked analysis of Tor as derived tensor. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Tor_n is the left derived functor of tensor product. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 28.15 (Ext as derived Hom diagnostic)**.** Give a rigorous worked analysis of Ext as derived Hom. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Ext^n is the right derived functor of Hom in the second variable or equivalently computed projectively in the first. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 28.16 (Universality diagnostic)**.** Give a rigorous worked analysis of Universality. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Derived functors are characterized by effaceability and universal delta-functor properties. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 28.17 (Long exact sequences diagnostic)**.** Give a rigorous worked analysis of Long exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Connecting morphisms package the failure of exactness into functorial long exact sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 28.18 (Acyclic resolutions diagnostic)**.** Give a rigorous worked analysis of Acyclic resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. One may compute derived functors using suitable classes of acyclic objects, not only projectives or injectives. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. □

Problem 28.19 (Tor is the left derived tensor functor proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For fixed N, the homology of a projective resolution tensor N gives the left derived functors of $- \otimes_A N$.

Solution. Projective resolutions are acyclic for tensor in positive resolution degrees and comparison maps make the construction functorial and independent of resolution; degree zero recovers ordinary tensor. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 28.20 (Ext is a derived Hom functor proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: For fixed N, cohomology of $\mathrm{Hom}(P_\bullet, N)$ gives derived functors extending $\mathrm{Hom}(-, N)$ contravariantly in the first variable.

Solution. Projective resolutions replace the first variable by Hom-acyclic objects. Comparison and homotopy yield functorial well-defined cohomology, and degree zero recovers Hom. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 28.21 (Derived functors produce long exact sequences proof dossier)**.** Prove the following structural statement and identify the hypothesis that makes the conclusion work: A short exact sequence in the appropriate variable induces a natural long exact sequence of derived functors.

Solution. Resolve compatibly or apply the functor to a short exact sequence of suitable resolutions. The resulting short exact sequence of complexes yields the standard long exact sequence in homology or cohomology via connecting morphisms. This proof isolates the reusable algebraic mechanism behind the result. □

Problem 28.22 (Chapter synthesis)**.** Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Derived functors organize Tor, Ext, and similar constructions as systematic corrections to nonexact functors. Resolutions make these corrections computable, while long exact sequences expose their functorial structure. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. □

28.14 Exercises with complete solutions

Exercise 28.23. State and apply the central principle behind Exact and derived behavior. Give one concrete algebraic consequence.

Hint. First identify whether the original functor is left exact or right exact; that determines whether right or left derived functors are needed. □

Solution. Left or right exact functors lose information at kernels or cokernels; derived functors measure that failure in higher degrees. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 28.24. State and apply the central principle behind Left derived functors. Give one concrete algebraic consequence.

Hint. For a right exact functor, resolve by projectives/flats and take homology after applying the functor. ☐

Solution. Projective or flat resolutions produce left derived functors of right exact functors. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 28.25. State and apply the central principle behind Right derived functors. Give one concrete algebraic consequence.

Hint. For a left exact functor, use injective resolutions or an equivalent acyclic class and take cohomology. ☐

Solution. Injective resolutions produce right derived functors of left exact functors. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 28.26. State and apply the central principle behind Tor as derived tensor. Give one concrete algebraic consequence.

Hint. Tor_n is obtained by deriving tensor product; check degree zero recovers ordinary tensor before interpreting higher degrees. ☐

Solution. Tor_n is the left derived functor of tensor product. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 28.27. State and apply the central principle behind Ext as derived Hom. Give one concrete algebraic consequence.

Hint. Ext^n is obtained by deriving Hom; track whether the projective resolution is in the contravariant first variable or an injective resolution in the second. ☐

Solution. Ext^n is the right derived functor of Hom in the second variable or equivalently computed projectively in the first. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 28.28. State and apply the central principle behind Universality. Give one concrete algebraic consequence.

Hint. Long exact sequences come from short exact sequences of complexes and the connecting homomorphism in homology/cohomology. ☐

Solution. Derived functors are characterized by effaceability and universal delta-functor properties. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 28.29. State and apply the central principle behind Long exact sequences. Give one concrete algebraic consequence.

Hint. To compare resolutions, use comparison maps and chain homotopies; invariance is not a statement that the resolutions are identical. ☐

Solution. Connecting morphisms package the failure of exactness into functorial long exact sequences. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Exercise 28.30. State and apply the central principle behind Acyclic resolutions. Give one concrete algebraic consequence.

Hint. The derived-category viewpoint localizes quasi-isomorphisms, turning resolution-dependent constructions into ordinary functorial objects up to the intended equivalence. ☐

Solution. One may compute derived functors using suitable classes of acyclic objects, not only projectives or injectives. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. ☐

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

28.15 Graded supplementary exercises

These exercises extend the protected chapter with a balanced set of computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 28.31 (Tensor derived)**.** What are the left derived functors of tensor product?

Hint. Use resolutions, exactness defects, and derived functors. ☐

Solution. They are the Tor functors. ☐

Exercise 28.32 (Hom derived)**.** What are the right-derived-type groups associated with Hom in this projective-resolution viewpoint?

Hint. Use resolutions, exactness defects, and derived functors. ☐

Solution. They are the Ext functors. ☐

Exercise 28.33 (Degree zero Tor)**.** Identify Tor_0.

Hint. Use resolutions, exactness defects, and derived functors. ☐

Solution. Tensor product. ☐

Exercise 28.34 (Degree zero Ext)**.** Identify Ext^0.

Hint. Use resolutions, exactness defects, and derived functors. ☐

Solution. Hom. ☐

Exercise 28.35 (Projective input)**.** What happens to positive derived groups on projective inputs?

Hint. Use resolutions, exactness defects, and derived functors. ☐

Solution. They vanish. ☐

Proofs

Exercise 28.36 (Tor is the left derived tensor functor proof)**.** Prove: For fixed N, the homology of a projective resolution tensor N gives the left derived functors of $- \otimes_A N$.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Projective resolutions are acyclic for tensor in positive resolution degrees and comparison maps make the construction functorial and independent of resolution; degree zero recovers ordinary tensor. ☐

Exercise 28.37 (Ext is a derived Hom functor proof)**.** Prove: For fixed N, cohomology of $\mathrm{Hom}(P_\bullet, N)$ gives derived functors extending $\mathrm{Hom}(-, N)$ contravariantly in the first variable.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Projective resolutions replace the first variable by Hom-acyclic objects. Comparison and homotopy yield functorial well-defined cohomology, and degree zero recovers Hom. ☐

Exercise 28.38 (Derived functors produce long exact sequences proof)**.** Prove: A short exact sequence in the appropriate variable induces a natural long exact sequence of derived functors.

Hint. Start from the theorem hypotheses and identify the decisive algebraic construction. ☐

Solution. Resolve compatibly or apply the functor to a short exact sequence of suitable resolutions. The resulting short exact sequence of complexes yields the standard long exact sequence in homology or cohomology via connecting morphisms. ☐

Exercise 28.39 (Structural synthesis)**.** Prove a corollary that genuinely uses both Tor is the left derived tensor functor and Ext is a derived Hom functor.

Hint. Apply the first theorem to obtain its structural description, then verify the second theorem's hypotheses. ☐

Solution. A correct proof explicitly invokes both results, verifies the common hypotheses, and derives a new consequence rather than merely restating either theorem. ☐

Counterexamples and hypothesis tests

Exercise 28.40 (Exact functor)**.** Test the claim: An exact functor has nonzero higher derived functors.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; higher derived functors vanish. ☐

Exercise 28.41 (Choice dependence)**.** Test the claim: Derived functors depend essentially on the chosen resolution.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; comparison theorems give canonical invariance. ☐

Exercise 28.42 (Short sequence only)**.** Test the claim: A short exact sequence yields only another short exact sequence after a nonexact functor.

Hint. Use the smallest concrete ring, module, or resolution that can decide the claim. ☐

Solution. False; derived functors organize the defect into a long exact sequence. ☐

Applications and investigations

Exercise 28.43 (Unified viewpoint)**.** How do Tor and Ext fit one framework?

Hint. Translate the question into resolutions, exactness defects, and derived functors. ☐

Solution. They are homology or cohomology of resolutions measuring failure of ordinary functors to be exact. ☐

Exercise 28.44 (Computation strategy)**.** Why are resolutions central to derived methods?

Hint. Translate the question into resolutions, exactness defects, and derived functors. ☐

Solution. They replace arbitrary modules by acyclic projective objects where the functor is easier to compute. ☐

Challenge problems

Exercise 28.45 (Local-global synthesis)**.** Connect 'Exact and derived behavior' with 'Algebra-geometry bridge' in one rigorous argument.

Hint. State the relevant ring map, module map, or complex before applying the structural theorem. ☐

Solution. The opening idea is: Left or right exact functors lose information at kernels or cokernels; derived functors measure that failure in higher degrees. The later viewpoint is: Derived tensor and derived Hom underpin intersection theory, deformation theory, and sheaf cohomology. A complete solution identifies the structural theorem that links these descriptions and checks its hypotheses. ☐

Exercise 28.46 (Hypothesis audit)**.** Choose one theorem from the chapter, remove one essential hypothesis, and exhibit or explain a concrete failure.

Hint. Use one of the chapter's explicit computations or counterexamples as the test case. ☐

Solution. The solution must name the removed hypothesis, show exactly which proof step breaks, and give a concrete algebraic object where the claimed conclusion fails. ☐